# The Global Allocative Efficiency of Deforestation

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**Abstract**

Deforestation is predominantly driven by expanding crop and cattle production. Different forests are differently emissions intensive to remove, and bear different agricultural returns. Is deforestation occurring on the right land? Using global high-resolution satellite data on agricultural returns and emissions, I estimate the jointly determined supply curve of deforestation $CO_2$ emissions, meat, and crops. I embed this supply curve in a general equilibrium trade model. A social planner who values carbon at \$190 per ton implements a Pigouvian tax on deforestation. The optimal policy avoids 56.2% of deforestation emissions since 2000 at an average cost of \$1.30 per ton per year. 25% of first-best welfare gains are achieved by purely reallocating deforestation to less emissions-intensive forest, holding food production and forest area fixed. Welfare costs of optimal policy are concentrated on tropical producers, so transfers are required sustain cooperation. Given the option to deviate, major forest basins individually rationally choose not to price deforestation. A bilateral contract mechanism between a high-income donor country and a forest-rich recipient achieves constrained-optimal welfare gains, but multilateral contracts unravel because recipient countries free-ride.

*Keywords*: deforestation, carbon tax, misallocation, trade, land use

*JEL codes*: Q2, Q56, Q57, Q15, D61, H23

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# 1 Introduction

Carbon dioxide emissions have risen precipitously over the last century. Without intervention, global temperatures are projected to exceed 1.5°C above pre-industrial levels before 2100. Climate policy has slowed emissions growth, primarily by targeting the manufacturing and transportation sectors (Gillingham and Stock 2018). However, policies targeting deforestation still struggle to achieve scale (Balboni et al. 2023). In the last decade, emissions from agricultural deforestation, the main driver of deforestation (Curtis et al. 2018), drove roughly one-tenth of annual global carbon dioxide emissions (Friedlingstein et al. 2023).[1] Deforestation policy in the coming decades must balance a fundamental trade off: more carbon sequestration, or more food production.

This paper offers the first quantitative assessment of how inefficient deforestation has been and what, if anything, can be done about it. Two complexities historically hindered an answer. First, tropical deforestation is concentrated — since 1982, 62% of global deforestation emissions occurred on 3.1% of global land — necessitating granular measurement of yields and carbon emissions. Second, measuring reallocation requires an equilibrium framework: agriculture from deforesting regions is traded on global, integrated markets.

To make progress, I assemble global data on agricultural yields, deforestation emissions, land use, and transportation costs. The data consists of 1,591,999 plots of land at 0.1 arc-degree resolution. I construct a measure of agricultural profitability: revenue (country-level price times "best-case" plot-level potential yields in 2000) less yield-scaled transport costs.[2] Pastoral yields are country-level meat tonnage per pastoral hectare. 79% of deforestation earned less in agricultural flow returns than the annualized social cost of $CO_2$ emissions, valued at a \$190/ton social cost of carbon (SCC, U.S. Environmental Protection Agency 2023). This fact hints at the scale of inefficiency. To understand *where* deforestation should instead occur, I need a framework which characterizes how agricultural prices and input costs would change under counterfactual production geographies.

I develop an equilibrium trade framework which characterizes the plot-level geography of deforestation and food production. Landowners allocate plots to crops, pasture, forest, or other natural use, trading off profits, use-specific amenities, and a labor wage. A constant-returns outside sector competes for labor. Clearing for crops or pasture emits $CO_2$ and ships goods through an intermediate transport sector. Agriculture and pasture compete with the outside sector for labor. Consumers with nested constant elasticity of substitution preferences substitute across origin countries and across crops, meat, and the outside good. A tax levied in one country moves prices abroad. With lump-sum transfers, a utilitarian planner optimally sets a Pigouvian $CO_2$ tax.

1. Carbon dioxide, or $CO_2$, emissions from deforestation come from burning trees. Trees, in the process of photosynthesis, take in carbon dioxide and release oxygen, storing atmospheric carbon as woody, carbon-rich, biomass.
2. Potential yields are best-case yields given climate and soil characteristics. They should be thought of as "in-laboratory" yields.

Two empirical objects quantify this benchmark. On the supply side, the elasticity of landowners to a change in profits determines the scale of food loss and $CO_2$ reductions. On the demand side, a tax raises food prices where deforestation falls and, through trade, abroad. The welfare cost of these price increases depends on how consumers substitute across origins and across goods.

On the supply-side, I estimate a multinomial logit land use decision using a method of moments estimator building on Souza-Rodrigues (2019). Landowners are differently elastic to profitability: price times yield less a transport cost. A carbon tax changes deforestation more in more elastic regions, but more elastic regions also respond more strongly to increasing output prices induced by a tax. Identification addresses simultaneity between within-country transport costs and plot-specific amenities using straight-line distances and steep slopes as instruments. Globally, a 1% increase in agricultural returns raises agricultural land shares by 1.46%. Pasture is less elastic.

Demand estimation follows Costinot, Donaldson, and Smith (2016) and Carleton, Crews, and Nath (2023). New to this paper, consumers substitute across country-specific varieties of meat and across meat and crops. Given fairly elastic landowners, estimates suggest that consumers bear much of the cost of a carbon tax, as they do not readily substitute away from tropical, high-emissions varieties. Crop-meat substitution is also inelastic, meaning price gaps *between* meat and crops similarly persist in equilibrium.

First, I quantify the welfare gains of pricing deforestation at scale. A global, utilitarian social planner values $CO_2$ emissions at an assumed \$3.80/t$CO_2$/year. They maximize welfare with two instruments: carbon taxes and global lump-sum transfers. The optimal policy reduces 56.2% of realized deforestation-related emissions since 2000, generating an annual welfare benefit of 0.04% of global GDP or \$2.50 per ton of reductions. Lowering emissions costs 1.75% of global food and 4.03% of global meat production. Major forest basins — the Amazon, the Congo, and Indonesia — reduce deforestation most. Remaining deforestation occurs on land with high yields and low emissions, including in Mexico, the Brazilian Atlantic Forest, the US, and China. Redistribution is large: gross transfers are 2.6 times larger than welfare gains, and 3.6 times larger than committed funding for land use initiatives to date (Climate Policy Initiative 2026). However, approximately half of the world, including both Indonesia and the Congo River Basin, would not individually rationally participate in this deforestation pricing regime.

Welfare gains and the geography of emissions reductions are robust to several extensions: nonhomothetic demand, alternative social planner weights, and including biodiversity losses as a cost. Removing transfers, the planner uses country-level taxes to achieve similar gains. Gross complementarity between meat and crops reallocates some pastoral loss to crop loss.

Welfare gains come in part from the planner's specific spatial *targeting* of deforestation. I characterize this targeting in two steps. First, fix the geography of emissions and let output adjust: constrain every country-biome to its status quo share of global emissions. This constrained planner still achieves 61% of first-best welfare gains. Of the missing 39%, approximately half come from reallocating emissions across countries, and the remainder within countries and across biomes. Even constrained, the planner always limits deforestation in the carbon-dense Amazonian and Indonesian forest interior. Second, fix output and reallocate emissions. Maintaining the same forest area, global crop output, and global meat production, the planner can reduce emissions by 17% and achieve 25% of first-best gains. Gains are purely from trading a $CO_2$-rich deforested hectare for a $CO_2$-poor deforested hectare. Relative to this constrained efficient geography, status quo deforests too much in major tropical basins — e.g., Brazil — and too little in basin neighbors — e.g., Peru — and the US, Canada, and Western Europe.

Two regularities emerge from the analysis: welfare gains are implementable even under sharp constraints, but tropical countries bear costs. The global expenditure-weighted consumption basket rises a modest 0.08% $(0.07, 0.09)$, but the burden is concentrated. 53% of global food losses fall on Indonesia, whose consumption basket price rises 3.46%; Brazil bears a third of global meat losses. With concentrated costs and diffuse benefits, tropical countries may not participate in global carbon taxation, and the policy could unravel.

To shed light on the appropriate policy solution, I exactly decompose the gap between non-cooperative optimal deforestation policy and the benchmark planner's policy into two coordination failures: free-riding and a lack of internalization of cross-country price spillovers. Assuming countries internalize global emissions at a rate proportional to GDP (Nordhaus 2015), I analyze the Nash equilibrium of a game where countries choose their own $CO_2$ tax rate. I contrast this with a "Nash-internalized" game, where each country values global emissions at global social cost, correcting for free-riding. Under Nash, taxes are all $\sim\$0/tCO_2$. Moving from uncoordinated to "Nash-internalized" recovers 92.2% of the planner's welfare gain. Empirically, most of the gap is *free riding*: tropical countries bearing concentrated costs enjoy insufficient benefits.

Finally, I explore contracts in which high-income, low-forest countries pay low-income tropical countries to preserve their forest, inspired by the Amazon Fund and the still-undersubscribed Tropical Forest Forever Facility. To solve the free riding problem, I model these contracts as a provision point mechanism (Bagnoli and Lipman 1989; Andreoni 1998). Under the mechanism, donors transfer funds to tropical countries on a per-hectare basis, and recipients use carbon taxes to earn higher payments. To solve donor-side free riding, donors must collectively meet a threshold, or no transfers are made. With GDP-based willingness to pay for $CO_2$, European transfers support the same emissions reductions as the global social planner limited to a Brazilian carbon tax instrument. However, the mechanism fails when multiple forested countries compete for a funding pot. Free riding across *recipient countries* reduces gains from trade to near 0. Given a fixed total budget, recipients can deforest and collect a higher rate per remaining forest. Donors get less carbon per dollar, reducing willingness to pay: the mechanism unravels.

In sum, there are large gains from repricing food from the forest frontier and reallocating emissions both within and across countries. Fewer emissions can be achieved without food loss. Yet, inequality implications are large: without transfers, the global coordinated tax is not individually rational. In non-cooperative policymaking, free riding incentives prevent progress. These same free riding incentives can undercut *multilateral* contracts for deforestation, requiring new solutions.

**Related Literature.** The economics of tropical deforestation, reviewed in Balboni et al. (2023), has no normative benchmark for *where* deforestation should occur. The agricultural opportunity cost of $CO_2$ abatement varies by orders of magnitude across space (Lagakos and Waugh 2013; Chen, Restuccia, and Santaeulàlia-Llopis 2023), and agriculture is a core economic driver in carbon-dense tropical regions (Tombe 2015; Nath 2025). I supply the first global, general-equilibrium benchmark for gauging the efficiency of observed deforestation, where existing global analyses evaluate conservation (Joppa and Pfaff 2011) and trade policy (Farrokhi et al. 2025). Dominguez-Iino (2026) considers the role of intermediary market power in South America in targeting a carbon tax.

I build on microeconomic evidence which suggests deforestation can be avoided at low cost with a general equilibrium treatment that characterizes how carbon pricing alters the spatial allocation of forests across multiple forest basins. Revealed-preference, partial equilibrium approaches suggest regional carbon taxes come at low producer-surplus costs (Souza-Rodrigues 2019; Araujo, Costa, and Sant'Anna 2025). Building on their estimation frameworks, I obtain global land use elasticities. In equilibrium, a carbon price reallocates deforestation and agriculture relative to business as usual, connecting to a debate on the extent of agricultural misallocation in the developing world (Gollin, Lagakos, and Waugh 2014; Adamopoulos and Restuccia 2014; Adamopoulos et al. 2022; Foster and Rosenzweig 2021). This is a rare setting where the wedge is directly observed at *plant* level (one other such setting is Asker, Collard-Wexler, and De Loecker 2019) as opposed to inferred (Hsieh and Klenow 2009; Bai, Jin, and Lu 2024).

There is scarce direct evidence on policies that protect forest while supporting development (Alix-Garcia et al. 2026). Randomized evaluations show payments for ecosystem services can couple poverty reduction with forest protection locally (Jayachandran et al. 2017; Jayachandran 2023). I quantify the tradeoff globally and find it distributional rather than aggregate: despite welfare wins, sustaining the first best demands large transfers which countries would not individually rationally commit to make. Globally, distributional concerns unravel cooperative policy.

Non-cooperative policymaking can fail to deliver even large aggregate gains. Work on international climate agreements and carbon-tax mechanisms examines participation gaps for oil and fossil fuel settings without tropical distributional concerns (Nordhaus 2015; Barrage 2020; Bourany 2024; Farrokhi and Lashkaripour 2025; Garcia-Lembergman et al. 2025). Ferguson, Staiger, and Yurukoglu (2026) highlights the importance of agricultural emissions in a broad multi-sector framework, but abstract from the within-country selection of land which drives deforestation emissions. Harstad (2024) shows conditional trade agreements can implement the first best but focuses on a single tropical entity; I show that free-riding *across* forest basins is first-order — the smallest donor coalition abates $\sim$0% of global emissions when all three major basins are eligible. Hsiao (2026) quantifies coordination and commitment in a dynamic deforestation mechanism. I develop a complementary static provision-point mechanism to study the distinct problem of heterogeneous recipient basins (Bagnoli and Lipman 1989; Andreoni 1998).

# 2 Data

## 2.1 Data sources

An observation is a 0.1 arc-degree by 0.1 arc-degree square grid, 11.1 $\times$ 11.1 kilometers at the equator. 1,591,999 such cells cover non-polar terrestrial land.

**Profitability measures.** Profitability of good $h$ (crop or meat) represents the annual flow profit ($/hectare/year) of a farmer or pastoralist. The profitability measure combines prices $p_i^h$, measured in country $i$, transportation costs $\tilde{c}(\omega)$, measured on pixel $\omega$, and yields $\eta^h(\omega)$:

$$[p_i^h - \tilde{c}(\omega)]\eta^h(\omega) \quad h \in \{P\text{asture}, A\text{griculture}\}$$

For crops, producer prices are from the Food and Agriculture Organization (FAO) (Food and Agriculture Organization of the United Nations 2026). Transportation costs combine data on freight rates (Herrera Dappe, Lebrand, and Stokenberga 2024) and travel times (Nelson et al. 2019). Yields $\eta^A(\omega)$ and prices are a production-weighted average of soy, wheat, maize, oilpalm, sugarcane, rice, potato, and sweet potato profits, chosen because they represent a large share of global calories (Costinot, Donaldson, and Smith 2016). There are two yield measures. Potential yield data comes from the FAO Global Agro-Ecological Zones (GAEZ) data (Fischer et al. 2021); and agricultural yields are from Monfreda, Ramankutty, and Foley (2008). In the model, all results use potential yields rescaled by a country-level factor to match value added data from Borchert et al. (2022), as actual yields are 0 on land with no status quo agricultural production. The rescaling procedure is described in Appendix B.2. This approach used in work to generate administrative supply use tables, ensures consistency across land use, macroeconomic moments, and trade data (Antrás and Chor 2022). The yield data determine the spatial distribution of returns within a country, which is the variation the estimation exploits, while this rescaling is primarily to reconcile levels. [3] Meat production data from the FAO. Pastoral area is from Ramankutty et al. (2008). For notational convenience, $\eta^P(\omega)$ represents *country*-level meat production per pastoral hectare.

3. Because trade values are recorded inclusive of processing and distribution margins, a ton of soy is valued at the average value added embodied in that ton. This is the correct valuation if downstream value added falls in proportion to upstream crop production, and overstates the cost of reduced crop output if it does not.

**Deforestation emissions** Emissions convert plot-level aboveground carbon density (tons per hectare, Spawn and Gibbs 2020) to $CO_2$. On plot $\omega$, deforestation emissions $d(\omega)$ per hectare are:

$$d(\omega) = \frac{44}{12}\text{aboveground carbon}(\omega), \tag{1}$$

where $\frac{44}{12}$ converts carbon into carbon dioxide (IPCC 2006). This models deforestation as slash-and-burn. IPCC (2006) carbon accounting varyingly additionally account for "roots and shoots" being burned or future regrowth: I present later results with these alternative assumptions in Appendix E.3. Descriptive statistics calculate emissions using Song et al. (2018) land use data 1982-2016 to provide a retrospective summary of deforestation emissions; the model uses a 2000 base year from ESA Land Cover CCI project team and Defourny (2019).[4]

**Other datasets.** Forest share and other land share data comes from the European Space Agency (Defourny et al. 2023). Trade data comes from the International Trade and Production Database for Estimation (ITPD-E, Borchert et al. 2022); macroeconomic moments from the Penn World Table (Feenstra, Inklaar, and Timmer 2015); tariffs from MacMAP (Guimbard et al. 2012); elevation data is from the Copernicus DEM (European Space Agency 2022); city locations from Akbar et al. (2023); and port locations and metadata from (National Geospatial-Intelligence Agency 2019). See Appendix B for more details.

## 2.2 Describing global forest cover

**Global forests vary substantially in carbon density.** The top panel of Figure 1 maps potential emissions from deforestation, derived from the carbon content of trees as in Equation (1). The bottom panel maps a production-weighted average of crop-level potential agricultural yields. While tropical regions support high carbon accumulation due to warmth and moisture, tropical soils are nutrient-poor without sustained fertilizer application (Sanchez 2019). Woody, biomass-rich tree crops are better adapted to grow in this tropical band (while dense rainforest is the primary natural beneficiary, oilpalm and other fruiting tree crops also grow well in this band). The three largest old-growth forest basins in the tropics are the Amazon River Basin (Brazil and surrounding countries), the Congo River Basin (primarily in the Democratic Republic of the Congo), and Southeast Asia (Potapov et al. 2017).

Interpreting the map: the carbon density in Equation (1) is a plot-specific productivity. Per hectare forested, land in the tropics tends to sequester more carbon per unit forest.[5] Corroborating this,

4. The key constraint on the recency of the modeling is sufficiently rich global pastoral land use data.

5. The longest-run, temporally consistent records on forest cover come from pollen records. Based on evidence from the Last Glacial Maximum, the carbon density of the Amazon could be altered by long-run climate change: I abstract from such endogeneity (Mayle et al. 2004).

flexible cubic spline terms in tree cover explain 58% of variation in carbon per hectare, indicating substantial residual variation in carbon density conditional on forest extent. Land use thus trades off two plot-specific productivities: *yield* of food with *carbon emissions*.

**The carbon intensity of tropical deforestation.** Next, I provide summary statistics on the global tradeoff of these two productivity measures. Table 1, Column (1) starts with a global view. Burning the aboveground carbon of an average hectare would emit 39.19 $tCO_2$. Land earns \$23.90 per ton of emitted $CO_2$. 44% of global crop and pastoral land area has lower profitability than the cost of emissions. In this table, and in the later model, I annualize a \$190 per $tCO_2$ social cost of carbon to a \$3.80 per $tCO_2$ per year annuity (U.S. Environmental Protection Agency 2023).

Limiting attention to the 3.1% of global surface which is deforested land, here defined as land which lost at least 10% of vegetation cover after 1982 in Song et al. (2018), carbon intensity is 55 tons per hectare. For context, U.S. Environmental Protection Agency (2024) equivalence tables suggest that 55 tons per hectare is the equivalent of $\sim$9 average US homes' electricity use for a year. Profitability is higher than the average global pixel — global land includes much unproductive desert and tundra area. However, the profitability differential is not as stark as the $CO_2$ differential: profits per ton of $CO_2$ are lower than global figures at \$9.89 per ton. The scope for inefficiency is much larger: nearly 80% of deforested land was less profitable per ton than a social breakeven of \$3.80 per $tCO_2$.

Column (3) refines attention to deforestation in the three regions with major forest basins: Brazil, Indonesia, and the Congo River Basin. These regions contribute one-third of deforested land over the focal period but half of emissions. 85% of their deforestation has profitability below \$3.80 per ton per year.

Appendix Table A.9 replicates Table 1 with Monfreda, Ramankutty, and Foley (2008) actual yield data for the year 2000. Globally, column (1) suggests actual crop profitability and potential profitability are similar per $tCO_2$. Potential profitability is understated in the upper tail of profitability, largely in the US. On deforested land, potential profitability is an upper envelope for actual profitability. Potential profitability acts as an upper bound on the cropland profitability of tropical deforested land and understates the degree to which large economies can increase food production under a carbon tax.

**Trade and deforestation intensity.** On the demand side, Figure 2 shows the relationship between trade and deforestation. High-deforestation countries — largely in the tropics — tend to source agricultural consumption domestically. Manufacturing goods are more readily sourced from abroad. Regardless of emissions intensity, crop self-trade is higher than manufacturing self-trade. Large relative domestic consumption can reflect true comparative advantage, preferences, or trade barriers. Tombe (2015) and Nath (2025) link large relative agricultural self-trade to higher *relative* agricultural barriers in poor countries.

Agricultural trade barriers and preferences are held fixed in this paper: two implications follow. High self-trade limits leakage — changing deforestation at home poses less risk of raising prices, and thus deforestation, abroad through prices. Second, trade barriers increase reliance on domestic agriculture regardless of comparative advantage, potentially requiring more deforestation per ton output (Farrokhi et al. 2025).

# 3 Model

**Economic Environment.** There are $N$ plots of land of equal area and $N$ workers, partitioned among $J$ countries, indexed by $i$. Land and workers are immobile across countries. Workers are mobile across sectors within countries, so labor markets clear at country level. Each plot, $\omega \in \Omega$, contains a unit measure of land parcels $p$, with each parcel owned by a distinct landowner. Country $i$ is comprised of plots $\Omega_i \subset \Omega$.

There are three private goods — agriculture (produced with land and labor), pasture (land and labor), and manufacturing (labor only) — and one public good: carbon sequestration from forests. International trade in private goods is subject to a transport wedge. Consequently, prices for traded goods differ across countries.

This section lays out production and trade (Section 3.1), demand (3.2), and equilibrium (3.3). Section 3.3 further summarizes optimal carbon taxation results derived in Appendix A.1.

## 3.1 Production and trade

**Agriculture and the landowner's problem** Each landowner allocates a parcel $p$ in plot $\omega$ to a use $h \in \{F, A, P, O\}$ — forest, agriculture, pasture, or other. Agricultural and pastoral production require one parcel $p$ and one worker hired at wage $w_i$. Production yields $\eta^h(\omega)$ tons of good $h$, sold at price $p_i^h$. Converting forest to agriculture or pasture releases $d(\omega)$ tonnes of $CO_2$, priced at the carbon tax $t_i$ ($t_i = 0$ in the baseline). The *dollar* net returns to agriculture and pasture are

$$v^h(\omega) \;=\; \big[p_i^h - w_i c(\omega)\big]\eta^h(\omega) \;-\; w_i \;-\; t_i\, d(\omega), \tag{2}$$

where $c(\omega)$ is a transportation cost denominated in labor units. Forest and other do not directly produce marketed output; they provide an amenity in *utility* units with average value over the whole *plot* $\omega$, $\xi^F(\omega)$ and $\xi^O(\omega)$. Pastoral land carries an amenity $\xi^P(\omega)$. Agricultural land carries no amenity (a normalization).

Landowners rank uses by utility: the monetary return is deflated into utility units by the deflator $P_i^C$, while amenities, being utility flows, are not.[6] With an idiosyncratic, utility-denominated

6. Deflating the monetary return makes the land-allocation block homogeneous of degree zero in nominal prices.

parcel-level shock $\epsilon_p^h(\omega)$ on each use, realized payoffs are

$$\pi_\epsilon^h(\omega) = \pi^h(\omega) + \epsilon_p^h(\omega) = \begin{cases} \frac{v^h(\omega)}{P_i^C} + \xi^h(\omega) + \epsilon_p^h(\omega),\ h \in \{A, P\} \\ \xi^h(\omega) + \epsilon_p^h(\omega),\ h \in \{O, F\} \end{cases} \tag{3}$$

The $\epsilon_p^h(\omega)$ are i.i.d. Type-I extreme value with scale $1/\gamma(\omega)$, observed by the landowner but not the econometrician.[7] Landowners allocate their entire parcel $p$ to the land use which maximizes payoffs. Aggregating over parcels $p$ within plot $\omega$, the expected return on plot $\omega$ is

$$\pi(\omega) \ :=\ \mathbb{E}_\epsilon \max_{h \in \{F,A,P,O\}} \pi_\epsilon^h(\omega) \ =\ \frac{1}{\gamma(\omega)}\Bigg(\aleph + \log \sum_{h \in \{F,A,P,O\}} \exp\big(\gamma(\omega)\pi^h(\omega)\big)\Bigg), \tag{4}$$

for Euler-Mascheroni constant $\aleph$. $\gamma(\omega)$, the scale parameter of the parcel-level shock $\epsilon_p$, is a key estimated parameter, as it governs landowner responsiveness to dollar-denominated profits. Total returns in country $i$ are $\Pi_i = P_i^C \sum_{\Omega_i} \pi(\omega)$. Conditional choice probabilities $\mu^h(\omega)$ are

$$\mu^h(\omega) = \frac{\exp\big(\gamma(\omega)\pi^h(\omega)\big)}{\sum_{k \in \{F,A,P,O\}} \exp\big(\gamma(\omega)\,\pi^k(\omega)\big)}, \qquad \log \frac{\mu^A(\omega)}{\mu^F(\omega)} = \gamma(\omega)\left[\frac{v^A(\omega)}{P_i^C} - \xi^F(\omega)\right]. \tag{5}$$

Total emissions are $D = \sum_{\omega \in \Omega} d(\omega)[\mu^A(\omega) + \mu^P(\omega)]$; country $i$ emits $D_i = \sum_{\omega \in \Omega_i} d(\omega)[\mu^A(\omega) + \mu^P(\omega)]$. Both can be written as a function of the carbon tax $t$, $D_i(t), D(t)$.

**The manufacturing sector.** A competitive manufacturing sector in each country $i$ uses labor and a fixed factor, with returns to scale governed by a parameter $\beta$:

$$Y_i^M = B_i(L_i^M)^\beta, \qquad \beta \in (0, 1]$$

The fixed factor is absorbed into $B_i$. Profit maximization by competitive firms yields a labor demand schedule as a function of the output price $p_i^M$.

$$w_i = p_i^M \cdot \beta B_i(L_i^M)^{\beta-1}.$$

Manufacturing disciplines an endogenous opportunity cost of land $w_i$ in agriculture: value add per worker in the rest of economy. I assume constant returns to scale $\beta = 1$.

**Trade and transportation.** Trade occurs on a hub-and-spoke network (Farrokhi and Pellegrina 2023). From (2), landowners pay a cost $c(\omega)w_i$ to get to export hubs. The transportation sector in a country is competitive and constant returns with intermediate value added $w_i L_i^T$ where $L_i^T =$

7. Expressing the landowners' problem in levels allows the land use elasticity of a parcel with respect to price to differ from that with respect to yields or transport costs; a Fréchet imposes equality.

$\sum_{h\in A,P}\sum_{\omega} c(\omega)\eta^h(\omega)\mu^h(\omega)$. Between international export hubs, trade costs are good-specific ad valorem wedges, $T_{ij}^h$. Manufacturing only faces international trade costs $T_{ij}^M$.

**Labor supply.** Total labor supply in country $i$ is perfectly inelastic at $N_i = L_i^A + L_i^T + L_i^M + L_i^P$. Labor is perfectly mobile across the manufacturing, pastoral, and agricultural sectors but immobile across countries. Manufacturing is the residual claimant of labor in the economy.

## 3.2 Demand

A representative agent in $i$ consumes country-specific varieties of agriculture, meat, and manufacturing with homothetic preferences given income $\widetilde{Y}_i$.[8]

**Preferences.** In each country $j = 1, 2 \ldots, J$, the representative consumer has a nested, constant elasticity of substitution utility. In the innermost nest, consumers choose between country-specific varieties of good $h$ with origin $i$-specific preference shifters $\theta_{ij}^h$ and a cross-origin elasticity of substitution.

$$U_j^h(X) = \left(\sum_{i=1}^{J} \theta_{ij}^h \left(X_{ij}^h\right)^{\frac{\sigma^h-1}{\sigma^h}}\right)^{\frac{\sigma^h}{\sigma^h-1}} \quad \text{for } h \in \{A, P, M\}. \tag{6}$$

Demand for good $h$ in destination $j$ is $X_j^h = \sum_{i=1}^J X_{ij}^h$. In a middle nest, consumers trade off crop and meat consumption. $K$ as an index refers to the total of meat and crop consumption.

$$U_j^K(X) = \left(\sum_{h\in\{A,P\}} \theta_j^h \, U_j^h\left(X_j^h\right)^{\frac{\sigma^K-1}{\sigma^K}}\right)^{\frac{\sigma^K}{\sigma^K-1}}. \tag{7}$$

In the outermost nest, consumers substitute across manufacturing and food with elasticity $\sigma$.

$$U_j(X) = \left(\sum_{h\in\{K,M\}} \theta_j^h \, U_j^h\left(X_j^h\right)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}. \tag{8}$$

**Budget constraint.** Expenditure on variety $i$ of good $h$ in destination $j$ is $E_{ij}^h = (1+T_{ij}^h)p_i^h X_{ij}^h$. The representative consumer faces an aggregate budget constraint on expenditures:

8. Aggregation with homothetic CES preferences follows the standard result in Mas-Colell, Whinston, Green, et al. (1995), §4.B. Appendix E explores results with nonhomothetic preferences.

$$\sum_{i=1}^{J}(E_{ij}^K + E_{ij}^M) = \sum_{i=1}^{J}(E_{ij}^A + E_{ij}^P + E_{ij}^M) = \widetilde{Y}_j.$$

**Expenditure within-sector, across origin.** In each sector, the representative consumer spends share $\lambda_j^{i|h} = \frac{E_{ij}^h}{E_j^h}$ of total expenditures $E_j^h$ on goods from $i$ and faces Dixit-Stiglitz price index $P_j^h$:

$$\lambda_j^{i|h} = \frac{(\theta_{ij}^h)^{\sigma^h}\left[(1+T_{ij}^h)p_i^h\right]^{1-\sigma^h}}{\sum_{n=1}^{J}(\theta_{nj}^h)^{\sigma^h}\left[(1+T_{nj}^h)p_n^h\right]^{1-\sigma^h}},\ P_j^h = \left(\sum_{n=1}^{J}(\theta_{nj}^h)^{\sigma^h}\left[(1+T_{nj}^h)p_n^h\right]^{1-\sigma^h}\right)^{\frac{1}{1-\sigma^h}} \tag{9}$$

**Total agricultural expenditures.** Consumers split total expenditure $E_j^K$ on meat and crops into shares $\lambda_j^{h|K}$, $h \in \{A, P\}$ with agricultural basket price index $P_j^K$,

$$\lambda_j^{h|K} = \frac{(\theta_j^h)^{\sigma^K}(P_j^h)^{1-\sigma^K}}{\sum_{k\in\{A,P\}}(\theta_j^k)^{\sigma^K}(P_j^k)^{1-\sigma^K}},\ P_j^K = \left(\sum_{k\in\{A,P\}} P_j^K = (\theta_j^k)^{\sigma^K}(P_j^k)^{1-\sigma^K}\right)^{\frac{1}{1-\sigma^K}}. \tag{10}$$

**Expenditure by sector.** Sectoral expenditure shares $\lambda_j^h$ depend on the sectoral price index:

$$\lambda_j^h = \frac{(\theta_j^h)^{\sigma}(P_j^h)^{1-\sigma}}{\sum_{k\in\{K,M\}}(\theta_j^k)^{\sigma}(P_j^k)^{1-\sigma}} \qquad h \in \{K, M\} \tag{11}$$

Write the share of income $\widetilde{Y}_j$ spent on each good: $\lambda_{ij}^h = \lambda_j^h \lambda_j^{h|K} \lambda_j^{i|h}$ for $h \in \{A, P\}$ and $\lambda_{ij}^M = \lambda_j^M \lambda_j^{i|M}$. The real price index is $P_j^C = \left(\sum_{k\in\{K,M\}}(\theta_j^k)^{\sigma}(P_j^k)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$.

## 3.3 Equilibrium, emissions, and social welfare

**Aggregation.** Aggregating over the set $\Omega_i$ of $N_i$ plots $\omega$ in $i$ gives an aggregate supply curve for meat and crops $Q_i^h = \sum_{\omega\in\Omega_i}\eta^h(\omega)\mu^h(\omega) \quad h \in \{A, P\}$. Agricultural value added is $\sum_{h\in\{A,P\}} p_i^h Q_i^h - w_i L_i^T$. Transportation value added is $w_i L_i^T$. Value add in the outside sector is $\frac{1}{\beta} w_i L_i^M$. Income is:

$$\widetilde{Y}_i \equiv \frac{1}{\beta} w_i L_i^M + \sum_{h \in \{A,P\}} p_i^h Q_i^h + \mathcal{T}_i + \mathcal{F}_i$$

where $\mathcal{F}_i$ represents any net transfers to $i$ and carbon tax revenues $\mathcal{T}_i = t_i D_i$ are lump-sum recycled domestically. I refer to $Y_i \equiv \widetilde{Y}_i - (\mathcal{T}_i + \mathcal{F}_i)$ as net-of-policy income.

**Market clearing.** Equilibrium pins down $3 \times J - 1$ endogenous prices: $2 \times J$ country-level prices $p_i^h$ and $J - 1$ country-level wages $w_i$ (the US wage is a numeraire). Goods market clearing equates supply-side value add (from the landowners' problem) to international expenditures (from the consumer choice problem), giving $J \times 3 - 1$ unique equilibrium conditions:

$$\sum_{j=1}^{J} E_{ij}^h = p_i^h Q_i^h. \qquad h \in \{A, P, M\} \tag{12}$$

Given manufacturing returns to scale $\beta$, demand elasticities $\sigma, \sigma^A, \sigma^P, \sigma^K, \sigma^M$, preferences $\theta$, trade costs $T$, land use elasticities $\gamma$, factor endowments $B_i, N_i$, and the distribution of land $\Omega, \{\eta, c, \xi, \epsilon, d\}$, an equilibrium is defined by a set of prices and wages such that (12) hold.[9]

**Social welfare.** For each country $j$, the social planner selects a carbon tax $t_j$ and net transfers $\mathcal{F}_j$. They place weight $\psi_j$ on country-specific utilities which value: (i) land amenities $A_i$ in utils, defined via the identity $\Pi_i = \sum_{h \in \{A,P\}} \left(p_i^h Q_i^h - w_i L_i^h\right) - w_i L_i^T - t_i D_i + P_i^C A_i$; (ii) the value $V_i(t, \mathcal{F})$ from utility maximization $V_i(t, \mathcal{F}) = A_i + \max_X U_i(X; \mathcal{F}_i)$ at a specified transfer scheme $\mathcal{F}_j$; and (iii) the social cost of deforestation $\sum_{j=1}^{J} \frac{\chi_j}{\psi_j} \kappa D = \kappa D$, where each country bears a share $\chi_i$ of social cost. Damages from emissions are represented as an annuity, so that $\kappa$ is foregone utility per ton per year. Welfare is:

$$\mathcal{W}^\star = \max_{\mathcal{F}_j, t_j} \sum_{j=1}^{J} \psi_j \left[ V_j(t, \mathcal{F}) - \frac{\chi_j}{\psi_j} \kappa D \right], \quad \sum_j \mathcal{F}_j = 0 \tag{13}$$

The utilitarian planner places weights $\psi_j = 1$ for all $j$, and is the benchmark used in the main text. Robustness checks consider an egalitarian planner and the (Negishi) weights $\psi_j$ such that $\mathcal{F}_j = 0$.

9. Existence and uniqueness are analytically nontrivial to verify. Relative to Allen, Arkolakis, and Li (2024), the supply elasticity matrix diverges as $P_i^C \to 0$. Relative to Alvarez and Lucas (2007), I allow for gross complementarity of the excess demand system $\sigma < 1$. Numerically, results converge to a unique equilibrium across a wide range of starting guesses.

Under non-cooperative policy, countries set a domestic carbon tax $t_j$ to maximize the selfish criterion:

$$\mathcal{W}_j^{unc} = \max_{t_j} V_j(t, 0) - \chi_j \kappa D \tag{14}$$

The benchmark utilitarian planner $\psi_j = 1$ maximizes the sum of selfish welfare.

Using Equation (13), Appendix A.1 Proposition 1 shows that the planner applies the global Pigouvian $CO_2$ tax when they have access to global lump-sum transfers: $t^\star = \kappa$. Without transfers in Proposition 2, the planner chooses country-specific tax rates to account for gaps in relative prices.

**The role of the emissions distribution.** Per (2), a marginal carbon tax shifts emissions by

$$\frac{\partial D}{\partial t} = -\sum_{\omega \in \Omega} \frac{\gamma(\omega)}{P_i^C} [d(\omega)]^2 \cdot [\mu^A(\omega) + \mu^P(\omega)] \cdot [1 - \mu^A(\omega) - \mu^P(\omega)]. \tag{15}$$

Emissions supply depends on the weighted variance of the global emissions distribution. Country-level emissions factors ignore this curvature, overstating emissions reductions in regions where agriculture and pasture co-locate with emissions-intensive land.

# 4 Empirical strategy

I estimate two sets of parameters: (i) the distribution of land use elasticities $\gamma(\omega)$, identified from land use shares via Equations (5), and (ii) the cross-country trade elasticity $\sigma^A$, identified from trade shares via Equation (9).

## 4.1 Landowner's problem

**Parameterization of the landowner's problem.** Estimation follows the logit inversion procedure of Berry (1994).[10] Define $Y^{hl}(\omega) = \log \mu^h(\omega) - \log \mu^l(\omega)$. From Equation (5):

$$Y^{hl}(\omega) := \log \mu^h(\omega) - \log \mu^l(\omega) = \gamma(\omega)[\pi^h(\omega) - \pi^l(\omega)]$$

Define $\nu^h(\omega) = \gamma(\omega)[\xi^h(\omega) - \xi^F(\omega) - w_i/P_i^C]$ as a composite unobservable, with $\xi^A(\omega) = 0$. Define

10. Appendix B.3 describes an Empirical Bayes procedure to round shares so that $\mu^h(\omega) \in (0, 1)$.

$\tilde{c}(\omega) = c(\omega)w_i$ as the total transport cost.

$$\begin{cases} Y^{AF}(\omega) & = \frac{\gamma(\omega)}{P_i^C}[p_i^A - \tilde{c}(\omega)]\eta^A(\omega) + \nu^A(\omega) \\ Y^{PF}(\omega) & = \frac{\gamma(\omega)}{P_i^C}[p_i^P - \tilde{c}(\omega)]\eta^P(\omega) + \nu^P(\omega) \\ Y^{OF}(\omega) & = \gamma(\omega)\xi^O(\omega) - \gamma(\omega)\xi^F(\omega) \end{cases} \tag{16}$$

I parameterize $\gamma(\omega)$ with a country-by-biome $b$ level intercept $\gamma_b$. Estimation of Equation (16) is biased if, within country-biomes, profitability $v^h(\omega)$ correlates with the unobservable $\nu^A(\omega)$. Recall that pastoral productivity $\eta^P(\omega)$ is measured at country level in the data.[11]

**Estimation and instruments.** I estimate the system (16) through a Generalized Method of Moments (GMM) estimator. Estimation of country-biome coefficients is prone to overfitting and statistically noisy (Dingel and Tintelnot 2026), so I adopt an Empirical Bayes (EB) approach to penalize imprecisely estimated country-biomes. I separately estimate the GMM criterion in every country-biome, giving a set of coefficients $\vec{\gamma}_b$. Denote $\bar{x}_b$ as the area-weighted mean of $x(\omega)$ in $\Omega_b$.

$$m_b(\gamma_b) = \mathbb{E}_{\Omega_b}\begin{bmatrix} (\nu^A(\omega) - \bar{\nu}_b^A)(\delta(\omega) - \bar{\delta}_b) \\ [(\nu^A(\omega) - \nu^P(\omega)) - (\bar{\nu}^A - \bar{\nu}_b^P)](\rho^{15}(\omega) - \bar{\rho}_b^{15}) \end{bmatrix} \tag{17}$$

where $\delta(\omega)$ is distance to city and $\rho^{15}(\omega)$ is the share of plot $\omega$ with a slope greater than or equal to 15 degrees. Distance to city shifts the supply of agricultural land through transportation costs (Souza-Rodrigues 2019).[12] Agriculture is less viable on slopes than pasture, so slopes shift *relative* land supply (Nunn and Puga 2012; Fischer et al. 2021). Within-country estimation can be biased otherwise: unobserved amenities $\xi^F(\omega)$ affect land use both directly and by disincentivizing roadbuilding $\tilde{c}(\omega)$, attenuating $\gamma_b$.

Next, to form a prior, define a global intercept $\tilde{\gamma}(\omega) = \gamma_0$. Define $\tilde{\nu}^h(\omega)$ as the residual from the restricted model. The pooled GMM criterion is:

$$m(\tilde{\gamma}) = \mathbb{E}_{\Omega}\begin{bmatrix} (\tilde{\nu}^A(\omega) - \bar{\tilde{\nu}}_b^A)(\delta(\omega) - \bar{\delta}_b) \\ [\tilde{\nu}^A(\omega) - \tilde{\nu}^P(\omega) - (\bar{\tilde{\nu}}^A(\omega) - \bar{\tilde{\nu}}_b^P)](\rho^{15}(\omega) - \bar{\rho}_b^{15}) \end{bmatrix}$$

I shrink the vector of country-biome-level estimates $\gamma_b$ towards the common prior $\tilde{\gamma}$ (Walters 2024). This differs from the standard EB approach of shrinking to the across-country-biome mean, instead imposing an explicit model-based prior and allowing for some bias as the objective is to reduce variance (dimensionality reduction). Exact shrinkage formulas are provided in Appendix C.3.

11. With panel variation in pastoral land use, one can invert $Y^{PF}(\omega) - Y^{AF}(\omega)$ to estimate $\hat{\eta}^P(\omega), \hat{\xi}^P(\omega)$.

12. Long-run instrument variation is essential, as land use change can involve transitional dynamics in response to short-run price variation (Scott 2013).

### 4.2 Demand parameters and trade barriers

I estimate $\sigma^A, \sigma^P, \sigma^K$ from agricultural trade flows. I calibrate the manufacturing cross-country elasticity of substitution to $\sigma^M = 4$ (Bernard et al. 2003; Simonovska and Waugh 2014), and the outer nest elasticity of substitution between agriculture and manufacturing to $\sigma = 0.5$ (a homothetic approximation of Comin, Lashkari, and Mestieri 2021).

**Cross-country elasticity of substitution.** Following 9, the logged expenditure share of country $j$ consumers on the country $i$ variety of good $h$ is given by:

$$\log \lambda_j^{i|h} = (1 - \sigma^h) \log p_i^h + \delta_j^h + \epsilon_{ij}^h \tag{18}$$

where trade barriers and preferences $(1 - \sigma^h) \log[1 + T_{ij}^h] + \sigma^h \log \theta_{ij}^h$ enter the residual, and the importer-sector fixed effect $\delta_j^h$ is proportional to the aggregate price index $P_j^h$ of country $j$.

An unobserved taste shock $\epsilon_{ij}^h$ in market $j$ favoring variety $i$ simultaneously bids up prices $p_i$ in $i$ and raises bilateral flows $\lambda_j^{i|h}$. OLS estimates are attenuated. For agriculture $h = A$, I use country-level average potential yields $\eta_i^A$ as a supply shifter for agricultural production. For meat $h = P$, I use exporters' crop prices, which shift importers' relative demand for $i$'s variety only through $i$'s own feed prices — a supply shifter.[13]

**Crop-meat substitution.** Price indices are calculated from Equation (18): $\hat{P}_j^h = \Big( \sum_i \exp\Big(\hat{\epsilon}_{ij}^h\Big) \cdot (p_i^h)^{1-\hat{\sigma}^h} \Big)^{\frac{1}{1-\hat{\sigma}^h}}$ (Costinot, Donaldson, and Smith 2016). The estimating equation is:

$$\log \lambda_i^h = (1 - \sigma^K) \log P_i^h + \delta_i + \epsilon_i^h \tag{19}$$

where $\delta_i$ is a fixed effect and $\epsilon_i^h$ captures relative tastes for crop as opposed to meat. I use GAEZ potential yields for wheat (agriculture) and grasses (pasture) as supply shifters. To take these regressions to the data, I pool across years with importer-by-year fixed effects.

13. Domestic demand shocks for meat in large importers could increase food prices of exporters through land competition, threatening exclusion.

# 5 Empirical Results

## 5.1 Land use model

**Deforestation elasticities.** Panel A of Table 2 presents the pooled global coefficient estimates. Standard errors are clustered at the second administrative unit level (ADM-2). Appendix C.1 shows first stages. Panel B shows the EB-shrunk, posterior return elasticities of agricultural and pasture share (an area weighted average of $\gamma(\omega)v^h(\omega)[1-\mu^h(\omega)]$). Interpreting the elasticity, a 1% increase in agricultural returns (per Table 1, \$2.36/ha/year) raises agricultural land use shares by 1.46%.[14] Pastoral land, where a 1% increase is \$0.98/ha/year, is less elastic at 0.76% (though, for the same absolute increase in profits, pasture is more elastic). Spatial differencing in Appendix C.2 reveals similar responses. Tying back to the identifying variation, a 1 p.p. increase in steep slope fraction drives a \$0.34 decrease in the relative returns to agriculture-versus-pasture (Appendix Table A.4), leading to a 0.14% decrease in the *relative* agricultural share, consistent with reduced form evidence in Appendix C.1.

**Decomposition of variance: how much do amenities matter?** Appendix Figure A.4 quantifies the within-country share of land use variation explained by profitability and $\gamma$ variation rather than the unobservables $\xi^h(\omega)$. Focusing first on cropland in A.4a: set $\xi^F(\omega)$ to a constant within-country such that total cropland share is constant and hold fixed other and pastoral land use. Reallocation thus occurs between crops and forest. Without within-country variation in amenities, I recompute agricultural land shares. 20% of within-country-biome variation in cropland share is profitability and elasticity variation. The slope suggests the model reproduces the scale of spatial dispersion within 12%.[15] Variation in price responsiveness $\gamma_b$ in Appendix Figure A.4b explains 2% of land use variation.

In Appendix Figure A.4c, the analogous decomposition without variation in $\xi^P(\omega)$, holding fixed crop and other share, explains 47% of variance. Recalling that pastoral TFP is cross-country, this high $R^2$ reflects that pasture is a "land use of last resort:" other land suitability predicts the pasture-forest tradeoff.

**Discussion.** With elastic supply, low carbon prices dramatically increase forest cover. Appendix Figure A.3 plots the partial equilibrium supply curve of global $CO_2$ as a function of a hypothetical carbon tax. A 50% increase in the long-run food price increases $CO_2$ emissions by 50% relative to status quo. Food production for the same price increase doubles.

14. Figure 4 of Souza-Rodrigues (2019) reports a 36% reduction in agricultural land in the Amazon under a \$42.50/ha tax. Given his reported average return of \$120/ha in the Amazon, this is a returns elasticity of 1.

15. Write the slope as $\rho_{\hat{\mu},\mu}\sigma_{\hat{\mu}}/\sigma_\mu$ where $\rho$ is the within-correlation across model $\hat{\mu}$ and data $\mu$ shares and $\sigma$ their respective within-standard deviations. $R^2 = \rho^2_{\hat{\mu},\mu}$, allowing me to recover a variance ratio of 0.88.

### 5.2 Demand estimation

Table 3 reports demand estimates. Panel A reports the elasticity of substitution across agricultural varieties, $\sigma^A$. OLS (1) and PPML (2) estimates are attenuated towards 0. The IV and PPML-IV column use potential yields as an instrument which shifts long-run crop supply. The first stage of prices on potential yields is negative with a strong $F$-statistic: high potential yield countries have lower prices in expectation (Costinot, Donaldson, and Smith 2016; Sotelo 2020). Because PPML-IV in Column (4) accounts for 0 trade flows, it is my preferred specification (Fally 2015), yielding a trade elasticity of $\hat{\sigma}^A = 2.02$. This trade elasticity is on the lower end of estimated agricultural trade elasticities: Costinot, Donaldson, and Smith (2016) estimates 5.4 using two-stage least squares (as opposed to PPML-IV): my approach replicates this elasticity in Column (3). The lower trade elasticity is driven in part by 0 flows rather than an entire consequence of aggregation of crops into a crop aggregate, which can attenuate the estimate (Imbs and Mejean 2015).[16]

Panel B shows the elasticity estimates for meat. The first-stage coefficient of exporters' meat prices on exporters' crop prices is large and positive, consistent with crops driving up meat prices through feed costs. Prior work which models meat substitution uses an identical elasticity for agricultural crops and meat (Gouel and Laborde 2021). In Panel E, I cannot reject equality of the preferred PPML-IV coefficients.

Finally, meat-crop substitution $\sigma^K$ is estimated in Panel C. In Panel E, the PPML-IV estimate is statistically indistinguishable from the elasticity of substitution across *crops* from Costinot, Donaldson, and Smith (2016). On *average*, consumers substitute between meat and crops as well as they substitute between, e.g., maize and soybeans.

**Discussion.** Meat and agriculture are equally substitutable across countries and with one another. However, these trade elasticities are relatively low. A $CO_2$ tax can change the relative price of meat and agriculture through different supply side incidence (driven by differing emissions intensities and estimated elasticities). Relatively inelastic consumers do not change sourcing of meat and agriculture, nor do they dramatically change the relative meat-crop composition of their consumption basket. These findings suggest price differences are passed on from elastic producers to inelastic consumers and price differences persist in equilibrium.

## 6 Gains from coordinated deforestation policy

Below I consider the benchmark, coordinated policy experiment: the global social planner with transfers and carbon taxes uses them to choose the allocation of deforestation. The benchmark is

16. In particular, by aggregating to a crop basket, my elasticity is a composite of cross-crop substitution and cross-country substitution. Cross-country substitution is traditionally estimated to be more elastic — in the range of 4-8 (Caliendo and Parro 2015), whereas cross-crop substitution closer to 2 (Sotelo 2020).

compared to the status quo, which is obtained from inverting the model at estimated parameters to match data on land use and trade. Appendix C.4 describes the standard model inversion procedure, the crux of which is setting baseline wages to match manufacturing value added. I set the social marginal emissions cost $\kappa$ of $CO_2$ to \$3.80, an annualized value of a \$190/ton social cost of carbon at a discount rate of 2% (U.S. Environmental Protection Agency 2023). At this price, a planner achieves large welfare gains in 6.1. However, food costs are concentrated and countries violate individual rationality by participating. Land use reallocates in 6.2, accounting for up to 39% of planners' emissions reductions and 45% of welfare gains in 6.3.

## 6.1 Welfare gains from pricing deforestation

A utilitarian planner maximizes social welfare through two instruments: international lump-sum transfers and a carbon tax. The carbon tax corrects prices to account for the social cost of deforestation, and reallocates land accordingly per Proposition 1; transfers correct for changes in the income distribution and leakage-related concerns per Proposition 2.[17] Results are in Panel A of Table 4.

Deforestation $CO_2$ reductions in general equilibrium are cheap and effective. Emissions from deforestation fall by 56.2%. Per (13), the social value of $CO_2$ reductions is \$3.80/t$CO_2$/year. In Table 4, the *net* social benefit is \$2.50 per ton per year. The gap, \$1.30/t$CO_2$/year, is the average abatement cost (proof in Lemma 1): foregone agricultural returns, pastoral returns, or manufacturing value added.

Abatement costs are half of welfare gains due to precise Pigouvian targeting and elastic supply. Per (15), the supply curve of emissions *reductions* is quadratic in starting emissions. Appendix E.7 confirms the causal role of this targeting: when the planner cannot accurately target plot-level emissions, two-thirds of the efficient abatement cannot be achieved.

The carbon tax reduces crop production by 1.75% and meat production by 4.03%. Losses are concentrated in the tropics. 53% of food losses are in Indonesia. Brazil drives cattle losses, accounting for one-third of global meat losses. China and Europe, in contrast, expand crop production to offset losses in trading partners. Even with nonhomothetic preferences, sensitivity analysis in Appendix E.10 does not find dramatically different *geography* of food losses, if a slightly smaller level.[18] Concentrated food losses are a supply story: $CO_2$ emissions are high on productive land in Indonesia and Brazil, and counterfactually usable land is not sufficiently low-emissions or high-return (e.g., due to other land uses or low yields) to meaningfully reduce food losses from the carbon tax.

17. Numerically, I verify the rule that the global optimal tax $t^\star$ with utilitarian weighting and lump-sum transfers is Pigouvian to 5 significant digits. The planner does not optimize over trade barriers in this paper.

18. Appendix E.9 also tests robustness to alternate trade elasticities.

**Utilitarian transfers.** The planner requires transfers 2.6 times the welfare gains to sustain first-best. Gross transfers $\sum_{i=1}^{J} \mid \mathcal{F}_i \mid /2$ are \$6.63/tCO$_2$/year. With elastic supply and relatively inelastic demand, the carbon tax opens cross-country gaps in consumption prices (and thus the marginal utility of income). The planner uses transfers to close these gaps, with gross transfers of \$74.1B $(72.2, 76.0)$/year. For comparison, the Climate Policy Initiative estimates a total of \$20.51 billion in annual committed climate finance contributed to date for Agriculture, Forestry, and other Land use (AFOLU) projects as of 2026 (Climate Policy Initiative 2026).

**Removing transfers.** Without lump-sum transfers, the planner cannot separate the corrective and the distributional effects of the carbon tax. By Proposition 2, the country-level tax deviates from the Pigouvian rate to account for redistribution and cross-border leakage. I find optimal country-level taxes are nearly unchanged from first-best. Panel B of Table 4 quantifies aggregate results. Welfare outcomes differ from the full solution by \$0.02 per tCO$_2$. Without transfers, the planner supports more agricultural production, raising emissions by $\sim$2 percentage points.

Geography again proves fundamental: countries with a high correlation between emissions and yields deforest more without transfers. The average ton of food is emissions intensive and there is less low-emissions land to substitute towards. In Appendix E.2, I isolate this channel by reordering yields within country against a fixed emissions distribution: under perfect negative rank dependence (highest yield implies lowest emissions), post-CO$_2$ tax, global prices are 73% lower.

**Individual rationality.** Under the utilitarian planner, 20 countries violate their individual rationality condition: they are worse off than in status quo. Focusing on the major basins, Indonesia is the largest welfare loser from deforestation pricing, while Brazil only violates individual rationality in 3% of bootstrap draws. Without transfers, 14 countries violate individual rationality because the planner cuts food production less.

Strikingly, there is sufficient surplus to compensate welfare losses. In Appendix E.6, all countries can feasibly experience a proportional annual welfare gain of 0.034% of GDP from the carbon tax — a Pareto improvement. Of course, this Pareto improvement requires welfare 'winners' like the US, Europe, and Mexico to cooperate and transfer to welfare 'losers', which would again violate individual rationality absent a coordination mechanism.

**Other environmental considerations.** Appendix E considers optimality accounting for other forest environmental services. Appendix E.3 incorporates belowground carbon and forest regrowth into emissions estimates. I quantify complementarities across emissions reductions and biodiversity conservation in Appendix E.4. Appendix E.5 tests gains at other values of the social cost of carbon.

## 6.2 Land use and producer surplus

Deforestation declines primarily in the three tropical deforestation frontiers — the Amazon, the Congo River Basin, and Indonesia. Figures 3a and 3b illustrate land use change under the carbon tax policy. Cropland and pasture expand in lower-carbon forests on basin peripheries. For example, within Brazil, deforestation falls in the Amazonian interior, but rises in the cerrado (tropical savannah) and Atlantic Forest regions, which are less carbon dense per ton agricultural output.

Pasture seeks land with low agricultural and other-land returns: often, hilly rangelands. For example, in the US, pastoral use increases in the mountain West, while cropland increases in the Grain Bowl of the Midwest. Other regions show similar separation in comparative advantage across the two outputs: China intensifies pasture in the West and North; Central Asia deepens cattle production.

Incidence on producers is closely linked to land use change. Figure 3c maps landowners' equivalent variation.[19] On balance, American and European landowners gain; tropical landowners lose. The average landowner loses \$0.48/ha/yr $(0.36, 0.59)$, while weighting by status quo emissions, the average landowner loses \$30.30/ha/yr $(28.85, 31.83)$. Cattle producers lose more than crop producers: weighting by initial land use, pasture-weighted losses are \$0.79/ha/yr $(0.51, 1.00)$. However, cropland-weighted, producers gain \$4.00/ha/yr $(3.74, 4.24)$. Emissions-intensive producers lose the most. Cattle producers transfer surplus to inframarginal crop producers, who gain from rising prices and consumer substitution away from meat.

## 6.3 Welfare gains from reallocating deforestation

In aggregate, pricing $CO_2$ confers large societal benefits: how much of this gain comes from correcting *where* we deforest? To explore this, I constrain the planner's objective (13) with one of several constraints: (1) shut down cross-country reallocation of emissions, $\frac{D_i(t)}{D(t)} = \frac{D_i(0)}{D(0)}$; (2) shut down cross-country-biome reallocation of emissions; (3) fix global crop production at current levels $\sum_{i=1}^{J} Q_i^A$; (4) fix meat production $\sum_{i=1}^{J} Q_i^P$; (5) fix forest area $\sum_{\omega \in \Omega} \mu^F(\omega)$; (6) both (3) and (4); and (7) all of (3), (4), and (5). The planner uses carbon taxes (or subsidies) to maximize welfare. These constraints come in two families. The first two fix the geography of emissions and let output adjust (1), (2); the rest fix global output and let emissions move wherever they like (3)-(7).

**What share of welfare gains from pricing deforestation comes from moving emissions across regions?** Nearly 40% of first-best welfare gains require resorting deforestation across countries and biomes. In Figure 5a and 5b, cross-country reallocation (1) delivers 16% of first-best abatement and 22% of the first-best welfare gain. Figure 4a plots cross-country-constrained

19. Appendix Figure A.5 further decomposes general equilibrium determinants of landowner surplus. Rising equilibrium output prices reduce costs of deforestation reductions relative to partial equilibrium.

emissions relative to emissions under the global optimal tax. More brightly-colored regions reduce $CO_2$ by less than first best. Russian Boreal forests, Indonesian forests, and the cerrado and Atlantic Forests in Brazil deforest more under the constraint.[20] Cross-country reallocation prevents these relatively higher-return — largely crop-driven, per Figure 5c — $CO_2$ emissions. When the planner further cannot reallocate deforestation within country and across biomes (2), the additional gain from carbon trade across-biomes and within-countries is 17% of first-best gains. Figure 4b maps increases in emissions in the entirety of the Congo River Basin, non-Brazilian Amazon, and Southeast Asia.

Notably in Figure 5e, while reducing the scope for reallocation costs aggregate welfare, it lowers incidence on the tropics by reducing their relative contribution to $CO_2$ reductions (note that these bars are equal by construction under the constraint).

**Holding fixed global food production, can the planner better target existing deforestation?** Following Figure 5, under the most constrained allocation (7), one-sixth of first-best emissions reductions can be described as pure misallocation, and reallocating these emissions is responsible for a quarter of first-best welfare gains. Under (7), the planner moves deforestation moves off of the highest-emissions land substitutes it one hectare-for-one hectare with deforestation on lower-emissions land until food losses are exactly offset. Beyond the direct cost to producer surplus, two costs discipline welfare gains in equilibrium: goods must travel to markets which demand them, incurring iceberg costs, and consumers are forced to reduce consumption of preferred country-specific varieties. Cattle and crop production reallocate within the tropics from forest interiors like the Amazon to their peripheral forests like the Atlantic Forest. The US, Russia, and Western Europe intensify cattle production; crops intensify in similar regions but also in the outlying Amazonian states, SE China, and Indonesia which have a comparative advantage in crop production once carbon is priced.

Two *level* effects are clear: society benefits from reducing global meat production and raising global forest area. The crop constraint (3) is nearly free. However, holding pasture fixed (4) would cost 43% of first-best welfare gains and holding forest area fixed (5) costs 73% of total welfare gains.

Turning to *allocative* effects: under every constraint the planner reduces emissions in the tropical band, but the regions that substitute for the lost production differ. Each constraint binds on a different margin and the planner recruits whichever land is a net provider of the constrained input (Appendix Figure A.6). Fixing forest cover, tropical forest loss is swapped for low-$CO_2$ Boreal and Mexican/Central American deforestation; fixing pasture, pastoral land moves to productive Southern Canada and Europe; fixing crop production, crops move to Western Europe, China, the US Midwest, and outlying parts of *every* tropical basin. The same logic determines the sign of the

20. These predictions align with prior evidence on trade and deforestation: Carreira, Costa, and Pessoa (2024) discuss the importance of the soy boom in carbon emissions from the Brazilian cerrado; Hsiao (2026) discusses the centrality of trade policy in palm oil production in Indonesia.

tax (Appendix Table A.10). Across every scenario the planner taxes deforestation in the Congo River Basin and Indonesia. To fix forest area (5)–(7), it subsidizes deforestation in Mexico, a net provider of low-$CO_2$ forest; under most scenarios (4)–(7), it subsidizes deforestation in China, a net provider of low-$CO_2$ pastoral land. When both cattle and crop production are held fixed (6), Brazil is taxed or subsidized depending on the exact draw of parameters.

# 7 Pricing deforestation in a non-cooperative world

Unlike the cooperative setting in Section 6, this section explores non-cooperative policy. I decompose the gap between the planner and the Nash equilibrium of a non-cooperative carbon tax pricing game in 7.1: most of the gap is free-riding. Section 7.2 considers a classic free-riding solution mechanism.

## 7.1 Decomposing the gap between status quo and planner optimal

A significant share of status quo emissions can be cut even without major reallocation. It is then surprising that no country prices forest carbon.[21] Global conservation solutions struggle to garner adequate participation (Balboni et al. 2023), suggesting that this lack of a $CO_2$ price is a non-cooperative equilibrium outcome (Farrokhi and Lashkaripour 2025). In this section, I quantify the two key disincentives for carbon taxes in a non-cooperative framework: free-riding and uninternalized cross-country spillovers.

Suppose countries $i$ are non-cooperative as in Equation (14). Consider two settings: the *non-cooperative* regime with $\chi_i = \frac{Y_i}{\sum_j Y_j}$ ($\chi_i$ is GDP share at baseline, following Nordhaus and Yang 1996; Golosov et al. 2014); a utilitarian *planner* solves (13) but has no transfers to be consistent with non-cooperative play. First-order conditions are:

$$\text{(non-cooperative)} \quad \frac{\partial V_i(t,0)}{\partial t_i} = \chi_i \, \kappa \, \frac{\partial D}{\partial t_i}, \qquad \text{(planner)} \quad \psi_i \frac{\partial V_i(t,0)}{\partial t_i} + \underbrace{\sum_{j \neq i} \psi_j \frac{\partial V_j(t,0)}{\partial t_i}}_{\Theta_i} = \kappa \, \frac{\partial D}{\partial t_i}.$$

The own-country margin $\partial V_i / \partial t_i$ enters both. For a marginal $CO_2$ tax, the welfare gap between a utilitarian and non-cooperative policy contains two wedges:

$$\underbrace{(1-\chi_i)\, \kappa \, \partial D / \partial t_i}_{\text{free-riding wedge}} \qquad \text{and} \qquad \underbrace{\Theta_i = \textstyle\sum_{j \neq i} \psi_j \, \partial V_j / \partial t_i}_{\text{cross-country spillovers}}.$$

21. Offset markets and payments for ecosystem services contracts enable selective, voluntary pricing of carbon, and some countries allow for the transaction of such offsets in their own domestic carbon markets (Macintosh et al. 2025). Brazil places fines on deforestation in per-area terms (Assunção, Gandour, and Rocha 2023).

$\chi_i < 1$ makes $i$ undervalue the marginal ton and *free ride* (Nordhaus 2015); $\Theta_i$ is the cross-border price spillover $i$ ignores (Farrokhi and Lashkaripour 2025).

Define an intermediate "Nash-Internalized" (welfare $\mathcal{W}^{\mathrm{NI}}$) equilibrium with $\chi_i = 1$, keeping individual best response. I exactly decompose the gap between the non-cooperative equilibrium and the planner into the two wedges:

$$\mathcal{W}^{\star} - \mathcal{W}^{\mathrm{unc}} = \underbrace{\left(\mathcal{W}^{\mathrm{NI}} - \mathcal{W}^{\mathrm{unc}}\right)}_{\text{free-riding}} + \underbrace{\left(\mathcal{W}^{\star} - \mathcal{W}^{\mathrm{NI}}\right)}_{\text{cross-country spillovers}} .$$

Results compute the two Nash equilibria and compare to the planner's solution.

Quantifying this approach, I find that uncoordinated Nash equilibrium taxes are zero. Only the Congo River Basin (\$0.038/ton/yr) and Peru (\$0.032) price $CO_2$ at all.[22] These small taxes reflect terms of trade gains: small carbon taxes raise the price of exports.[23] Terms of trade do not, however, generate meaningful conservation, reducing ∼0.02% of baseline emissions.

In Figure 6a, the non-cooperative solution without freeriding $\mathcal{W}^{NI}$ recovers 92.2% of welfare gains from Pigouvian taxation. Uninternalized cross-country spillovers comprise the residual 7.8%.

In Figure 6b, free riding is the dominant deterrent to carbon taxation in the tropics, while spillovers dominate in high-income countries. Part of this is arithmetic: $\chi_i$ is GDP share, so free-riding is smallest in the largest economies. However, the tropics free ride due to concentrated cost rather than low $\chi_i$ alone: in Appendix E.8, free riding remains dominant even as tropical countries internalize larger shares of the social cost of carbon. The figure shows three countries with large spillover shares as well: Mexico, Malaysia, and Thailand hold some forest carbon. All have global emissions shares of at most 0.4%. In general, countries which leverage a carbon tax as a terms of trade instrument do not have large emissions tax bases.

**Leakage.** In Nash-internalized $\mathcal{W}^{NI}$, countries also internalize carbon leakage — spillovers of emissions into unregulated markets. To demonstrate its role separately, I consider a deviation where $\chi_i = \frac{D_i}{D}$: countries value their domestic emissions at full social cost. The welfare gap between this case and $W^{NI}$ is plotted as "leakage" in 6a. It is small. Low leakage is a feature of demand: consumers do not substitute agricultural varieties and consume most food at home per Figure 2, so there is limited scope for prices to rise abroad.

22. Desert countries such as Niger are indifferent across tax rates; I fix them to \$0 here and to $\kappa$ in the planner's solution.

23. Even with $\chi_i = 0$, the non-zero tax survives in the Congo River Basin and Peru.

### 7.2 Evaluating a candidate participation mechanism

Can bilateral or multilateral transfers for forest reduce free riding-induced misallocation? I implement a provision point mechanism (Bagnoli and Lipman 1989; Andreoni 1998): *donor*, high-income countries pay *recipient* tropical forest basins. Payments are conditional on forest area in the recipients, mirroring payment rules used by the Amazon Fund and Tropical Forests Forever Facility. Recipients compete for the donation pot using domestic carbon taxes. To make the transfer donor-side individually rational, I assume that the mechanism unravels to status quo if donations do not exceed a threshold $\bar{F}$. Bilateral donations support welfare-improving trades, but multilateral settings unravel.

**Environment.** Partition countries $\{1, 2, \ldots, J\} = \mathcal{P} \cup R$ into participants (donors) $\mathcal{P}$ and recipients $R$. Recipients set a $CO_2$ tax $t_r \geq 0$ and receive transfers; donors contribute $\mathcal{F}_p \geq 0$ to a common pool and set no tax.[24] Write $\mathcal{F} = (\mathcal{F}_p)_{p \in \mathcal{P}}$ and let

$$a(\mathcal{F}) \;=\; \mathbf{1}\Big\{\textstyle\sum_{p \in \mathcal{P}} \mathcal{F}_p \geq \bar{F}\Big\}$$

indicate whether contributions clear a threshold $\bar{F}$. The mechanism $\{R, \bar{F}\}$ operates as follows: if $a(\mathcal{F}) = 1$ it activates, disburses exactly $\bar{F}$, and recipients tax; if $a(\mathcal{F}) = 0$, the status quo holds. Contributions above $\bar{F}$ are held by the fund.

Let $\Phi_i(t) = \sum_{\omega \in \Omega_i} \mu^F(\omega)$ denote forest mass in $i$ and $\widetilde{Y}_i(t, \mathcal{F})$. On activation $a(\mathcal{F}) = 1$ a recipient receives $f_r(t, \mathcal{F}) = \frac{\bar{F}}{\sum_{r \in R} \Phi_r(t)}\, \Phi_r(t)$ and a donor pays $\mathcal{F}_p$, so total income and real income are

$$Y_i(t, \mathcal{F}) = \frac{1}{\beta} w_i L_i^M + \sum_{h \in \{A,P\}} p_i^h Q_i^h + + \underbrace{f_r(t, \mathcal{F})\, \mathbf{1}\{i \in R\}}_{\text{received}} - \underbrace{\mathcal{F}_i\, \mathbf{1}\{i \in \mathcal{P}\}}_{\text{contributed}}, \qquad V_i(t, \mathcal{F}) = \frac{Y_i(t, \mathcal{F})}{P_i^C(t, \mathcal{F})}.$$

Country $i$'s provision-point payoff is

$$W_i(t, \mathcal{F}) = a(\mathcal{F}) \Big[ V_i(t, \mathcal{F}) - \chi_i \kappa D(t) \Big] + \big(1 - a(\mathcal{F})\big) \Big[ \tfrac{\widetilde{Y}_i(\mathbf{0})}{P_i^C(\mathbf{0})} - \chi_i \kappa D(\mathbf{0}) \Big],$$

where $\mathbf{0}$ denotes the no-tax allocation. Under non-activation in the second bracket, every contribution is returned, so a failed attempt is payoff-equivalent to the status quo. Finally, define $t^\star(\bar{F})$ as the recipients' best response assuming $a(\mathcal{F}) = 1$. Donors contribute because the activation $a(\mathcal{F}) = 1$ mechanism captures the full inframarginal surplus between the funded point and the status quo, overcoming the free riding incentive on the margin from Section 7.1.

24. Restricting donors from taxing shuts down donor-country abatement and isolates the transfer margin.

**Provision point.** Define donor $p$'s *real activation benefit*—its willingness to pay for the move from status quo to the funded allocation—as

$$b_p \equiv \left[\frac{\widetilde{Y}_p(t^\star,\mathcal{F})}{P_p^C(t^\star,\mathcal{F})} - \frac{\widetilde{Y}_i(\mathbf{0})}{P_p^C(\mathbf{0})}\right] - \chi_p\kappa\big[D(t^\star) - D(\mathbf{0})\big],$$

with nominal counterpart $B_p \equiv P_p^C(t^\star)\, b_p$. The first bracket is $p$'s general-equilibrium income change from tropical taxation; the second is $p$'s value of emissions reductions. Holding $B_p$ at their model-implied values, the funding game is quasilinear, and Proposition 3 in Appendix A.3 provides conditions under which non-zero contributions are an equilibrium of the provision point game. In general equilibrium, payoffs are not quasilinear—contributions are income transfers that move prices—so I compute the funding equilibrium numerically (similar to Ossa 2014), verifying that no donor has a profitable unilateral deviation at the reported profile.

**The mechanism supports a non-empty *bilateral* coalition.** In Table 5, I report results for the minimum-size non-empty coalition as a conservative lower bound on delivered abatement. In Column (1), transfers finance a Brazilian $CO_2$ tax of \$3.80 per $tCO_2$— exactly marginal damages — and reduce global deforestation $CO_2$ emissions by 7.6%. The mechanism achieves the same emissions reductions and tax rate as a planner with access to only a Brazilian $CO_2$ tax. Europe supports Brazilian abatement, transferring \$1.34 per ton. Their willingness to pay $\chi_i\kappa$ for carbon alone is ~\$1.30 per ton: the (small) gap represents willingness to pay from improving terms of trade. Brazilian food gets more expensive, so Brazilian consumers buy more food from Europe.

**But free riding among recipients undercuts *multilateral* contracts.** Column (4) shows results with all three major forest basins as recipients. Recipients are strategic and can free-ride on each other's land tax policy to get more funding. The Nash outcome is complete unraveling: the minimal non-empty coalition has only the US as a payer, and finances negligible total abatement.[25] Indonesia, in particular, free rides and sets a tax rate of \$0.005 per $tCO_2$. The contest over a pot is the core design flaw, and makes free riding optimal: Indonesia can set a lower carbon tax, lower the total forest over which payments are disbursed, and thus collect a higher payment per hectare forest remaining.[26]

If recipients either commit to a common tax in column (2) or commit to maximize joint surplus in column (3), the mechanism dominates the Brazil-alone recipient mechanism *and* the US continues to support abatement. A cooperative multilateral solution among forest basins *supports* cooperation among donors. However, Indonesia would not individually rationally participate in (4).

25. In a dynamic setting, the scope for unraveling worsens in this mechanism because participants have an option to delay their payments (Harstad 2016).

26. Converting the per-hectare payment to a per-$tCO_2$ payment thus has a similar pitfall.

## 8 Discussion

This paper provides a quantitative framework highlighting inefficient deforestation and diagnoses why that deforestation persists. In equilibrium, deforestation is cheap to prevent even with limited ability to trade in carbon across regions. Part of why deforestation is cheap is geography: forest loss on low-yield land can be avoided with well-targeted policy, and reallocation to lower-emissions land makes up for some food loss. An important corollary: a hectare of deforestation in two different forest basins is not tradeable one-to-one.

Correcting deforestation has large equity implications, and two of the major forest basins would not participate in a utilitarian planners' deforestation policy. However, a Pareto improvement is possible with coordinated transfers. In reality, deforestation policy struggles to achieve scale, suggesting coordination is difficult and putting emphasis on non-cooperative policy where free riding is a major concern.

Turning to policy, I evaluate international contracts for forest conservation. The provision point mechanism achieves a non-empty coalition. In pure bilateral settings, the provision point mechanism achieves the same gains as the planner without requiring broad coordination. Multilateral transfers are undercut by free-riding. Future work could devote attention to assessing alternative, yet-unfunded deforestation conservation mechanisms like the Tropical Forest Forever Facility, and whether their pricing rules are subject to the same free riding risk among recipient countries. Another direction of progress is to use dynamic agreements to enforce commitment with multiple competing recipient countries.

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## Tables and Figures

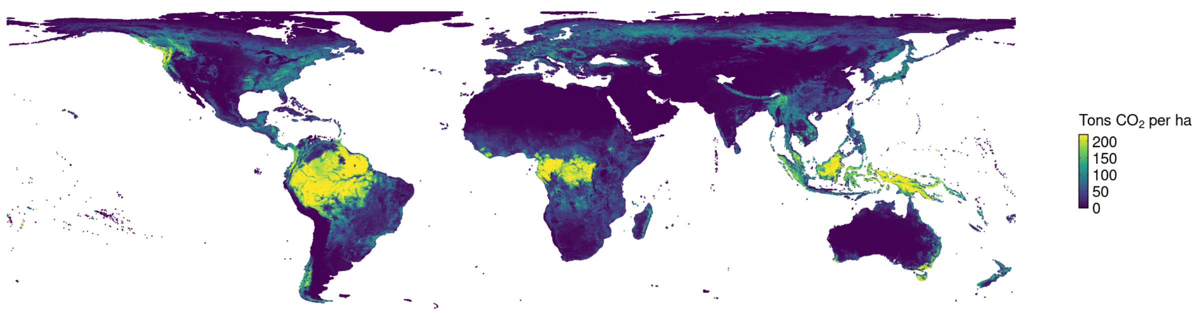
Table 1: Profitability and carbon on deforested land.

| | (1) Global | (2) Deforested | (3) Focal |
|---|---|---|---|
| Potential $CO_2$, t/ha | 39.19 (0.051) | 55.04 (0.183) | 64.43 (0.353) |
| Travel time to city, hours | 19.49 (0.026) | 7.63 (0.072) | 5.31 (0.111) |
| Profitability, USD/ha/yr | 236.53 (0.342) | 308.36 (1.150) | 277.17 (0.824) |
| Profitability per $tCO_2$, USD/$tCO_2$/yr | 23.90 (0.052) | 9.89 (0.106) | 7.33 (0.068) |
| Forest share, % | 30.29 (0.031) | 48.40 (0.159) | 44.76 (0.263) |
| Cropland share, % | 11.50 (0.018) | 6.36 (0.058) | 5.24 (0.077) |
| Pasture share, % | 20.62 (0.025) | 25.75 (0.144) | 26.88 (0.224) |
| Parcels | 1,588,481 | 41,077 | 13,734 |
| Share of global land, % | 100.00 | 3.10 | 1.20 |
| Share of global emissions, % | 100.00 | 62.15 | 31.72 |
| Crop profitability per $tCO_2$ | 49.75 (73.47) | 22.02 (49.65) | 7.52 (9.00) |
| Pastoral profitability per $tCO_2$ | 10.05 (26.27) | 1.32 (2.30) | 1.80 (1.50) |
| Share below \$3.80/$tCO_2$/yr, % | 44.04 | 79.24 | 85.26 |

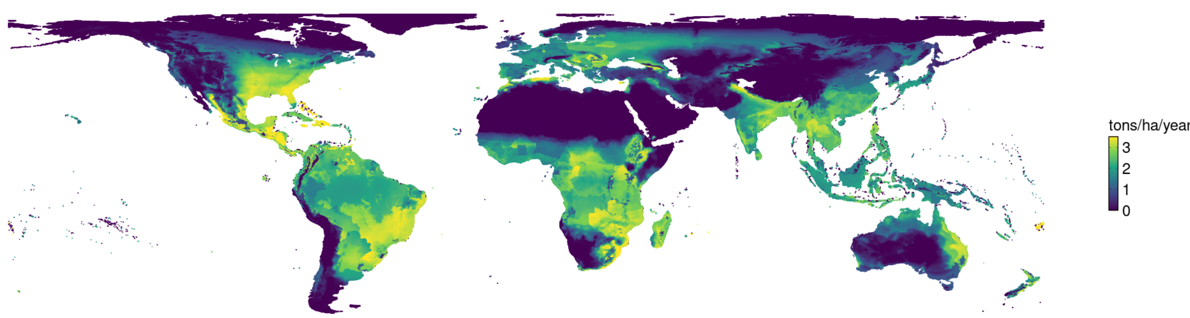
NOTES: Each cell reports the area-weighted mean and, in parentheses, its standard error. Columns (2)–(3) limit attention to land whose vegetation cover falls by more than 10% between 1982 and 2016 in Song et al. (2018); column (3) further limits to the focal regions of Brazil, Indonesia, and the Congo River Basin (Cameroon, the Central African Republic, the Republic of the Congo, the Democratic Republic of the Congo, Equatorial Guinea, and Gabon). Profitability is price times yield less freight cost times yield, production-weighted across crops, in USD per hectare per year. Yields are measured by potential yields (Fischer et al. 2021) scaled by a country-level multiplier which equalizes agricultural value add with total value added from the same crop categories in ITPDE data (Borchert et al. 2022): see Appendix B.2. Freight cost per ton is defined in Appendix B. Profitability per $tCO_2$ is annual profitability per ton of potential emissions, computed only on parcels emitting at least one ton of $CO_2$ per hectare so the ratio does not diverge on near-zero-carbon land. Land-use shares are country-biome shares of parcel area and need not sum to one. The final panel reports crop profitability weighted by cropland area and pastoral profitability weighted by pasture area, with standard deviations rather than standard errors in parentheses; the share below \$3.80/$tCO_2$/yr pools the two margins in proportion to their land-use weight, and so is a share of cropland and pasture rather than of all land.

Figure 1: Map of global productivity measures

(a) $CO_2$ emissions from deforestation

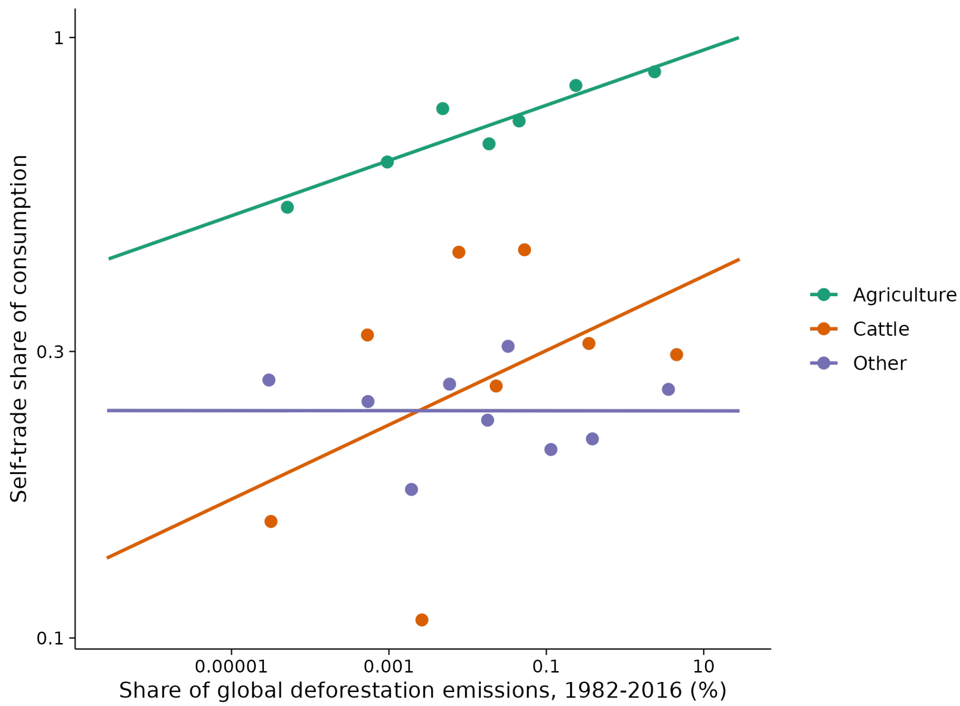



(b) Production-weighted average potential yields




NOTES: Top figure plots potential $CO_2$ emissions from deforestating one hectare of forest (a stock loss) based on the ORNL-DAAC data in 2010. Bottom figure plots production-weighted average potential yields over staple crops (a flow benefit) discussed in Appendix B.

Figure 2: Binned regression of self-trade share, pooling 2000-2020, against share of global emissions 1982-2016.




NOTES: Calculates self-trade share in Agriculture, Cattle, and Other industries using the ITPD-E. Emissions shares convert land use change from Song et al. (2018) into $CO_2$ emissions. Trade pools across years 2000-2020. Lines represent binned regressions with no fixed effects.

Table 2: Land use elasticity estimates for (17)

| **Panel A: Model estimates** | | | | |
|---|---|---|---|---|
| Parameter | Coefficient (Std. Error) | | $N$ cells | $N$ clusters |
| $\tilde{\gamma}_0$ | 0.0074 (0.0012) | | 1,475,462 | 35,220 |
| **Panel B: Continental returns elasticities** | | | | |
| Region | Ag. Elasticity [95% CI] | Pasture Elasticity [95% CI] | $N$ cells | $N$ clusters |
| Global | 1.4658 [1.3895, 1.5463] | 0.7555 [0.7090, 0.8051] | 1,511,828 | 45,666 |
| Africa | 1.2498 [1.0685, 1.4618] | 0.4554 [0.3676, 0.5642] | 261,974 | 6,713 |
| Americas | 1.6435 [1.5256, 1.7706] | 0.7149 [0.6515, 0.7843] | 470,809 | 16,674 |
| Asia | 1.9750 [1.7771, 2.1948] | 1.3498 [1.2087, 1.5073] | 315,747 | 11,565 |
| Europe | 0.8446 [0.7527, 0.9477] | 0.6679 [0.5905, 0.7554] | 382,328 | 9,660 |
| Oceania | 1.2168 [1.0350, 1.4304] | 0.0828 [0.0714, 0.0961] | 80,970 | 1,054 |

NOTES: Standard errors are clustered at ADM2 level. Estimates (17). Panel A shows global pooled prior coefficient $\tilde{\gamma}_0$. Panel B shows summary-statistics for posterior country-biome-level $\gamma_b$, rescaled to be elasticities using the logit formula. Panel B: elasticity w.r.t. per-hectare profit (returns). $N$ cells: Panel A reports the global pooled fit's own estimation sample, which drops countries with 0 price reporting. Panel B keeps these countries and shrinks them completely towards global prior.

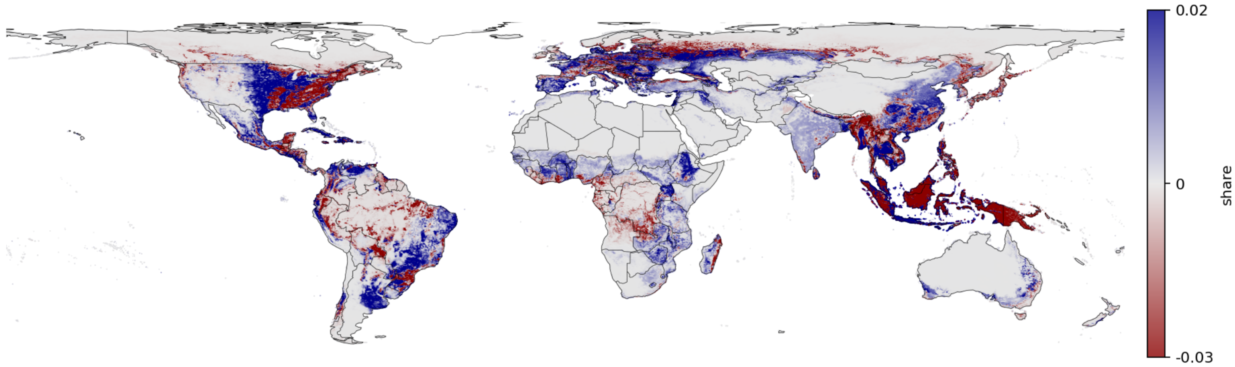
Table 3: Estimates of international agricultural demand parameters

| | (1) OLS | (2) Poisson | (3) IV | (4) PPML-IV |
|---|---|---|---|---|
| *Panel A: Elasticity of substitution, crops* | | | | |
| Crops, $\log(p)$ | -0.692 | -0.298 | -4.214 | -1.018 |
| | (0.242) | (0.109) | (3.148) | (0.368) |
| Observations | 89,877 | 313,234 | 89,523 | 312,064 |
| First-stage F | — | — | 1884.8 | — |
| First-stage Coefficient | — | — | -0.089 | -0.089 |
| *Panel B: Elasticity of substitution, meat* | | | | |
| Cattle, $\log(p)$ | -0.174 | -0.026 | -3.410 | -1.697 |
| | (0.375) | (0.207) | (1.456) | (0.373) |
| Observations | 115,096 | 334,590 | 101,433 | 287,417 |
| First-stage F | — | — | 18147.5 | — |
| First-stage Coefficient | — | — | 0.358 | 0.358 |
| *Panel C: Elasticity of substitution, meat vs. crops* | | | | |
| Price index, $\log(P)$ | -0.229 | -0.104 | -1.528 | -1.175 |
| | (0.023) | (0.028) | (0.280) | (0.717) |
| Observations | 5,172 | 5,172 | 5,140 | 5,140 |
| First-stage F | — | — | 54.4 | — |
| First-stage Coefficient | — | — | 0.055 | 0.055 |
| *Panel D: Structural Parameters* | | | | |
| $\sigma^A$ | 1.692 | 1.298 | 5.214 | 2.018 |
| $\sigma^P$ | 1.174 | 1.026 | 4.410 | 2.697 |
| $\sigma^K$ | 1.229 | 1.104 | 2.528 | 2.175 |
| *Panel E: Statistical tests* | | | | |
| $H_0: \sigma^P = \sigma^A$ | $z = 1.295$, $p = 0.195$ | | | |
| $H_0: \sigma^K = 2.82$ | $z = -0.899$, $p = 0.369$ | | | |

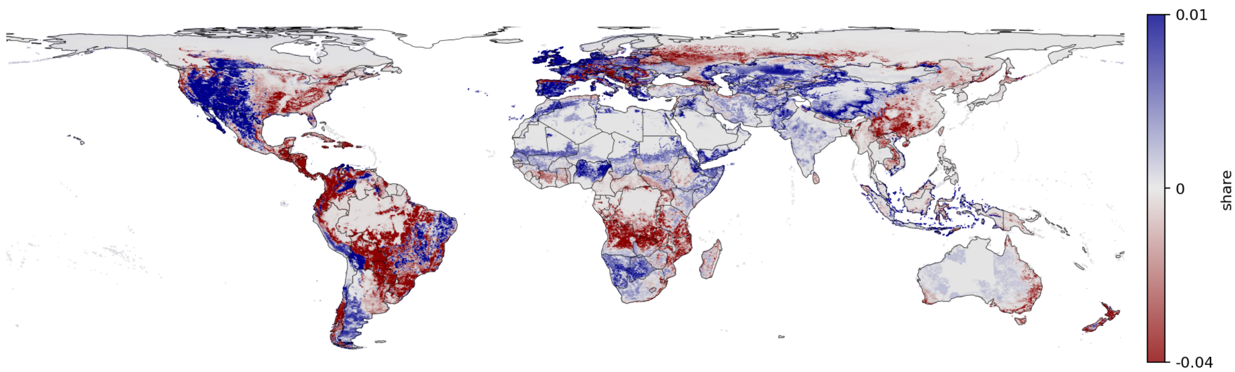
NOTES: Dependent variable is trade share (log for OLS/IV, level for Poisson/PPML-IV). Panel A and B specifications include importer×year fixed effects; Panel C importer + year. All cover years 2001–2016. OLS and Poisson estimate via least squares with log price (raw data in A, B; CES index in C) as regressor. IVs are: (A) log potential yields, (B) log crop price, and (C) log of potential wheat yields (crop) and grasses (cattle). PPML-IV estimates via GMM. F statistics report Cragg-Donald first-stage F-statistic for the relevant endogenous regressor. Standard errors clustered by exporter and importer (two-way) in Panels A and B, importer in C. The target parameter, $\sigma^A$, is given by $1 - \beta$ where $\beta$ is the estimated coefficient. Panel E reports two-sided $z$-tests based on the PPML-IV estimates: $\sigma^P = \sigma^A$ (treating the two estimates as independent). The second row tests $\sigma^K = 2.82$, the estimate of the cross-crop elasticity of substitution from Costinot, Donaldson, and Smith (2016). $z$-statistic and $p$-value shown.

Figure 3: Changes in land use and landowner welfare from optimal deforestation carbon tax.

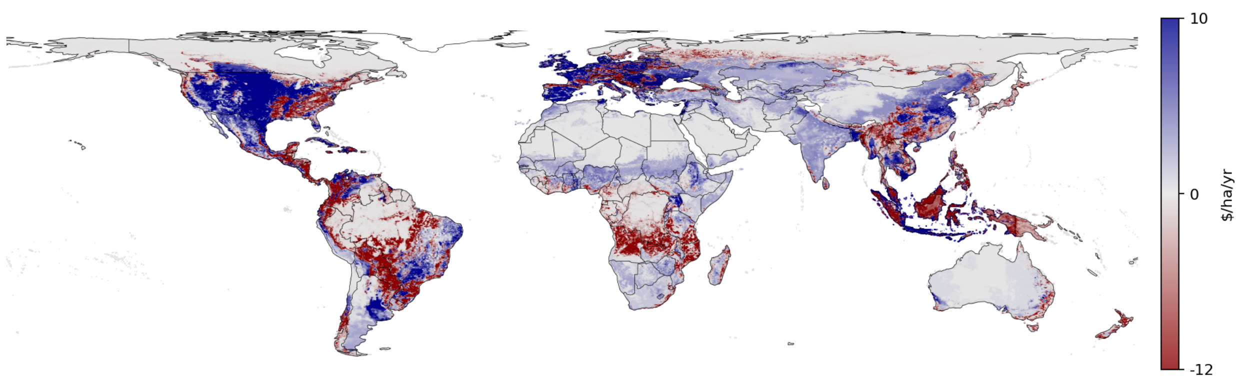
(a) Change in cropland land use shares under optimal tax




(b) Change in pastoral land use shares under optimal tax



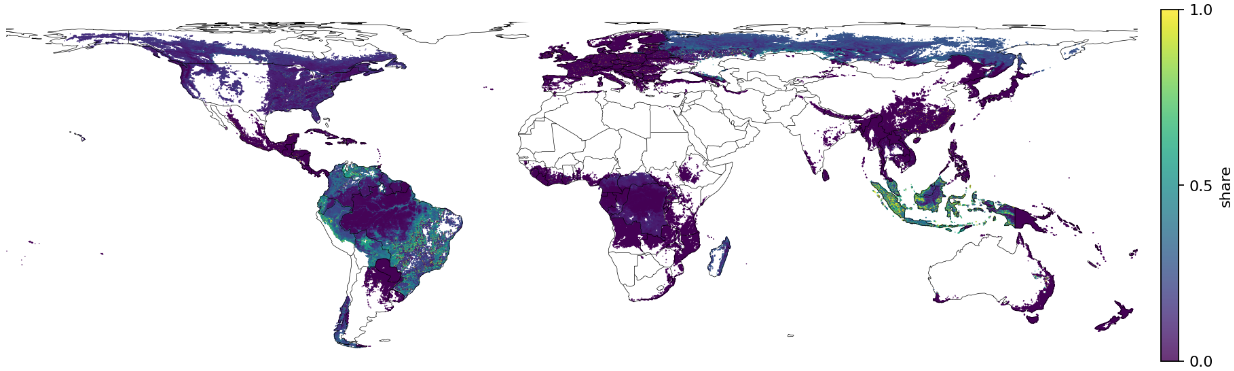

(c) Landowner welfare: change in inclusive value of land uses



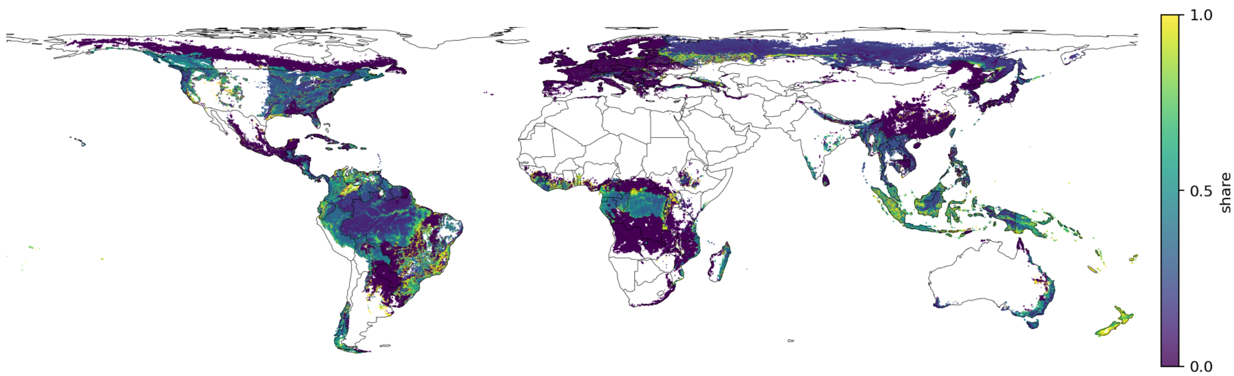

Notes: Figure illustrates changes in cropland share (top), pasture share (middle), and changes in landowner inclusive value (bottom) in the taxed equilibrium relative to business-as-usual (BAU). Inclusive values are deflated by country-level real price index $P_i^C$. Color scales are winsorized at the 5th and 95th percentiles of changes for visual clarity.

Figure 4: Share of optimal emissions reductions which cannot be achieved without reallocation.

(a) Cross-country reallocation shut down

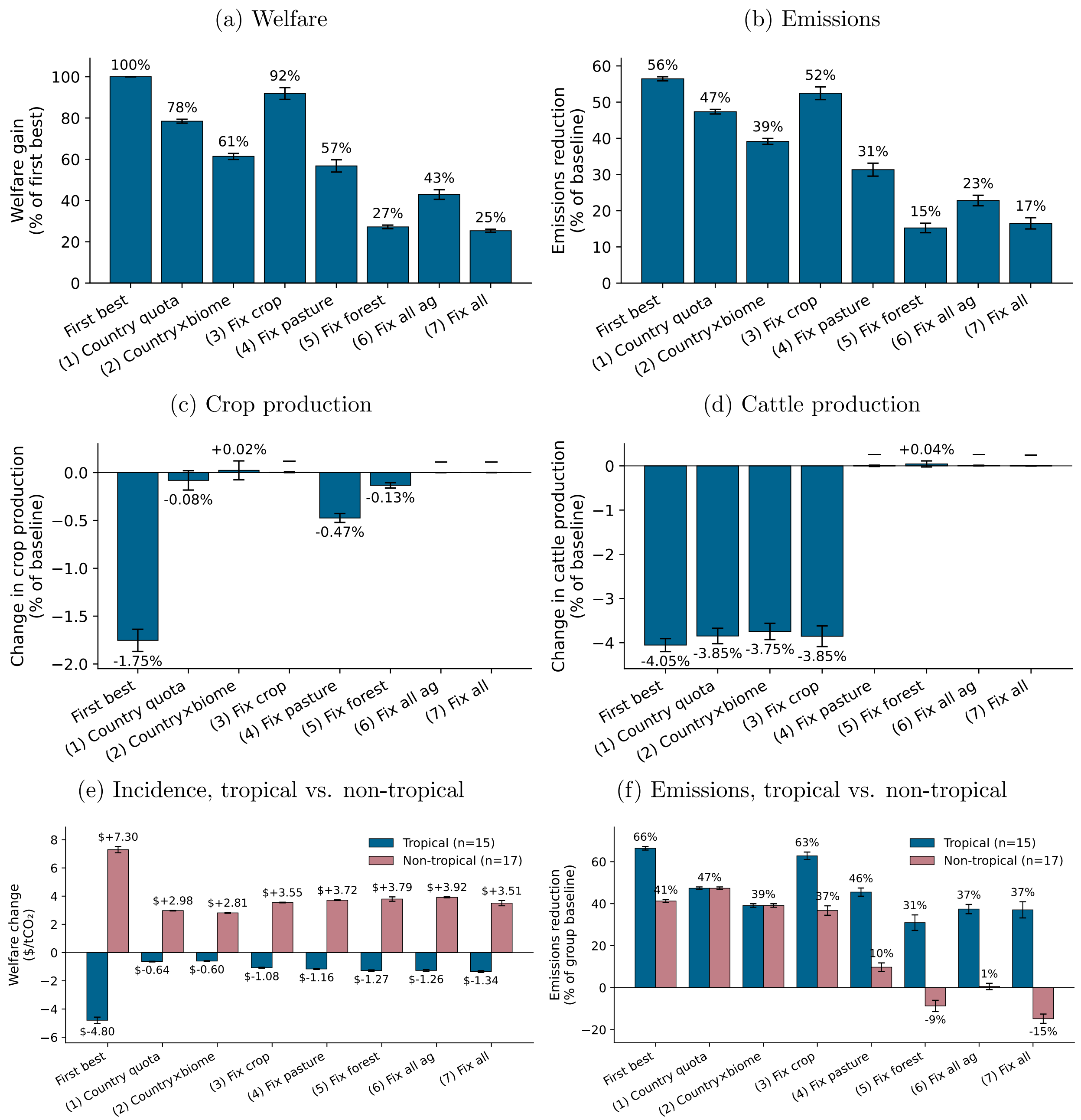



(b) Cross-country-biome reallocation shut down




NOTES: No cross-country case 4a maximizes social welfare with a country-level tax rate subject to the constraint that each country emits the same share of global deforestation emissions as under the status quo. Cross-country-biome 4b repeats, shutting down reallocation across biomes even within the same country. A higher share means *fewer* emissions can be avoided in the no-reallocation equilibrium. Regions which avoid less than 0.01 t / ha under the global optimum are recoded as NA for visual clarity.

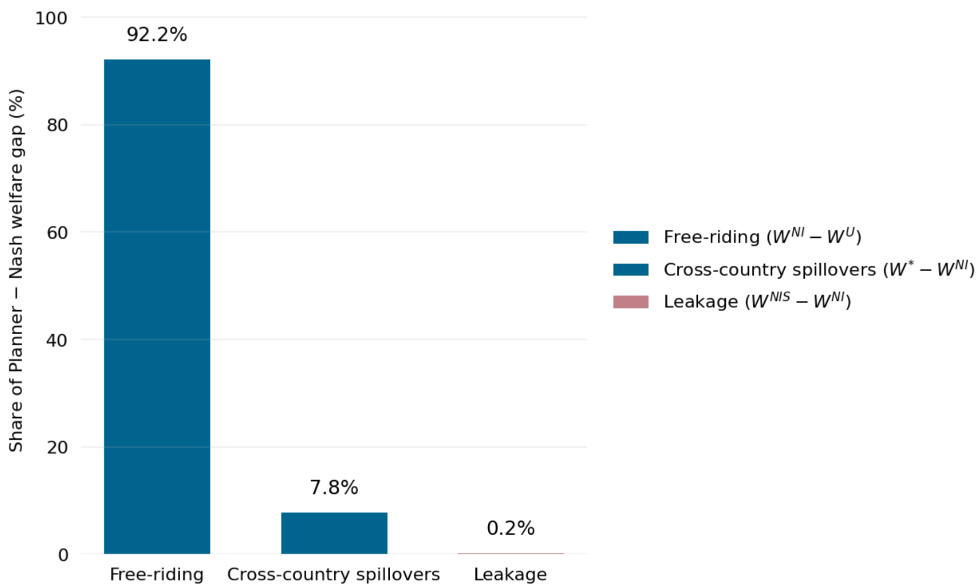
Figure 5: Constrained planner optimal policy results, Section 6.3.



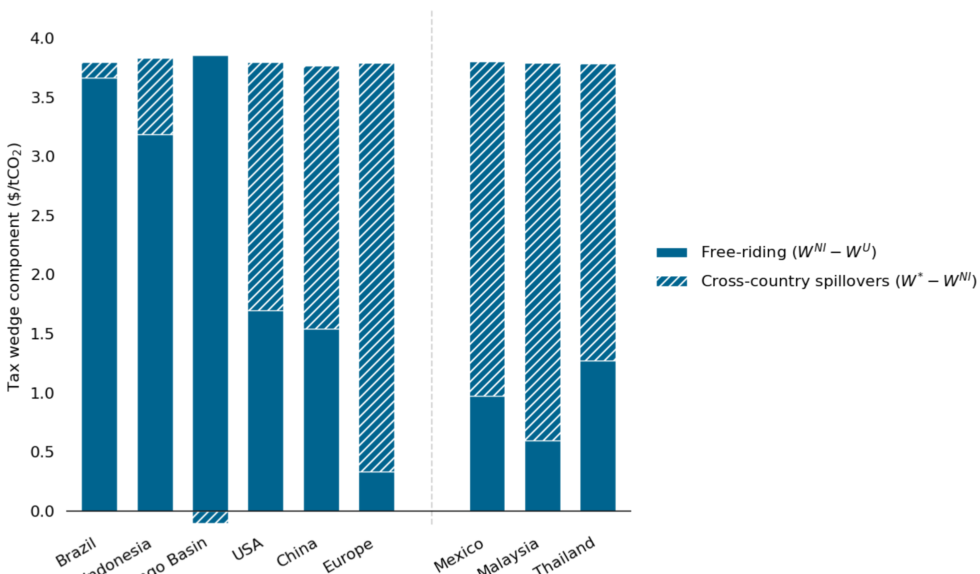

NOTES: Each scenario (1)-(7) runs a constrained planner optimization problem where the planner maximizes welfare with country-specific carbon prices. (1) requires all countries to maintain status quo shares of global emissions; (2) all country-biomes to maintain their share of global emissions; (3) fixes global crop production to status quo; (4) fixes global cattle production to status quo; (5) fixes total forest area to status quo; (6) implements constraints (3) and (4); (7) implements (3), (4), and (5). First-best is solution to (13) when planner has access to carbon taxes and lump-sum transfers. Reported values are bootstrap estimates of global aggregates and 95% confidence intervals over 100 draws. Tropics include all countries whose area-weighted plot centroid latitude is $\leq 23.5$ in absolute value. Rest-of-World and India (a marginal case) are moved to extra-tropical. "—" indicates a value is pinned by a constraint. Each bootstrap draws a vector of land use elasticities from the posterior distribution reported in 2 and the demand elasticities reported in 3 assuming independence across both sets of elasticities. Demand draws with $\sigma^A < 1$ or $\sigma^P < 1$ are redrawn as infeasible.

Figure 6: Decomposing sources of inefficiency in status quo deforestation pricing.

(a) Decomposing sources of welfare gains from planner's pricing into terms of trade and coordination gaps.




(b) Decomposing the Nash-planner gap in optimal carbon taxes for select countries




NOTES: 6a decomposes the gap between planner's welfare $\mathcal{W}^{\star}$ under country-level optimal taxes with no lump-sum transfers against two scenarios: "Nash-Internalized" (NI) where countries set tax rates non-cooperatively but value global carbon at the SCC and "Nash-uncoordinated" (U) where they value global carbon according to their GDP share $\times$ the SCC. These two bars add up to 100%. Leakage is a separate decomposition: it shows the gap between NI and a selfish Nash-internalized equilibrium where countries only price according to their countries' domestic emissions, ignoring global emissions in welfare. 6b shows the same decomposition in terms of the optimal planner's tax within country. Hatched bars show the portion of the tax which is absent when terms of trade distortions operate, and solid when countries play the uncoordinated game.

Table 4: Deforestation policy counterfactual simulation results

| Metric | Estimate | 95% CI |
|---|---|---|
| **Panel A: Global optimum (with lump-sum transfers)** | | |
| Welfare gain (%) | 0.04 | [0.04, 0.04] |
| Welfare gain (\$/ton $CO_2$) | 2.498 | [2.47, 2.536] |
| Crop production change (%) | -1.75 | [-2.00, -1.57] |
| Cattle production change (%) | -4.03 | [-4.26, -3.74] |
| Emissions change (%) | -56.20 | [-57.36, -55.14] |
| Gross transfers per forested ha (\$/ha) | 17.84 | [17.38, 18.29] |
| Gross transfers per $tCO_2$ abated (\$/$tCO_2$) | 6.63 | [6.45, 6.82] |
| **Panel B: Country-level optimum (per-country tax, no transfers)** | | |
| Welfare gain (%) | 0.03 | [0.03, 0.04] |
| Welfare gain (\$/ton $CO_2$) | 2.474 | [2.446, 2.513] |
| Crop production change (%) | -0.27 | [-0.47, -0.11] |
| Cattle production change (%) | -3.75 | [-3.96, -3.47] |
| Emissions change (%) | -54.07 | [-55.23, -52.99] |

NOTES: Panel A is global optimal carbon tax when planner has access to transfers; Panel B is global optimal set of carbon taxes absent transfers. Reported values are bootstrap estimates and 95% confidence intervals over 100 draws. Percentages represent a percent of the status quo quantity. Each bootstrap draws a vector of land use elasticities from the posterior distribution reported in 2 and the demand elasticities reported in 3 assuming independence across both sets of elasticities. Demand draws with $\sigma^A < 1$ or $\sigma^P < 1$ are redrawn as infeasible.

Table 5: Outcomes of a provision point mechanism

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| Recipient(s) | BRA | BRA, IDN, CRB | BRA, IDN, CRB | BRA, IDN, CRB |
| Instrument | Per-t$CO_2$ | Per-t$CO_2$ | Per-t$CO_2$ | Per-t$CO_2$ |
| Mode | Nash | uniform | joint | Nash |
| Area-weighted avg tax rate | 3.800 | 3.800 | 3.717 | 0.012 |
| Global $\Delta$Emissions (%) | -7.56 | -24.78 | -24.58 | -0.11 |
| Emissions abated (Mt $CO_2$) | 1503 | 4929 | 4889 | 22 |
| Min-size Nash coalition (payers) | EUR | USA + EUR | USA + EUR | USA |
| $k^\star$ (# of payers) | 1 / 31 | 2 / 29 | 2 / 29 | 1 / 29 |
| Transfers total ($B) | 2.009 | 7.607 | 7.412 | 0.002 |
| Transfers per ha forest ($/ha) | 4.48 | 8.70 | 8.48 | 0.00 |
| Transfers per t$CO_2$ abated ($/t$CO_2$) | 1.34 | 1.54 | 1.52 | 0.09 |
| Welfare gain, total ($B) | 3.795 | 13.257 | 13.260 | 0.088 |
| payer net ($B) | 0.026 | 6.376 | 6.448 | 0.029 |
| free-rider net ($B) | 3.769 | 6.882 | 6.812 | 0.059 |
| Welfare gain per ha forest ($/ha) | 8.47 | 15.16 | 15.17 | 0.10 |
| Welfare gain per t$CO_2$ abated ($/t$CO_2$) | 2.52 | 2.69 | 2.71 | 3.98 |
| *Recipient welfare (net, post-transfer, $B):* | | | | |
| Brazil | +0.00 | +1.73 | +1.68 | +0.00 |
| Indonesia | — | -4.09 | -3.97 | -0.00 |
| Congo Basin | — | +2.36 | +2.30 | +0.00 |
| Total (coalition) | +0.00 | -0.00 | -0.00 | +0.00 |

NOTES: Recipients indicate countries which receive per-forest-hectare transfers from remaining participant countries. Recipients can set a tax per agricultural hectare to maximize welfare. BRA = Brazil, IDN = Indonesia, CRB = Congo River Basin. Emissions reductions relative to status quo. Min-size Nash coalition indicates the smallest number of countries who would voluntarily contribute transfers conditional on their incentive compatibility constraint under the provision point mechanism. Modes: uniform — a planner sets a uniform $CO_2$ tax within recipients; joint — a planner sets country-level $CO_2$ taxes to maximize joint surplus among the recipients; and Nash — countries maximize their own welfare criterion by setting their $CO_2$ tax.

# Appendix: The Global Allocative Efficiency of Deforestation

## A Theoretical appendix

### A.1 Social welfare

Define $V_i(t, \mathcal{F})$ as the social planner's value function in consumption $X$ in Equation (13). Global social welfare is characterized by the following optimization problem over an $J \times 1$ dimensional vector of $CO_2$ taxes $t_j$ and country-level lump sum transfers $\mathcal{F}_i$. Define an amenity- and transfer-adjusted surplus measure denominated in nominal dollars $\widetilde{Y}_i = Y_i + \mathcal{T}_i + \mathcal{F}_i$. Amenities $A_i$ are in utility units. Then $V_i(t, \mathcal{F}) = \frac{\widetilde{Y}_i}{P_i^C} + A_i$ (the equivalence between the CES value function and real income can be found in e.g., Arkolakis, Costinot, and Rodríguez-Clare 2012).

$$\mathcal{W}^\star = \max_{t_i, \mathcal{F}_i} \sum_{i=1}^{J} \psi_i V_i(t, \mathcal{F}) - \chi_i \kappa D \tag{A.1}$$

$$\mathcal{T}_i = t_i \sum_{\omega \in \Omega_i} d(\omega)[\mu^A(\omega) + \mu^P(\omega)] \tag{A.2}$$

$$\sum_{i=1}^{J} \mathcal{F}_i = 0 \tag{A.3}$$

I ex ante restrict the planner to be unable to change tariff rates across countries to focus ideas on the tax as the key allocative mechanism.

#### A.1.1 Recovering a global Pigouvian tax.

**Proposition 1.** *A single, Pigouvian tax equal to marginal damages $t^\star = \kappa$ maximizes global welfare under the following conditions: (i) unobserved, idiosyncratic amenities $\epsilon_p(\omega)$ are distributed Type-I extreme value; (ii) access to global lump sum transfers; and (iii) $\beta = 1$ constant returns to manufacturing labor.*

*Proof.* Taking the first order condition with respect to a global tax rate $t_j = t$, which Proposition 2 will show to be without loss:

$$\frac{\partial \mathcal{W}^\star}{\partial t} = \sum_{i=1}^{J} \psi_i \frac{\partial V_i(t, \mathcal{F})}{\partial t} - \chi_i \kappa \frac{\partial D}{\partial t}$$

**Step 1. Nominal welfare response to a marginal carbon tax.** Denote the set $\mathcal{K} = \{A, P\}$ as the deforesting land uses. Define nominal dollar-denominated profits $\Pi_i = \sum_{h \in \mathcal{K}} (p_i^h Q_i^h - w_i L_i^h) - w_i L_i^T - tD_i + P_i^C A_i = P_i^C \sum_{\omega \in \Omega_i} \pi(\omega)$. It is useful to rewrite an accounting identity relating this object and national income: $Y_i = \Pi_i - P_i^C A_i + w_i \sum_{h \in \mathcal{K}} L_i^h + \frac{1}{\beta} w_i L_i^M + w_i L_i^T$. Applying the envelope theorem to $\Pi_i$,

$$\frac{d}{dt}\Pi_i = \sum_{\omega \in \Omega_i} \sum_{h \in \mathcal{K}} \mu^h(\omega) \frac{d}{dt} v^h(\omega) + \frac{dP_i^C}{dt} A_i$$

A direct total derivative of the profit term returns the following expression:

$$\frac{d}{dt}\Pi_i = \sum_{\omega \in \Omega_i} \sum_{h \in \mathcal{K}} \mu^h(\omega) \frac{d}{dt} v^h(\omega) + \sum_{\omega \in \Omega_i} \sum_{h \in \mathcal{K}} v^h(\omega) \frac{d}{dt} \mu^h(\omega) + P_i^C \frac{dA_i}{dt} + \frac{dP_i^C}{dt} A_i$$

Thus, as a direct implication of the envelope condition $P_i^C \frac{dA_i}{dt} = -\sum_{\omega \in \Omega_i} \sum_{h \in \mathcal{K}} v^h(\omega) \frac{d}{dt} \mu^h(\omega)$. Tracing through the original profit definition,

$$\begin{aligned} &\sum_{\omega \in \Omega_i} \sum_{h \in \mathcal{K}} [(p_i^h - c(\omega) w_i) \eta^h(\omega) - w_i - td(\omega)] \frac{d\mu^h(\omega)}{dt} \\ &\qquad = \sum_{h \in \mathcal{K}} \left( p_i^h \frac{dQ_i^h}{dt} - w_i \frac{dL_i^h}{dt} \right) - w_i \frac{dL_i^T}{dt} - t \frac{dD_i}{dt} = -P_i^C \frac{dA_i}{dt}. \end{aligned}$$

Next, we apply this result to aggregate income $Y_i$ to obtain:

$$\begin{aligned} \frac{dY_i}{dt} &= \frac{d\Pi_i}{dt} + \frac{dw_i}{dt} \left( \sum_{h \in \mathcal{K}} L_i^h + L_i^T + \frac{1}{\beta} L_i^M \right) + w_i \frac{1-\beta}{\beta} \frac{dL_i^M}{dt} - P_i^C \frac{dA_i}{dt} - \frac{dP_i^C}{dt} A_i \\ &= \sum_{h \in \{A,P,M\}} \frac{dp_i^h}{dt} Q_i^h - D_i - P_i^C \frac{dA_i}{dt} \end{aligned}$$

Combining the previous derivation with the tax rebate derivative $dT_i/dt = D_i + t\, dD_i/dt$ and the transfer derivative $d\mathcal{F}_i/dt$, total derivative of $V_i(t, \mathcal{F})$ decomposes as:

$$\frac{dV_i(t,\mathcal{F})}{dt} = \underbrace{\frac{d}{dt}\left(\frac{1}{P_i^C}\right)\cdot(Y_i+\mathcal{T}_i+\mathcal{F}_i)}_{\substack{\text{price-level effect} \\ \text{(monetary income only)}}} + \underbrace{\frac{1}{P_i^C}\sum_{h\in\{A,P,M\}}\frac{dp_i^h}{dt}\,Q_i^h}_{\text{general equilibrium}} + \underbrace{\frac{1}{P_i^C}\left[t\,\frac{dD_i}{dt} + \frac{d\mathcal{F}_i}{dt}\right]}_{\text{direct effects}} \tag{A.4}$$

Both amenity terms cancel: $P_i^C\frac{dA_i}{dt}$ in $\frac{dY_i}{dt}$ cancels against $\frac{dA_i}{dt}$ in $\frac{dV_i}{dt}$, and $\frac{dP_i^C}{dt}A_i$ cancels between the envelope and direct derivatives of $\Pi_i$. The carbon payment $-D_i$ borne by landowners cancels against the rebate $+D_i$.

**Step 2. Simplifying general equilibrium effects.** In this section, I demonstrate that the general equilibrium effects and the price level effect cancel. For convenience, define $\widetilde{Y}_i = Y_i+\mathcal{T}_i+\mathcal{F}_i$ is monetary income inclusive of the carbon rebate $\mathcal{T}_i = tD_i$ and the global transfer $\mathcal{F}_i$ Two applications of Shephard's lemma for a CES demand function give:

$$d\log P_i^C = \sum_h \lambda_i^h\, d\log P_i^h, \qquad d\log P_i^h = \sum_k \lambda_{ik}^h\, d\log p_k^h.$$

Combining these decompositions,

$$\sum_i \widetilde{Y}_i\,\frac{d\log P_i^C}{dt} = \sum_i \widetilde{Y}_i \sum_h \lambda_i^h \sum_k \lambda_{ik}^h\,\frac{d\log p_k^h}{dt} = \sum_h\sum_k \frac{d\log p_k^h}{dt}\underbrace{\sum_i \lambda_i^h\,\lambda_{ik}^h\,\widetilde{Y}_i}_{\substack{\text{global expenditure} \\ \text{on variety }(h,k)}}.$$

Market clearing for variety $(h,k)$ equates global expenditure on it to country $k$'s producer revenue,

$$\sum_i \lambda_i^h\,\lambda_{ik}^h\,\widetilde{Y}_i = p_k^h\,Q_k^h \qquad h\in\{A,P,M\}$$

Substituting,

$$\sum_i \widetilde{Y}_i\,\frac{dP_i^C}{dt} = \sum_h\sum_k p_k^h\,Q_k^h\,\frac{d\log p_k^h}{dt}$$

Under this result, Equation (A.4) simplifies to the direct effect alone.

**Step 3. Deriving Pigou.** Define the multiplier on the constraint for total transfers, Equation (A.3), as $\varrho$. Consider allocations at which the marginal utility of income is equalized across countries at the multiplier's value, $\psi_i/P_i^C = \varrho$ for all $i$ (characterized in Step 4). By Step 2, the price blocks cancel from every condition. The direct effect for the tax in Equation (A.4) is:

$$\text{Direct effect} = \frac{1}{P_i^C}\left[\frac{\partial \mathcal{F}_i}{\partial t} + t\frac{\partial D_i}{\partial t}\right]$$

Combining this result with the equalization, and $\sum_{i=1}^{J} \frac{\partial \mathcal{F}_i}{\partial t} = 0$ given the adding-up constraint:

$$\therefore t^{\star} = \left[\sum_{i=1}^{J} \frac{\psi_i}{P_i^C}\left(\frac{\partial D_i}{\partial t}\right)\right]^{-1}\left(\sum_{i=1}^{J} \chi_i \kappa \frac{\partial D}{\partial t}\right) = \frac{\kappa}{\varrho}$$

where $\sum_{i=1}^{J} \chi_i = 1$ by assumption. Each transfer condition likewise retains only direct effects:

$$\sum_{i=1}^{J} \frac{\psi_i}{P_i^C}\left[t\,\frac{\partial D_i}{\partial \mathcal{F}_j} + \frac{\partial \mathcal{F}_i}{\partial \mathcal{F}_j}\right] - \kappa\,\frac{\partial D}{\partial \mathcal{F}_j} = \varrho \implies (\varrho t - \kappa)\,\frac{\partial D}{\partial \mathcal{F}_j} = 0,$$

using $\partial \mathcal{F}_i/\partial \mathcal{F}_j = \mathbf{1}\{i = j\}$ — satisfied at the same tax $t^{\star} = \kappa/\varrho$.

**Step 4. Marginal utility of income.** It remains to characterize the allocations imposed in Step 3. Under flexible lump-sum transfers, the social planner will always choose to redistribute such that the marginal utility of income is equalized across countries: with the weights held fixed, $\psi_i/P_i^C = \varrho$ comprises $J$ conditions determining the $J-1$ free transfers and the multiplier $\varrho$. Because $\psi$ is defined only up to scale I normalize $\varrho = 1$, which fixes the utils-per-dollar conversion between $\kappa$ and $t$. Therefore $t^{\star} = \kappa$. □

The Pigouvian tax is a more general result than Proposition 1, up to a normalization. Per Step 3 of the proof, the optimal tax is uniform and Pigouvian, $t^{\star} = \kappa/\varrho$, equal to $\kappa$ under the scale normalization $\varrho = 1$. This hold under any strictly positive weight vector $\{\psi_i\}_{i=1}^{J}$ in (A.1) and conditions (i)–(iii). Welfare weights determine the income distribution through $P_i^C$.

Lemma 1 relates general equilibrium welfare gains from the Pigouvian tax to the marginal abatement cost — the foregone welfare for a marginal ton of emissions reductions (abatement).

**Definition of marginal and average abatement cost.** Start with the planner's objective (13). Write reductions $R(t)$ as avoided emissions at tax $t$ relative to $t = 0$ and abatement costs

$C(t)$ as the loss in economic surplus $S(t)$:

$$\mathcal{W}(t) = S(t) - \kappa D(t), \qquad S(t) \equiv \sum_{i=1}^{J} \psi_i V_i(t, \mathcal{F}), \qquad R(t) \equiv D(0) - D(t), \qquad C(t) \equiv S(0) - S(t).$$

Define marginal and average abatement cost as

$$\text{MAC} \equiv -\frac{dS}{dR}, \qquad \overline{\text{AC}}(t) \equiv \frac{C(t)}{R(t)}.$$

**Lemma 1.** *Under the conditions of Proposition 1 — Type-I extreme value amenities, utilitarian weights, and access to global lump-sum transfers —* $\text{MAC}(t) = t$ *for every* $t$*, and* $dW/dR = \kappa - \text{MAC}$*. Consequently* $t^\star = \kappa$*, average cost is the abatement-weighted mean of the tax path,*

$$\overline{\text{AC}}(t) = \frac{\int_0^t sR'(s)ds}{\int_0^t R'(s)ds} = \mathbb{E}\big[s \mid s \le t\big],$$

*and the planner's gain is the wedge between the social cost of carbon and average cost,*

$$\Delta\mathcal{W} \equiv \mathcal{W}(\kappa) - \mathcal{W}(0) = R(\kappa)\big[\kappa - \overline{\text{AC}}(\kappa)\big], \qquad \kappa - \overline{\text{AC}}(\kappa) = \frac{1}{R(\kappa)} \int_0^\kappa R(t)dt.$$

*Proof.* Steps 2 and 3 of the proof of Proposition 1 leave $dS/dt = tD'(t)$, so $\text{MAC}(t) = -tD'(t)/[-D'(t)] = t$ and, from $\mathcal{W} = S - \kappa D$, $dW/dR = \kappa - t$, so $t^\star = \kappa$. Then $C(t) = -\int_0^t S'(s)ds = \int_0^t sR'(s)ds$, the stated conditional mean under the density $A'$, and

$$\Delta\mathcal{W} = \kappa R(\kappa) - C(\kappa) = \int_0^\kappa (\kappa - t)R'(t)dt = \int_0^\kappa R(t)dt$$

by parts. Dividing by $R(\kappa)$ gives both stated forms. □

### A.2 Country-level carbon taxes without international transfers

Removing the transfer instrument sets $\mathcal{F}_i \equiv 0$ and deletes the constraint $\sum_i \mathcal{F}_i = 0$ together with its multiplier $\varrho$; the within-country carbon tax rebate $T_i = \sum_{\omega\in\Omega_i} t_i\, d(\omega)[\mu^A(\omega) + \mu^P(\omega)] = t_i D_i$ is retained. Writing monetary income as $\widetilde{Y}_j = \Pi_j - P_j^C A_j + w_j N_j + \mathcal{T}_j$, the planner solves

$$\mathcal{W}^\star = \max_{\{t_i\}_{i=1}^J} \sum_{j=1}^{J} \psi_j V_j(t, 0) - \kappa D, \qquad V_j = \frac{\widetilde{Y}_j}{P_j^C} + A_j, \qquad D = \sum_j D_j.$$

Define country $j$'s price-induced real income response to a marginal tax in $i$,

$$\Theta_j^{(i)} \equiv \underbrace{\frac{\partial(1/P_j^C)}{\partial t_i}\widetilde{Y}_j}_{\text{price level}} + \underbrace{\frac{1}{P_j^C}\left[\sum_{h\in\{A,P,M\}} Q_j^h \frac{\partial p_j^h}{\partial t_i}\right]}_{\text{general equilibrium effects}}$$

and the redistribution wedge $\mathcal{R}_i \equiv \sum_{j=1}^{J} \psi_j \, \Theta_j^{(i)}$.

**Proposition 2** (Country-level tax, no transfers)**.** *Consider a planner with no access to international transfers for redistribution; ($\mathcal{F}_i \equiv 0$). We continue the assumptions of (i) Type-I extreme value amenities and (ii) constant returns to manufacturing scale $\beta = 1$. The welfare-maximizing country-level tax vector is Pigouvian but adjusts for a redistribution wedge and leakage effect.*

$$\boxed{t_i^\star = \frac{\kappa P_i^C}{\psi_i} - \frac{P_i^C}{\psi_i}\left(\frac{\partial D_i}{\partial t_i}\right)^{-1}\left[\mathcal{R}_i + \sum_{j\neq i}\left(\frac{\psi_j}{P_j^C}t_j - \kappa\right)\frac{\partial D_j}{\partial t_i}\right].} \tag{A.5}$$

*For countries where $\frac{\partial D_j}{\partial t_i} = 0$, any carbon tax can be justified.*

*Proof.* The first-order condition in $t_i$ is $\sum_j \psi_j \, \partial V_j(t,0)/\partial t_i = \kappa \, \partial D/\partial t_i$. Decomposing $\partial V_j/\partial t_i$ as in Step 1 of the proof of Proposition 1, the amenity terms cancel and the carbon payment $-D_j$ borne by landowners cancels against the rebate $+D_j$ in $\partial \mathcal{T}_j/\partial t_i$ when $j = i$; both are absent when $j \neq i$. What remains is the price block $\Theta_j^{(i)}$ and the mechanical revenue effect

$$\sum_j \psi_j \Theta_j^{(i)} + \sum_j \frac{\psi_j}{P_j^C} t_j \frac{\partial D_j}{\partial t_i} = \kappa \sum_j \frac{\partial D_j}{\partial t_i}.$$

Substituting the definition of $\mathcal{R}_j$,

$$\mathcal{R}_i + \sum_j \left(\frac{\psi_j}{P_j^C}t_j - \kappa\right)\frac{\partial D_j}{\partial t_i} = 0.$$

Isolating the $j = i$ term and solving for $t_i$ gives the boxed expression. When transfers are reintroduced, the planner will again set $\mathcal{R}_i = 0$. Further, $\frac{\psi_j}{P_j^C} = \varrho$ everywhere, which we normalize to 1, so $t_j = \kappa \, \forall j$ — exactly the global Pigouvian tax. □

Proposition 2 shows that, without access to international transfers, there are two new considerations in setting a first-best tax. The first, classically: the planner adjusts the optimal carbon tax because they cannot freely redistribute income across countries. A country can have a low planner weight $h_i$ either because its price level is high or its Pareto weight small. Low weight countries require a larger dollar tax to represent the same utility damage. Through the redistributive wedge $\mathcal{R}_i$, taxes are higher in countries whose terms-of-trade consequences accrue to countries the planner weights above their marginal utility of income, and lower where those consequences fall on countries it weights below. The second country-level adjustment is a leakage adjustment. Countries whose food supply is more readily diverted to deforestation abroad should be taxed less. The weights $\frac{\psi_i}{P_i^C}$ are the net social marginal value of income from Diamond (1975).

### A.3 Nash Equilibrium of Provision-point mechanism

The following proposition provides conditions under which a Nash Equilibrium is guaranteed for the provision point mechanism in 7.2. The main simplification required to provide a simple analytic characterization of the Nash equilibrium is quasilinearity on the donor side, which does not hold in full general equilibrium.

**Proposition 3.** *Fix the mechanism $\{R, \bar{F}\}$ and the recipient best response $t^\star(\bar{F})$, and assume donor payoffs are quasilinear: on activation donor $p$ receives $B_p - \mathcal{F}_p$ and under non-activation $0$, with the valuations $(B_p)_{p\in\mathcal{P}}$ invariant to the contribution profile.*

*(a) The zero-contribution profile $\mathcal{F} = \mathbf{0}$ is a Nash equilibrium if and only if $\max_{p\in\mathcal{P}} B_p \leq \bar{F}$.*

*(b) A Nash equilibrium with a non-empty contributor set exists if and only if $\sum_{p\in\mathcal{P}} \max\{B_p, 0\} \geq \bar{F}$; every such equilibrium has $\sum_p \mathcal{F}_p = \bar{F}$ exactly, with $\mathcal{F}_p \leq B_p$ for each contributor.*

*(c) If $\max_{p\in\mathcal{P}} B_p > \bar{F}$, then $\mathcal{F} = \mathbf{0}$ is not a Nash equilibrium and* every *Nash equilibrium activates the mechanism.*

*Proof.* For any profile with $\sum_p \mathcal{F}_p > \bar{F}$, a contributor can lower $\mathcal{F}_p$ while activating, strictly saving cost (under the quasilinearity assumption). So every activating equilibrium has $\sum_p \mathcal{F}_p = \bar{F}$. Given exact financing each contributor is pivotal: deviating downward yields the status quo, preferred iff $\mathcal{F}_p > B_p$. Nor is any upward deviation profitable: raising $\mathcal{F}_p$, or entering with $\mathcal{F}_p > 0$, preserves activation and strictly lowers $B_p - \mathcal{F}_p$ (again, under quasilinearity). Hence the activating equilibria split $\bar{F}$ with $\mathcal{F}_p \in [0, B_p]$, and one exists iff $\sum_p \max\{B_p, 0\} \geq \bar{F}$, giving (b). From $\mathbf{0}$, a deviation below $\bar{F}$ is payoff-equivalent to staying. A deviation to $\mathcal{F}_p \geq \bar{F}$ activates. Donor best response is then exactly $\bar{F}$, strictly so iff $B_p > \bar{F}$. $\mathbf{0}$ is Nash iff $\max_p B_p \leq \bar{F}$, giving (a). If $\max_p B_p > \bar{F}$ then $\mathbf{0}$ is not Nash. The donor with $B_{p^\dagger} > \bar{F}$ gains by contributing $\bar{F} - \sum_{q\neq p^\dagger} \mathcal{F}_q \leq \bar{F} < B_{p^\dagger}$. Activating equilibria exist by (b), so every equilibrium activates, giving (c). □

When $\sum_p B_p > \bar{F}$ there is a continuum of non-empty equilibria. As a toy illustration, suppose the US, Europe, and China each value a given mechanism activation at \$0.60 apiece. At an activating threshold $\bar{F}$ of \$1, one equilibrium has the United States, the European Union, and China each contribute \$0.333; another has the United States and the European Union each contribute \$0.50. The all-zero profile is also an equilibrium unless some donor's benefit alone exceeds \$1. The main text presents the minimum-size non-empty coalition of each provision point game.

Because the game is static, it abstracts from commitment to the threshold and from the timing of contributions and clearing; Hsiao (2026) models commitment in trade agreements in Indonesia.

**Provision point algorithm.** Given a set of candidate payer countries (each with a known willingness-to-pay for the mechanism) and recipients, I outline an algorithm for finding the minimum-size Nash-viable payer coalition if any exists.

1. Initialize a guess for recipients' tax vector.
2. Solve the world equilibrium at the current rate vector and calculate the breakeven transfer which supports the current recipient tax guess.
3. Sweep through the recipients one at a time. For each recipient, holding the others' current rates fixed, calculate the best response tax rate.
4. After sweeping through all recipients, return to step 2 with the updated rates. This recomputes the compensation floor at the new rates and initiates another sweep.
5. Stop when a complete sweep changes no rate by more than a pre-specified tolerance. Calculate payers' willingness to pay at the tax vector relative to non-cooperative Nash.
6. Sequentially add payers by highest willingness to pay for $CO_2$ to the payer coalition until the compensation floor is met.

# B Data appendix

## B.1 Sketch of data construction process

There are multiple types of data which must be synthesized to complete this project. This includes remote-sensing data, the bulk of which is presented in raster format, administrative data on boundaries, roadways, and waterways which is presented in a vector (geometric) format, and economic data which is often tied to an administrative unit.

To harmonize all of these data sources, I define plots using a rectangular grid overlaid on the world map. Table A.7 lists key raster data sources and describes their resolution.

I extract the data in 504, 10 $\times$ 10 arc-degree tiles in a WGS84 projection. Visualization and areas are computed using a CEA (equal-area) projection. Each tile is overlayed with 10,000 0.1 arc-degree sub-tiles. Finally, sub-tiles are intersected with country terrestrial boundaries from the Global Administrative Boundaries database so that every remaining grid fragment is firmly within a country boundary.

The key step in this data construction procedure is a concordance between plot-level data on yields and land use with national accounting data on value add and trade. I summarize this algorithm below.

1. Every pixel gets assigned a cropland share and pasture share from Earthstat data. Forest shares and other shares come from ESACCI data. All land use data is shrunk using an Empirical Bayes procedure: Appendix B.3.

2. I choose a set of crops: oilpalm, rice, maize, soybeans, wheat, white potatoes, sweet potatoes, sugarcane and calculate yields $\eta^A(\omega)$ at the plot level weighted by country-level production. I calculate a production-weighted output price $p_i$ for these crops as well. Robustness checks use alternative weighting schemes.[1]

3. I harmonize total value-added from agricultural activities from the land use data with bilateral trade data: Section B.2

This procedure results in an internally consistent dataset of prices, yields, trade, and land use shares which exactly rationalize aggregate value added in agriculture.

**Land use data.** The European Space Agency Climate Change Initiative, or ESACCI data, covers a 300 meter $\times$ 300 meter grid annually from 1992-2019. ESACCI is, at the time of drafting, the only global publicly available panel of land use at global scale which includes cropland, forest cover, grassland, and other land use categories (see A.6 for details on the classification scheme).

**Emissions.** I use a global cross-sectional map of aboveground and belowground carbon density for the year 2010 from NASA's Oak Ridge National Laboratory Distributed Active Archive Center (abbreviated ORNL-DAAC; Spawn and Gibbs 2020). This dataset is derived from LiDAR-calibrated satellite imagery and reports carbon density, measured in megagrams of carbon per hectare (MgC/ha), at a native spatial resolution of 300 meters by 300 meters.

**Potential yields.** Agricultural productivity is measured using the FAO GAEZ maps, which have a 9-by-9 km resolution. I restrict attention to the subset of crops used to construct agricultural productivity and output measures in the paper: wheat, rice, maize, oil palm, soybeans, white potatoes, sweet potatoes, and sugar cane. Rice comes in two varietals in the GAEZ: dryland and wetland. I take their maximum in a pixel to compute a single rice productivity. Used extensively in prior economic research, this dataset provides information on the biophysical suitability of different

1. Earthstat contains planted area data for 175 crops, and the FAO similarly has 78 crops with at least one country that has price data. However, price data outside of staples (as well as national accounts of planted area) are missing in many more countries. I focus on these crops because they account for much of global consumption and are well-represented in the data.

crops and livestock in different regions of the world, as well as information on climate, soil, and other environmental factors (Adamopoulos et al. 2022; Farrokhi and Pellegrina 2023; Costinot, Donaldson, and Smith 2016) The GAEZ reports potential yields for ∼30 crops at two input (high- and low-input) and water availability (irrigated and rainfed) levels over 30-year epochs. Data for the main results of this paper come from high-input, rainfed yields for the climate epoch 1980-2010.

**Crop prices.** I aggregate the Food and Agriculture Organization series on producer prices for individual crops. Following Costinot, Donaldson, and Smith (2016), missing crop prices are imputed in log space using country and crop fixed effects estimated on years prior to 2005, with residual missing cells filled by the log-price average of adjacent countries. I construct a country-level agricultural output price as

$$p^A_{it} = \sum_k p^k_{it}, w^k_{i,2000}$$

where I test weights $w^k_{i,2000}$ equal to harvested area (hectares) and total production (tonnes) by crop in 2000. The aggregate itself is separately imputed in log space using country and year fixed effects, then neighbor-country averaging, then a final continent-level, area-weighted backstop; only the year-2000 aggregate is retained.

The element-wise maximum of FAO "Sugar cane" and "Sugar beet" values was taken for each country-year-element cell, and the result was labeled "Sugarcane." I thus assign each country the higher of its cane or beet production, area, and yield in each year.

**Pastoral output and prices.** The country-year cattle dataset combines FAO production and price statistics, ITPD-E bilateral trade flows, and gridded pasture-area measures into a single panel indexed by country (ISO3) and year.

Physical output is drawn from the FAOSTAT Production of Crops and Livestock series. I retain only the following cattle-related items: Meat of cattle with the bone (fresh or chilled), Meat of goat, Meat of sheep, and Beef and Buffalo Meat (primary) — all excluding aggregate ("Total") rows. Observations without a valid ISO3 match are dropped, and production (in tonnes) is summed to the country-year level.

Producer prices come from the FAOSTAT Producer Prices series, restricted to the Producer Price (USD/tonne) element. I retain only the cattle items "Meat, cattle" and "Meat live weight, cattle"; where the live-weight price is available it takes precedence as cattle are sold live from the farm-gate, otherwise the carcass price is used. Tonnage-weighted across countries with both prices, live weight prices are 55.21% of carcass prices per ton in most countries, so I deflate carcass prices by this value. This approach may misstate value added of downstream slaughterhouses, whose value add (and potential markup) accrues to downstream producers in the manufacturing sector. Over 1991-2020, this conversion factor is relatively stable, with range [47.4%, 62%]. Prices are merged to ISO3 codes as above and collapsed to one price per country-year. Missing prices are imputed using continent-year means.

Combining these data with land use from Ramankutty et al. (2008), I compute production per hectare of pasture. Missing TFP is imputed by a continent-year mean.

**Transportation costs.** I use freight cost estimates (Herrera Dappe, Lebrand, and Stokenberga 2024) and travel time data (Nelson et al. 2019) to estimate the cost of freight travel per ton of goods. I assume a traveling speed of 60 km/hr:

$$\text{freight cost, \$/ton}, c(\omega)w_i = \text{travel time, hours} \times 60\text{km/hr} \times \text{\$ cost per ton-km}_i.$$

**Trade.** Trade flows come from the International Trade and Production Database for Estimation, or ITPD-E (Borchert et al. 2022), a product of the United States International Trade Commission. Trade flows are reported in value terms for 1991-2021 at an annual frequency, between 265 countries. Pastoral outputs cover USITC codes 17, 20, 36, and 40 (live cattle, other meats, livestock products, live animals, processing/preserving of meat, and dairy products). Crop outputs include USITC codes 1, 2, 3, 6, 7, 9, and 13 (wheat, rice, maize, soy, other oilseeds, sugar crops, and vegetables). All other indsutry codes are mapped to the outside (manufacturing) sector.

**Pasture land and crop-level realized yields.** The Earthstat project downscales administrative data on land footprints and agricultural production/yields using satellite data. The resulting datasets are consistent with administrative data vintages closest to 2000. Crop yields are from Monfreda, Ramankutty, and Foley (2008) and pastoral land is from Ramankutty et al. (2008).[2]

**Ad valorem tariffs.** I rely on the Market Access Map, or MacMAP data, to construct a time-varying measure of tariffs in terms of their ad valorem equivalents. The data is provided at the HS6 level. I aggregate it up to the HS4 code for each of my crops by taking the maximum across HS6-level applied tariffs. Details are provided in Appendix Table A.1.

**Population access.** Define access to population nearby as:

$$\text{Population access}(\omega) = \frac{\sum_{\omega' \neq \omega} \text{population}(\omega')\delta(\omega, \omega')^{-2}}{\sum_{\omega' \neq \omega} \delta(\omega, \omega')^{-2}}$$

2. A feasible model extension considers a crop choice nest as in Costinot, Donaldson, and Smith (2016). This paper abstracts from crop-specific choice to focus on the extensive decision between land uses.

Table A.1: Mapping of major crops to HS4 codes.

| **Crop name** | **HS4 code** |
|---|---|
| Wheat | 1001 |
| Rice | 1006 |
| Maize | 1005 |
| Oilpalm | 1207 |
| Soybeans | 1201 |
| White potatoes | 0701 |
| Sweet potatoes | 0714 |
| Sugar cane | 1212 |

where $\delta(\omega, \omega')$ measures the geodesic distance in kilometers between the two grid cells and 2 is a common distance elasticity (Duranton, Morrow, and Turner 2014). I omit own-plot population from the sum to avoid a mechanical relationship between population access and land shares. For computational reasons, I truncate the sum at 100 km. This variable is used in the first stage, Appendix C.1.

### B.2 Trade-Production Harmonization

Model inversion uses a harmonized dataset of trade flows and crop/cattle production. Harmonization follows biproportional fitting (Deming and Stephan 1940; Bacharach 1970), a method used to construct globally consistent input-output tables (Antrás and Chor 2022). The final dataset satisfies:

$$\sum_j X_{ij}^A = p_i Q_i^A, \qquad \sum_j X_{ij}^P = p_i^P Q_i^P, \qquad \sum_j X_{ij}^M = Y_i - p_i Q_i^A - p_i^P Q_i^P, \tag{A.6}$$

$$\sum_j \sum_s X_{ij}^s = Y_i = \sum_j \sum_s X_{ji}^s \quad \forall\, i, s \in \{A, P, M\}. \tag{A.7}$$

Let $Y_i \equiv \sum_s \sum_j X_{ji}^s$ denote absorption in $i$, the destination-side total of raw ITPD-E flows inclusive of domestic sales.[3] Production values are $p_i \tilde{Q}_i^A = \sum_{\omega \in i} p_i\, \eta^{A,\mathrm{pot}}(\omega)\, \tilde{\mu}^A(\omega)\, \mathrm{area}(\omega)$, using potential yields from the GAEZ (Fischer et al. 2021), FAO production-weighted prices, and the smoothed cropland shares of Appendix B.3; the pastoral analogue $p_i^P \tilde{Q}_i^P$ replaces potential yields with cattle output per pastoral hectare and cropland with pasture share.

The three identities are reached by different routes. Agricultural trade is left at its ITPD-E values and *productivity* is rescaled to meet it, through the region-level utilization shifter $m_i \equiv E_i^A / p_i \tilde{Q}_i^A$

3. This is gross absorption at basic values rather than value added, and exceeds national-accounts GDP for most regions. A single $Y_i$ serves as both row and column target, so income equals expenditure by construction.

where $E_i^A = \sum_j X_{ij}^A$. The model inverts $\eta^A(\omega) = m_i\, \eta^{A,\text{pot}}(\omega)$, so the first identity holds with $Q_i^A$ the utilization-adjusted potential production. Pastoral trade is instead deflated by a uniform per-origin constant $a_i \equiv E_i^P / p_i^P \tilde{Q}_i^P$, which imposes the second identity and leaves each origin's composition of sales unchanged. Manufacturing is the residual sector and the only one for which both marginals are imposed, with targets $r_i^M = Y_i - \sum_j X_{ij}^A - \sum_j X_{ij}^P$ and $c_j^M$ defined analogously by column; global mass balance $\sum_i r_i^M = \sum_j c_j^M$ holds identically, since both subtract the marginals of the same two matrices from the same vector $Y$.

**Mathematical formulation.** The manufacturing matrix is the Kullback–Leibler projection of $X^M$ onto the affine set defined by the prescribed marginals:

$$X^{M*} \;=\; \arg\min_{M \geq 0} \sum_{i,j} M_{ij} \log \frac{M_{ij}}{X_{ij}^M} \quad \text{s.t.} \quad \sum_j M_{ij} = r_i^M, \;\; \sum_i M_{ij} = c_j^M. \tag{A.8}$$

The dual solution takes the biproportional form $X_{ij}^{M*} = u_i v_j\, X_{ij}^M$, with $\{u_i, v_j\}$ obtained by alternating marginal rescaling

$$M_{ij}^{(k+\frac{1}{2})} = \frac{r_i^M}{\sum_{j'} M_{ij'}^{(k)}}\, M_{ij}^{(k)}, \qquad M_{ij}^{(k+1)} = \frac{c_j^M}{\sum_{i'} M_{i'j}^{(k+\frac{1}{2})}}\, M_{ij}^{(k+\frac{1}{2})}, \tag{A.9}$$

iterated until $\max\big\{ \sup_i \big| \sum_j M_{ij} - r_i^M \big|, \sup_j \big| \sum_i M_{ij} - c_j^M \big| \big\} < 10^{-12}$ million USD. The pastoral deflation above is the degenerate case of the same program with only the row constraint imposed, whose closed form is $u_i = r_i^P / \sum_j X_{ij}^P$ with $v_j \equiv 1$. No column constraint is imposed on the agricultural or pastoral matrices: the destination-side composition of those sectors is inherited from the ITPD-E.

**Other data sources** Appendix Table A.7 summarizes key raster and geospatial datasets, provides the spatial resolution of these data, and gives a brief description of each.

## B.3 Empirical Bayes for land use shares

I use an empirical Bayes (EB) procedure to smooth land use shares. The EB approach serves two purposes: first, it limits the influence of outliers arising from noisy satellite measurement, and second, it ensures interior land use shares in $(0, 1)$, a requirement for logit inversion (Berry 1994). The procedure follows prior work by Li (2023) but extends her Beta–binomial formulation to multiple land uses. Intuitively, the procedure treats the observed land use composition of a grid cell as draws from a "Pólya urn" whose composition is governed by the land use mix of the surrounding country-biome. Grid cells that are anomalously far from that mix (e.g., cells with zero shares) or small in area are shrunk toward the country-biome mix. For satellite products that also report a confidence or precision score, the procedure could shrink low-confidence cells more aggressively.

Index grid cells by $\omega$, country-biomes by $i$, and land uses by $h \in \{F, A, P, O\}$. The data report the area of cell $\omega$ under land use $h$, denoted $\hat{\mu}^h(\omega)$ and measured in square kilometers, with total classified area $N(\omega) \equiv \sum_h \hat{\mu}^h(\omega)$. I treat these areas as pseudo-counts, so that each square kilometer constitutes one draw of a land use classification; I return to this convention below. The object of interest is the cell's latent share vector $\boldsymbol{\mu}(\omega) \equiv \big(\mu^F(\omega), \mu^A(\omega), \mu^P(\omega), \mu^O(\omega)\big)$. The hierarchy is

$$\hat{\boldsymbol{\mu}}(\omega) \mid \boldsymbol{\mu}(\omega) \sim \text{Multinomial}\big(N(\omega),\ \boldsymbol{\mu}(\omega)\big),$$
$$\boldsymbol{\mu}(\omega) \sim \text{Dirichlet}\big(\boldsymbol{\alpha}_i\big), \qquad \omega \in i,$$

where $\boldsymbol{\alpha}_i \equiv (\alpha_i^F, \alpha_i^A, \alpha_i^P, \alpha_i^O)$ with $\alpha_i^h > 0$ and $\alpha_{i0} \equiv \sum_h \alpha_i^h$. The prior mean $\alpha_i^h/\alpha_{i0}$ is the country-biome land use mix, and the concentration $\alpha_{i0}$ governs how tightly cells cluster around it. By Dirichlet–multinomial conjugacy, the posterior over the latent shares is

$$\boldsymbol{\mu}(\omega) \mid \hat{\boldsymbol{\mu}}(\omega)\ \sim\ \text{Dirichlet}\big(\alpha_i^F + \hat{\mu}^F(\omega),\ \alpha_i^A + \hat{\mu}^A(\omega),\ \alpha_i^P + \hat{\mu}^P(\omega),\ \alpha_i^O + \hat{\mu}^O(\omega)\big).$$

I estimate the four hyperparameters separately for each country-biome by maximum likelihood on the marginal distribution of the observed areas, integrating out the latent shares. Treating cells within a country-biome as independent draws given $\boldsymbol{\alpha}_i$, the marginal likelihood is Dirichlet–multinomial (Pólya):

$$f\big(\{\hat{\boldsymbol{\mu}}(\omega)\}_{\omega\in i} \mid \boldsymbol{\alpha}_i\big) = \prod_{\omega\in i} \binom{N(\omega)}{\hat{\mu}^F(\omega), \hat{\mu}^A(\omega), \hat{\mu}^P(\omega), \hat{\mu}^O(\omega)} \frac{\Gamma(\alpha_{i0})}{\Gamma(N(\omega)+\alpha_{i0})} \prod_h \frac{\Gamma(\hat{\mu}^h(\omega)+\alpha_i^h)}{\Gamma(\alpha_i^h)},$$

where the multinomial coefficient is extended to real-valued areas as $\Gamma(N(\omega)+1)/\prod_h \Gamma(\hat{\mu}^h(\omega)+1)$. Because this coefficient does not involve $\boldsymbol{\alpha}_i$, it drops out of the estimation problem:

$$\hat{\boldsymbol{\alpha}}_i = \text{argmax}_{\boldsymbol{\alpha}_i>0} \sum_{\omega\in i} \left[ \log\Gamma(\alpha_{i0}) - \log\Gamma\big(N(\omega)+\alpha_{i0}\big) + \sum_h \Big( \log\Gamma\big(\hat{\mu}^h(\omega)+\alpha_i^h\big) - \log\Gamma(\alpha_i^h) \Big) \right].$$

The smoothed land use share is the posterior mean,

$$\tilde{\mu}^h(\omega)\ \equiv\ \mathbb{E}\big[\mu^h(\omega) \mid \hat{\boldsymbol{\mu}}(\omega); \hat{\boldsymbol{\alpha}}_i\big]\ =\ \frac{\hat{\mu}^h(\omega) + \hat{\alpha}_i^h}{N(\omega) + \hat{\alpha}_{i0}}$$

a convex combination of the cell's raw share and the estimated country-biome mix. Two properties follow immediately. First, because $\hat{\alpha}_i^h > 0$, the smoothed share $\tilde{\mu}^h(\omega)$ lies strictly in $(0, 1)$ even when $\hat{\mu}^h(\omega) = 0$, so the logit inversion is well defined. Second, the weight on the data is increasing in cell area, so smaller cells are shrunk more aggressively toward the country-biome mix.

The procedure is degenerate in one tile of the African Sahara in which all land is classified as bare: the likelihood is maximized as $\alpha_i^h \to 0$ for the never-observed land uses, leaving their posterior shares at zero. In this tile I assign a share of 0.001 to each remaining land use and renormalize.

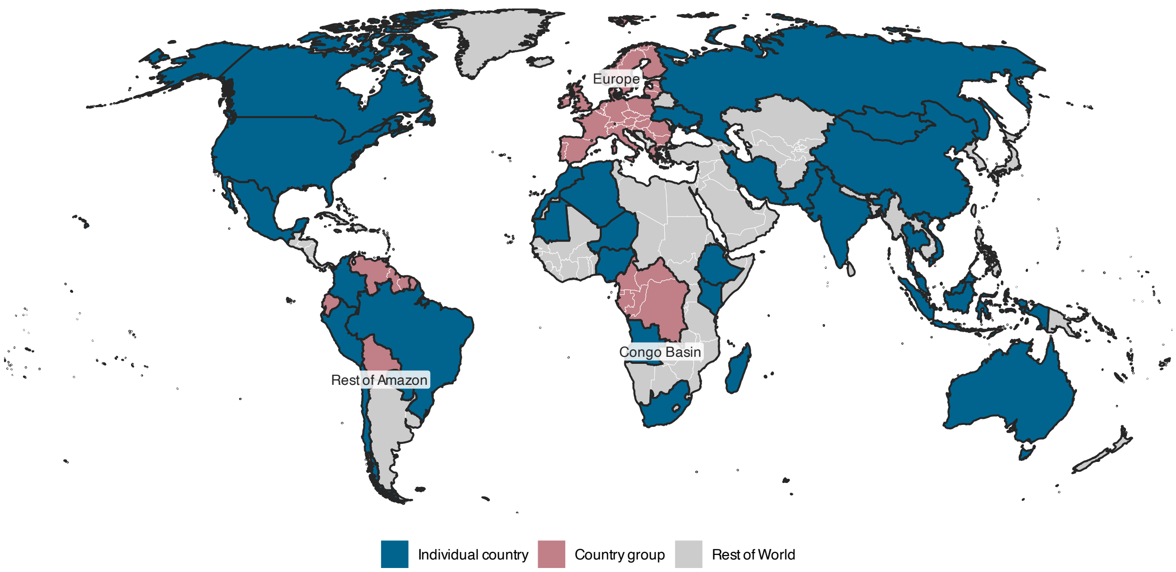
### B.4 Regional aggregation

The overall country grouping is visualized in Figure A.1. I construct an aggregated Europe which includes the formal European Union, the United Kingdom, Switzerland, Norway, and Serbia. I aggregate the Congo River Basin due to inconsistent data availability within the Basin: this is Cameroon, the Central African Republic (CAR), the Democratic Republic of the Congo (DRC), the Congo, Equatorial Guinea, and Gabon. The CAR and DRC have very limited data coverage, so this grouping imputes macroeconomic aggregates for these countries with regional averages. I similarly aggregate a Rest of Amazon region—Bolivia, Venezuela, Ecuador, Guyana, and Suriname—which preserves Amazon-basin deforestation policy variation outside of Brazil, Peru, and Colombia for countries that would otherwise be too small or too data-poor to model individually.

I perform two strict exclusions, meaning that the associated land area is dropped from the final analysis dataset. First, I exclude all GADM ISO3 codes corresponding to airports, military bases, research sites, and other non-sovereign territories. These are: XAD, XCL, XKO, XPI, ZNC, and Z01–Z09. Excluded non-sovereigns have total physical area 306,153 sq. km. Next, I exclude the three countries in the GADM boundaries data that report 0 crop area in the ESACCI data as of 2000: Monaco, St. Pierre & Miquelon, and Tokelau, with combined area of 241 sq. km.

169 countries are then grouped into a "rest of world" aggregate for one of four reasons: small, often island, countries which have bang-bang equilibrium behavior; countries with zero domestic absorption of crops or cattle in the ITPD-E; countries with 0 positive potential yields on at least 90% of land; and countries where, even after country-crop imputation, no crop price can be derived.

Figure A.1: Analysis sample of country groups.




NOTES: Blue are countries which enter the data as themselves; rose are country groups; gray is rest of world. All groups enter model and statistical analysis except hard exclusions described in Section B.4. Labels apply to country groups.

# C Land use estimation appendix

## C.1 Instrument first stages

**Distance instrument.** Agricultural profitability is measured both with potential yields and with actual yields from Monfreda, Ramankutty, and Foley (2008). The first stage of the agricultural moment condition in (17) is:

$$\hat{v}^A(\omega) = \beta_1 \Delta(\omega) + \beta_2 X(\omega) + u(\omega)$$

where $\Delta(\omega)$ is the instrument, distance to city, in kilometers, $X(\omega)$ are controls and $u(\omega)$ is the residual. I control for population access (omitting $\omega$ own-population), terrain/roughness, and ADM-2 (county) fixed effects.

Appendix Table A.2 shows results using potential yields to construct $\hat{v}^h(\omega)$. Starting in column (1), a 1 km increase in distance to a city results in \$0.033 fewer dollars per hectare on average. In column (2), I add the slope instrument for illustration. A 1 percentage point increase in the area share above 15 degrees results in a \$0.34 reduction in potential profitability, equivalent to a 10 km increase in distance to city. The magnitude and sign of both instruments is robust to controlling for potential confounders (3)-(5). By column (5), the identifying variation focuses on two grid cells in the same ADM-2 (a county) with the same local population characteristics, port accessibility, and land use share in bare and urban areas. The land use share and population density of the own tile are bad controls and are not used in analysis, but they serve as clear tests: the first stage does not load on these drivers of *selection* (proxies for $\xi^F(\omega), \xi^P(\omega)$) — instead, the coefficients are markedly stable.

Table A.3 repeats the exercise using actual yields from Monfreda, Ramankutty, and Foley (2008) in place of potential yields. The purpose of this table is validation of the economic channel rather than identification: Table A.2 alone cannot rule out that the first stage partly reflects the suitability model rather than realized economics. Actual yields are measured from production data and confirm that both instruments shift returns as realized by farmers: distance to city and steep slopes carry the same signs, with larger magnitudes. Two caveats prevent a structural reading of these magnitudes. First, actual yields are observed only on cells with positive planted area, and the instruments themselves predict planted area, so the sample is endogenously selected. Second, actual yields embed the composition of land use as well as suitability, so the larger coefficients partly reflect selection of better land into cultivation rather than a steeper profitability gradient. For both reasons, estimation uses the potential-yield measure, which is selection-neutral.

**Slope instrument.** The relative profitability of agriculture and cattle is shifted by steep slopes. Figure A.2a shows a visual reduced form for the slope instrument as a shifter of relative pastoral profitability. A greater fraction of steep slopes predicts a near-linear decline in agricultural shares above a fraction of ∼2%. In contrast, after the same threshold, pastoral land shares are flat. The first stage is:

$$\hat{v}^A(\omega) - \hat{v}^P(\omega) = \beta_1 \rho^{15}(\omega) + \beta_2 X(\omega) + u(\omega)$$

Why a 15 degree cutoff? Figure A.2a shows the visual reduced form fitted with a cubic spline over the average slope in a grid cell with controls for ADM-2 fixed effects, soil characteristics (sand, clay, nitrogen content, and other chemical descriptions listed in Table 1), elevation, long-run soil water balance, and bedrock depth. Pastoral land is essentially flat in the slope, while cropland declines markedly. Thus, even after controlling for agronomic shifters which are likely correlated with steep slopes, a marginal degree of steepness is more costly to food production than pasture.

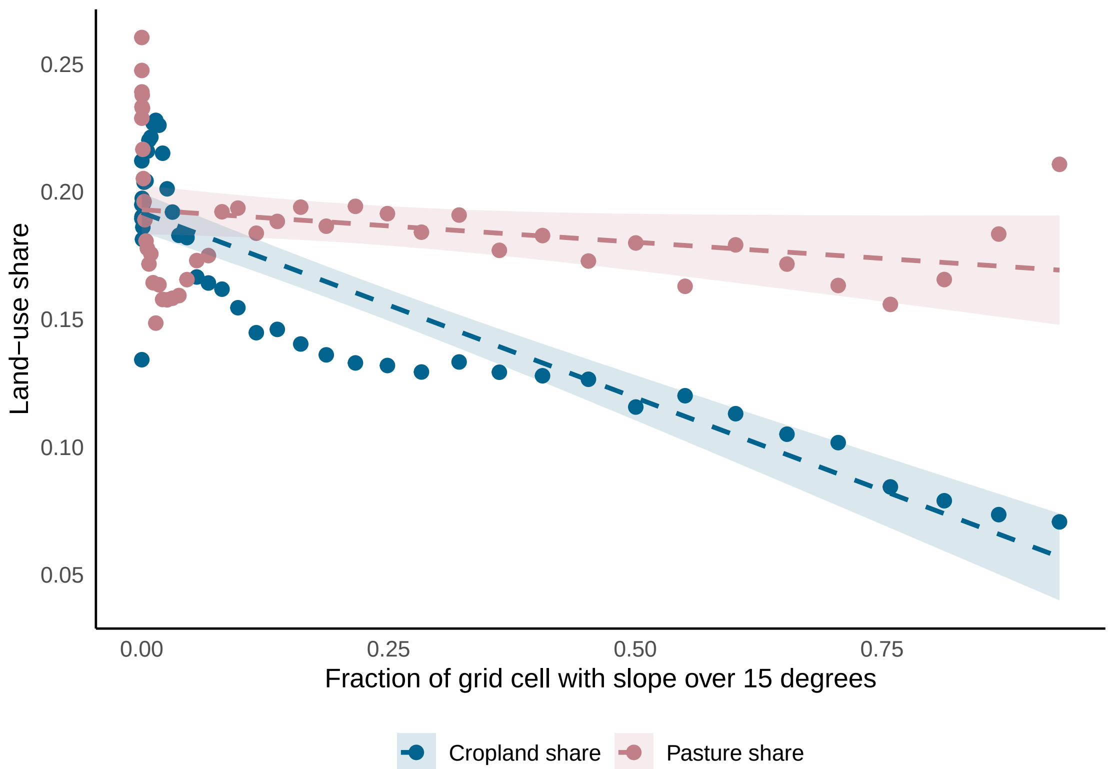
Table A.4 shows this first stage with the potential yield-defined agricultural return $\hat{v}^A(\omega)$.[4] A 1% increase in slope fraction drives a \$0.34 decrease in relative agricultural profits. Put differently, a 1% increase in steep slopes has the same cost to agriculture as being ∼10 km further away from a port per column (2). The magnitude of the coefficient is identical to the agriculture-only first stage. Per the reduced form in Figure A.2a, slopes sharply reduce agricultural profitability but pastoral returns are hardly affected. Slopes thus separate long-run profitability variation from relative amenities $\nu^A(\omega) - \nu^P(\omega)$ in the GMM of (17).

The first stage on relative agricultural profits is in Figure A.2c. Actual yields are everywhere decreasing in slopes. Potential yields, however, only strictly decrease after a slope of 15 degrees. Under the GAEZ Fischer et al. (2021) framework, the gap between actual yields and potential yields within the cross section reflects selection: e.g., yields are lower because there is less cropland as illustrated in A.2b, not because agronomic suitability is necessarily worse.

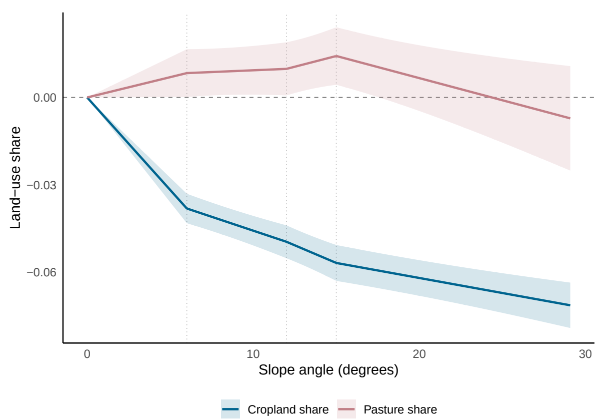
4. I do not include the distance to city instrument in this regression, but the core steep slope coefficient is quantitatively invariant to inclusion, merely more precisely estimated.

Figure A.2: Steep slope visual reduced form and first stage.

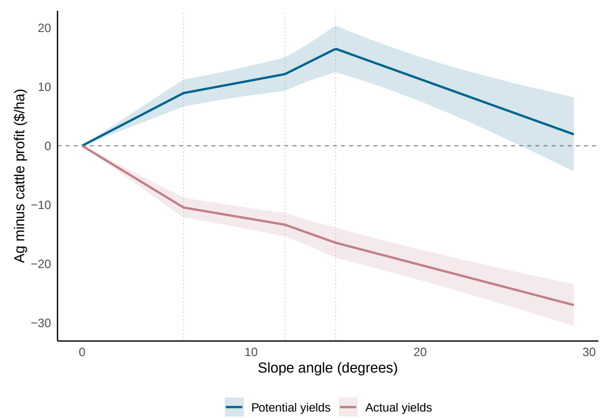
(a) Steep slope share and land use (reduced form).




(b) Average slope and land use (reduced form).




(c) Average slope and relative profitability (first stage).




Notes: A.2a is a binned regression on a stratified random sample of 1000 grid cells by country. Land use shares are shrunk to country-biome means, exactly as used in later analysis. A.2b and A.2c estimate a spline in average cell-level slope with knots at 0 degrees, 6 degrees, 12 degrees, and 15 degrees. Both control for soil characteristics, bedrock depth, long-run soil water depth, and elevation and ADM-2 fixed effects. $y$-axis of first stage in A.2c represents profitability measure: color shows profitability based on GAEZ potential yields (blue, Fischer et al. 2021) and actual yields (rose, Monfreda, Ramankutty, and Foley 2008).

Table A.2: First stage of agricultural returns on Souza-Rodrigues (2019) instruments using rescaled Fischer et al. (2021) yields.

| | Value of agricultural production ($/ha) | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Distance to nearest city (km) | -0.0333*** | -0.0328*** | -0.0294*** | -0.0284*** | -0.0322*** | -0.0214*** |
| | (0.0031) | (0.0032) | (0.0032) | (0.0031) | (0.0034) | (0.0036) |
| Fraction grid with slope $\geq$ 15 degrees | | -33.78*** | -34.11*** | -31.62*** | -32.15*** | -34.83*** |
| | | (3.704) | (3.710) | (3.556) | (3.595) | (3.701) |
| Port distance (km) | | | -0.0323*** | -0.0312*** | -0.0320*** | -0.0307*** |
| | | | (0.0049) | (0.0048) | (0.0050) | (0.0051) |
| Population access index | | | | 1.299*** | 1.262*** | 1.209*** |
| | | | | (0.2873) | (0.2965) | (0.2898) |
| 1975 Population density | | | | 0.0458 | 0.0234 | 0.0211 |
| | | | | (0.0293) | (0.0365) | (0.0355) |
| other | | | | | 28.05 | 20.63 |
| | | | | | (18.60) | (17.71) |
| Soil quality controls | | | | | | ✓ |
| $R^2$ | 0.96486 | 0.96513 | 0.96518 | 0.96542 | 0.96531 | 0.96686 |
| Observations | 1,525,725 | 1,523,304 | 1,523,304 | 1,522,927 | 1,502,947 | 1,491,433 |
| country fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| state fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| county fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

NOTES: Observations are grid cells with a valid dollar value of transportation costs. Standard errors are Conley standard errors with a 100 km spherical cutoff. Country, state, and county fixed effects reflect ADM-0, ADM-1, and ADM-2 fixed effects. The final column adds soil and soil-water-balance controls in levels (bedrock depth, soil water balance, clay, sand, soil organic carbon, nitrogen, pH, and CEC), indicated by the Soil quality controls row.

Table A.3: First stage of agricultural returns on Souza-Rodrigues (2019) instruments using Monfreda, Ramankutty, and Foley (2008) yields.

| | Value of agricultural production ($/ha) | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Distance to nearest city (km) | -0.2622*** | -0.2622*** | -0.2747*** | -0.2685*** | -0.2696*** | -0.2698*** |
| | (0.0538) | (0.0541) | (0.0556) | (0.0556) | (0.0556) | (0.0551) |
| Fraction grid with slope $\geq$ 15 degrees | | -53.59*** | -52.78*** | -51.38*** | -52.74*** | -56.93*** |
| | | (13.06) | (13.04) | (12.94) | (12.97) | (12.86) |
| Port distance (km) | | | 0.0764 | 0.0788 | 0.0779 | 0.0773 |
| | | | (0.0579) | (0.0579) | (0.0580) | (0.0577) |
| Population access index | | | | 0.9310*** | 1.153*** | 1.119*** |
| | | | | (0.3593) | (0.3572) | (0.3496) |
| 1975 Population density | | | | -0.1772** | -0.0078 | -0.0165 |
| | | | | (0.0811) | (0.0529) | (0.0526) |
| other | | | | | -221.0*** | -226.1*** |
| | | | | | (45.33) | (44.50) |
| Soil quality controls | | | | | | ✓ |
| $R^2$ | 0.90080 | 0.90088 | 0.90090 | 0.90092 | 0.90097 | 0.90181 |
| Observations | 735,746 | 733,501 | 733,501 | 733,495 | 733,481 | 732,788 |
| country fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| state fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| county fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

NOTES: Observations are grid cells with a valid dollar value of transportation costs and a positive planted area. Standard errors are Conley standard errors with a 100 km spherical cutoff. Country, state, and county fixed effects reflect ADM-0, ADM-1, and ADM-2 fixed effects. The final column adds soil and soil-water-balance controls in levels (bedrock depth, soil water balance, clay, sand, soil organic carbon, nitrogen, pH, and CEC), indicated by the Soil quality controls row.

Table A.4: First stage of the agricultural-cattle returns differential on Souza-Rodrigues (2019) instruments using Fischer et al. (2021) yields.

| | Agricultural minus cattle value of production ($/ha) | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Fraction grid with slope $\geq$ 15 degrees | -33.90*** | -34.22*** | -31.79*** | -32.13*** | -35.07*** |
| | (3.725) | (3.736) | (3.586) | (3.628) | (3.715) |
| Port distance (km) | | -0.0380*** | -0.0368*** | -0.0381*** | -0.0342*** |
| | | (0.0051) | (0.0050) | (0.0052) | (0.0053) |
| Population access index | | | 1.286*** | 1.248*** | 1.201*** |
| | | | (0.2891) | (0.2988) | (0.2928) |
| 1975 Population density | | | 0.0459 | 0.0214 | 0.0196 |
| | | | (0.0294) | (0.0368) | (0.0359) |
| other | | | | 30.96 | 23.05 |
| | | | | (18.90) | (18.04) |
| Soil quality controls | | | | | ✓ |
| $R^2$ | 0.96914 | 0.96920 | 0.96941 | 0.96926 | 0.97063 |
| Observations | 1,514,441 | 1,514,441 | 1,514,102 | 1,494,492 | 1,484,873 |
| country fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| state fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| county fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |

NOTES: Observations are grid cells with a valid dollar value of transportation costs. Standard errors are Conley standard errors with a 100 km spherical cutoff. Country, state, and county fixed effects reflect ADM-0, ADM-1, and ADM-2 fixed effects. The final column adds soil and soil-water-balance controls in levels (bedrock depth, soil water balance, clay, sand, soil organic carbon, nitrogen, pH, and CEC), indicated by the Soil quality controls row.

### C.2 Robustness: spatial differences

Section C.1 verifies that selection on rich observables does not compromise the validity of instruments. Landowners may still select land based on an unobservable $\xi^h(\omega)$ correlated with the instruments, compromising identification of Equation (17). Additionally, spatially correlated instruments like distance may themselves have overstated $F$-statistics. I introduce an alternative "spatial quasi-differences" strategy which weakens conditions for identification (extending Druckenmiller and Hsiang 2019).

Define $\{x_\omega, y_\omega\}$ as the coordinates for $\omega$ in Cartesian space. Define $d\omega$ as the neighboring plot at $\{x_\omega, y_\omega\} + \vec{e}$ for a step $\vec{e}$ in some direction (e.g., in the $x$-direction, $\vec{e} = \{1, 0\}$). Neighbors share a spatially persistent component of unobservable returns, $\xi^h(\omega)$, which obeys an AR(1) process with coefficient $\rho$:

$$\xi^h(\omega) = \rho \xi^h(d\omega) + \vartheta^h_{\vec{e}}(\omega)$$

where the innovation $\vartheta_{\vec{e}}(\omega)$ in direction $\vec{e}$ is as-good-as-random. Spatial quasi-differencing removes the persistent unobservable by assumption, leaving only the innovation process $\vartheta_{\vec{e}}(\omega)$. Quasi-differencing provides overidentifying restrictions for $(\rho, \gamma)$: the same parameters should fit the quasi-difference of a focal cell and the cell immediately to its west, north, south, and east (potentially even diagonals, but I limit to four directions to prevent having too many collinear moments). Each direction $\vec{e}$ supplies 3 moments yielding a total of 12 moments, of which 10 are overidentifying in the global parameterization. Defining the spatial quasi-difference in direction $\vec{e}$ as $\breve{x}_{\vec{e}}(\omega) = x(\{x_\omega, y_\omega\}) - \rho x(\{x_\omega, y_\omega\} + \vec{e})$, the GMM moments from direction $\vec{e}$ are:

$$g_{\vec{e}}(\gamma, \rho) = \begin{bmatrix} \breve{\nu}^A_{\vec{e}}(\omega)(\delta(\omega) - \bar{\delta}_b) \\ [\breve{\nu}^A_{\vec{e}}(\omega) - \breve{\nu}^P_{\vec{e}}(\omega)](\rho^{15}(\omega) - \bar{\rho}^{15}_b) \\ \vartheta^h_{\vec{e}}(\omega)\xi^h(d\omega) \end{bmatrix}$$

There are three overidentifying restrictions in this model. First, $\gamma$ must jointly rationalize the cattle and agricultural land use decisions. Second, $\rho$ must jointly rationalize all directions $\vec{e}$. Third, the spatial persistence $\rho$ is common across the land use-specific unobservables $\xi^h(\omega)$, $h \in \{A.P\}$.[5]

This spatial quasi-differencing strategy relaxes the requirement that the distance-to-city and terrain-slope instruments satisfy the exclusion restriction conditional on observables alone. Suppose the placement of infrastructure $c(\omega)$ is correlated with unobserved local fundamentals, such as labor

5. This last restriction is particularly difficult to relax without richer within-country variation in cattle profitability separate from agricultural profitability.

market productivity, which survive controls. Identification assumes the unobservable changes more slowly locally than profitability (a second derivative condition per Druckenmiller and Hsiang 2019).

Appendix Table A.5 reports results. First, the spatial autocorrelation parameter is high but statistically below unity: $\hat{\rho}^d \approx 0.97$. This reflects the high spatial persistence of unobserved returns $\xi^F(\omega), \xi^P(\omega)$. Second, the returns coefficient is precisely estimated at $\tilde{\gamma}_0^d \approx 0.008$, fairly similar to the preferred estimate in Table 2. I conclude that the focal estimate is robust to spatially correlated unobservables. The $J$-statistic in Panel A rejects a single $(\gamma, \rho)$ which rationalizes global data, lending support for a model with heterogeneous elasticities.

Table A.5: Land use elasticity robustness check: spatial quasi-differencing

| **Panel A: Global spatial quasi-differenced GMM (pooled across 4 directions)** | | | |
|---|---|---|---|
| $\tilde{\gamma}_0$ | 0.0085 (0.0010) | | |
| $\hat{\rho}$ | 0.9651 (0.0031) | | |
| $N$ pairs | 5,267,938 | | |
| $N$ clusters | 33,341 | | |
| Hansen-$J$ $(\chi^2_{10})$ | 532.31 | | |
| **Panel B: Continental summaries (EB-shrunk posterior)** | | | |
| Region | Ag. Elasticity [95% CI] | Pasture Elasticity [95% CI] | $N$ pairs |
| Global | 1.7761 [1.7086, 1.8462] | 1.0047 [0.9690, 1.0417] | 4,707,429 |
| Africa | 1.4591 [1.2910, 1.6490] | 0.5278 [0.4516, 0.6168] | 840,752 |
| Americas | 2.0529 [1.9543, 2.1565] | 1.3290 [1.2629, 1.3986] | 1,390,264 |
| Asia | 2.3861 [2.2060, 2.5809] | 1.2889 [1.2047, 1.3789] | 1,114,867 |
| Europe | 1.0822 [1.0071, 1.1628] | 1.0313 [0.9635, 1.1039] | 1,068,032 |
| Oceania | 1.2834 [1.1356, 1.4504] | 0.0849 [0.0761, 0.0947] | 293,514 |

NOTES: Standard errors are clustered at ADM-2 level. Panel A reports the global spatial quasi-differenced model coefficients, pooling across moments comparing four directional differences (e.g., east–west differences, north–south differences). These give 4 directions × 3 moments per direction = 12 moments. The Hansen-$J$ row is the canonical Hansen overidentification statistic, testing whether a single $(\gamma, \rho)$ is consistent with all moments. Panel B reports area-weighted country-biome-level EB-shrunk posterior elasticities. Uses crop-level RAS-scaled potential yields weighted by production. $N$ cells: Panel A reports the global pooled fit's own estimation sample, which drops countries with 0 price reporting. Panel B keeps these countries and shrinks them completely towards global prior.

### C.3 Shrinkage of $\gamma_b$

Panel B of Table 2 reports posterior country-biome elasticities. This appendix states the shrinkage procedure. This approach is distinct from classical EB (Walters 2024): it penalizes *overfitting* in small, imprecisely estimated regions by pulling these towards a precisely estimated global prior, a dimensionality reduction approach in the style of Dingel and Tintelnot (2026). I will show this approach is exactly a ridge-penalized GMM. Such approaches trade off some bias in estimates to improve precision.

Let $\hat{\gamma}_b$ denote the unrestricted GMM estimate of (17) in country-biome $b = 1, \ldots, B$, and let $\tilde{\gamma}_0$ denote the pooled estimate of the restricted model reported in Panel A. Both are asymptotically normal:

$$\hat{\gamma}_b \sim N\big(\gamma_b,\ \hat{s}_b^2\big), \qquad \tilde{\gamma}_0 \sim N\big(\gamma_0,\ v_0^2\big),$$

where $\hat{s}_b^2$ and $v_0^2$ are the respective sampling variances, clustered at ADM-2.

I take the sampling distribution of the pooled estimate as the prior for each country-biome parameter, $\gamma_b \sim N(\tilde{\gamma}_0, v_0^2)$. With the normal likelihood, the posterior is normal with precision-weighted mean (Walters 2024):

$$\boxed{\hat{\gamma}_b^{\text{post}} = \omega_b\,\hat{\gamma}_b + (1-\omega_b)\,\tilde{\gamma}_0, \qquad \omega_b = \frac{v_0^2}{v_0^2 + \hat{s}_b^2}, \qquad \operatorname{Var}\big(\gamma_b \mid \hat{\gamma}_b\big) = \frac{v_0^2\,\hat{s}_b^2}{v_0^2 + \hat{s}_b^2}}$$

Cells estimated precisely relative to the prior ($\hat{s}_b^2$ small) retain their own estimate; cells estimated imprecisely are pulled toward $\tilde{\gamma}_0$. Spatial first differences estimates shrink the joint $(\rho_b, \gamma_b)$ towards their global prior using the analogous multivariate normal shrinkage recipe in Walters (2024). As intuition, the estimator can be interpreted as the solution to a ridge-penalized GMM. Let $\hat{Q}_b(\gamma) \equiv n_b\, m_b(\gamma)'\,\hat{\Omega}_b^{-1}\, m_b(\gamma)$ denote the GMM criterion formed from the moments in (17), with $\hat{\Omega}_b$ the ADM-2–clustered variance of the moments and $\hat{G}_b = \partial m_b/\partial \gamma_b$. Because $\nu^h(\omega)$ is linear in $\gamma_b$, the criterion is exactly quadratic, $\hat{Q}_b(\gamma) = \hat{Q}_b(\hat{\gamma}_b) + (\gamma - \hat{\gamma}_b)^2/\hat{s}_b^2$, with $\hat{s}_b^2 = \big(n_b \hat{G}_b' \hat{\Omega}_b^{-1} \hat{G}_b\big)^{-1}$. Hence

$$\hat{\gamma}_b^{\text{post}} = \arg\min_{\gamma}\ \hat{Q}_b(\gamma)\ +\ \frac{(\gamma - \tilde{\gamma}_0)^2}{v_0^2},$$

a ridge-penalized GMM criterion with country-biome-specific penalty $\lambda_b = \hat{s}_b^2/v_0^2$ on deviations from the pooled solution. The posterior variance is the inverse Hessian of the penalized program.

## C.4 Model inversion

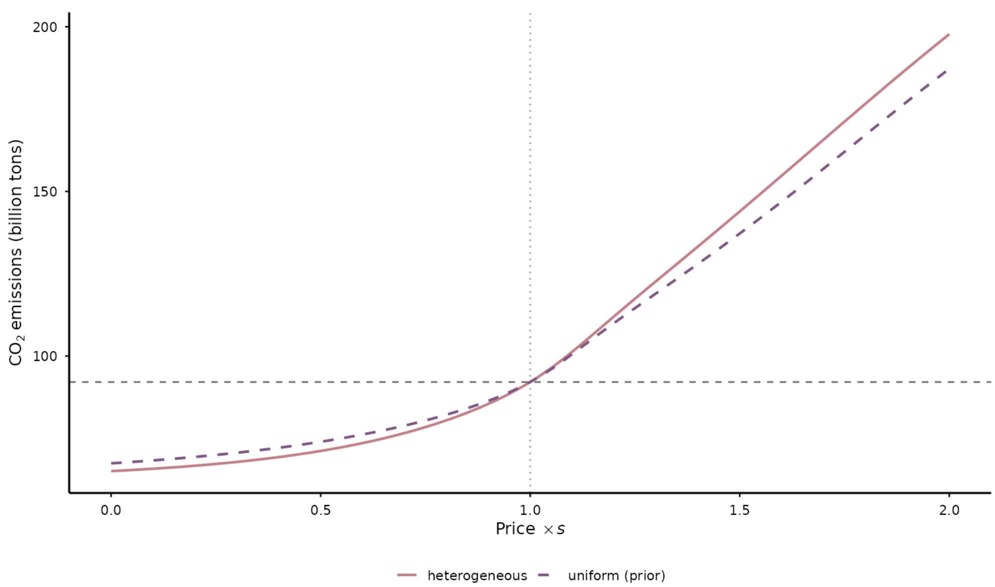
Counterfactual exercises require: value-added data, emissions, yields and pastoral TFP, baseline agricultural and pastoral prices, land use shares, consumer preferences and trade barriers. Value-added in raw trade data will not align exactly with the value added of the agricultural sector implied by land use data, either due to measurement error or to differences in production definition (e.g., wheat at HS2 level trade data is not the same as wheat cultivation in land use data). I resolve this data problem following the same proportional fitting procedure used to reconcile supply use tables (surveyed in Antrás and Chor 2022, see Appendix B.2). To invert, I follow the below steps.

**Step 1. land use data.** Given prices, $\gamma(\omega)$, yields, and transportation costs, modeled land use exactly matches the data through the Berry (1994) inversion.

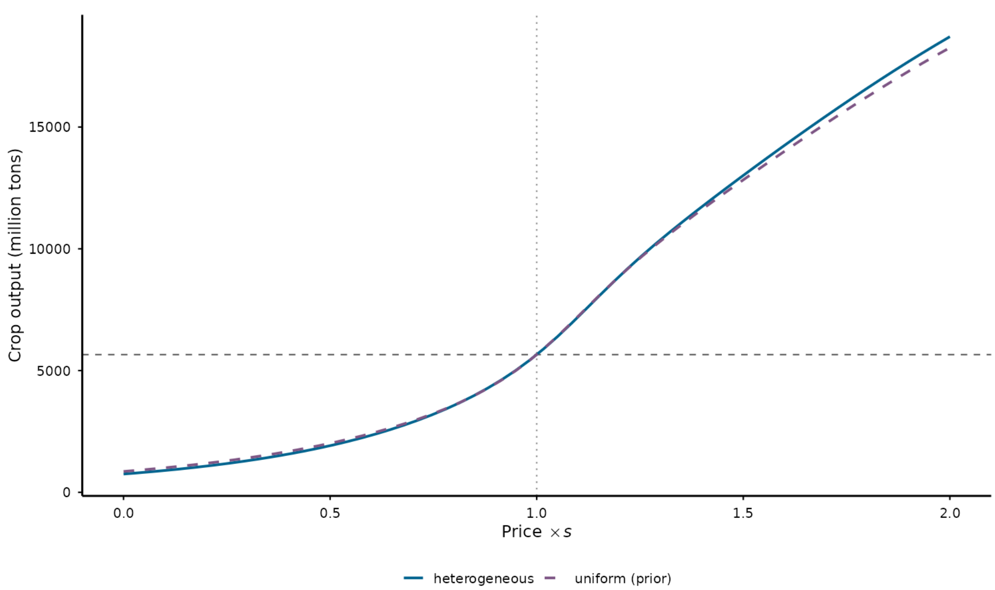
**Step 2. Inverting for wages.** Wages represent the per-hectare value add of the non-agricultural sector:

Figure A.3: Partial equilibrium supply curve of $CO_2$ emissions.

(a) $CO_2$ emissions



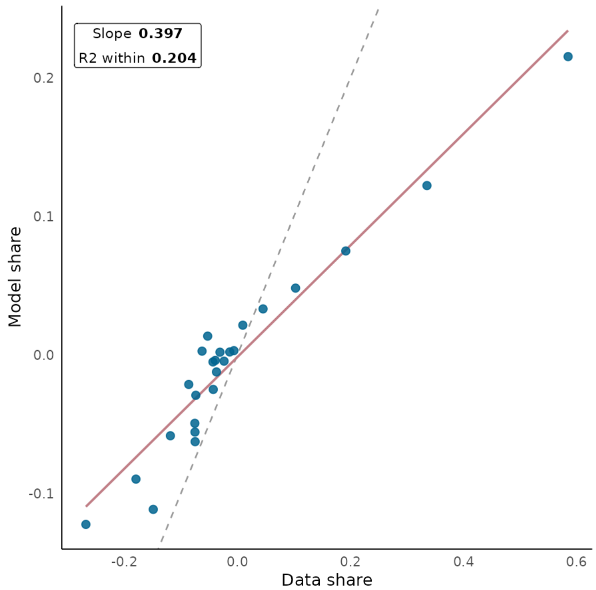

(b) Agricultural output



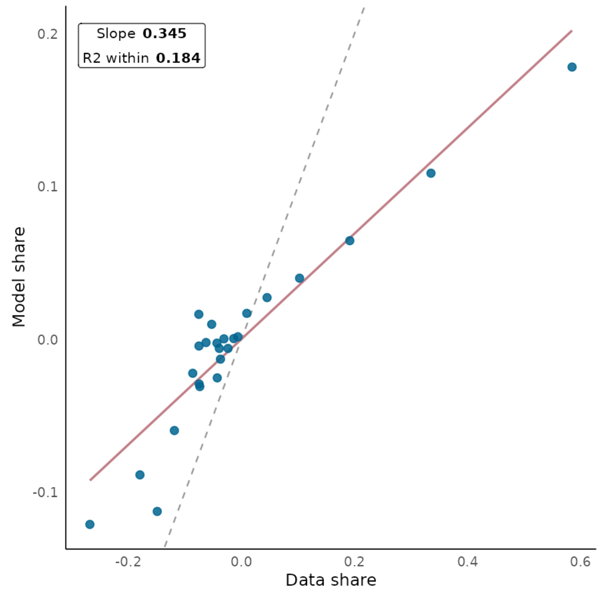

NOTES: Uses estimates from Table 2. Uniform shows results for global prior, heterogeneous shows results for Empirical Bayes-shrunk estimates with country-biome level heterogeneity in land use elasticity. $x$-axis scales country-level prices of crop output by a factor $s$, so that at $s = 1$ the status quo is returned.

Figure A.4: Supply model decomposition of variance

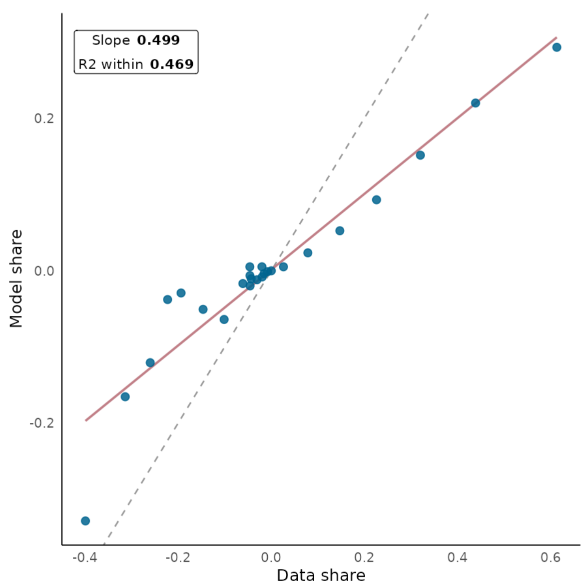
(a) Cropland Shares




(b) Cropland Shares, No $\gamma_b$ Variation




(c) Pasture Shares




NOTES: Uses estimates from Table 2. Cropland A.4a experiment solves for a $\xi^F(\omega) = \bar{\xi}_i^F$ such that, in each country, crop is equal to the country-level share of cropland. I thus test the degree to which the model can recover the correct allocation of cropland decisions without plot-level guidance. Figure A.4a shows binned regression of model- and data-implied cropland shares after a within-country transformation. Figure A.4b repeats the exercise but drops country-biome-specific land use parameters $\gamma_b$ and fixes them at the global prior. Figure A.4c does the same instead solving for reallocation under a fixed pasture residual $\xi^P(\omega) = \bar{\xi}_i^P$. Slope is slope of rose binned regression. Dashed line represents 45 degree line.

$$w_i = \frac{\sum_{j=1}^{J} \lambda_{ij}^M \cdot Y_j}{L_i^M}$$

**Step 3. trade and preferences.** I invert equations (6) and (8) at observed trade shares, prices, and wages to get preference parameters. Country pairs with 0 bilateral trade flows have destination-side preference parameters $\theta_{ij}^h = 0$.

## C.5 Additional documentation

Table A.6: Reclassification schema for European Space Agency Climate Change Initiative land use labels.

| ESACCI | | This Paper | |
|---|---|---|---|
| Code(s) | Description | Description | Code |
| 0 | No data | No data | NA |
| 10 | Cropland, rainfed | Cropland | 2 |
| 11 | Herbaceous cover | Other | 4 |
| 12 | Tree or shrub cover | Other | 4 |
| 20 | Cropland, irrigated or post-flooding | Cropland | 2 |
| 30 | Mosaic cropland (> 50%) / natural vegetation | Cropland | 2 |
| 40 | Mosaic natural vegetation | Forest | 1 |
| 50-90 | Tree cover | Forest | 1 |
| 100 | Mosaic tree and shrub (> 50%) / herbaceous cover (< 50%) | Other | 4 |
| 110 | Mosaic herbaceous cover (> 50%) / tree and shrub (< 50%) | Other | 4 |
| 120 | Shrubland | Other | 4 |
| 121 | Shrubland evergreen | Other | 4 |
| 122 | Shrubland deciduous | Other | 4 |
| 130 | Grassland | Pasture | 3 |
| 140 | Lichens and mosses | Other | 4 |
| 150 | Sparse vegetation | Other | 4 |
| 151 | Sparse tree (< 15%) | Other | 4 |
| 152 | Sparse shrub (< 15%) | Other | 4 |
| 153 | Sparse herbaceous cover (< 15%) | Other | 4 |
| 160 | Tree cover | Forest | 1 |
| 170 | Tree cover | Forest | 1 |
| 180 | Shrub or herbaceous cover | Other | 4 |
| 190 | Urban areas | Other | 4 |
| 200 | Bare areas | Other | 5 |
| 201 | Consolidated bare areas | Other | 5 |
| 202 | Unconsolidated bare areas | Other | 5 |
| 210 | Water bodies | No data | NA |
| 220 | Permanent snow and ice | No data | NA |

Table A.7: Spatial data sources.

| Data source | Resolution | Description |
|---|---|---|
| European Space Agency Climate Change Initiative (ESA Land Cover CCI project team and Defourny 2019) | 300 m | Globally estimated land cover maps provided annually, 1992-2019 |
| Vegetation Continuous Fields (Song et al. 2018) | 250 m | Continuous tree-canopy, short-vegetation, and bare-ground cover; level in 2016 and net change since 1982. Used for model validation. |
| Food and Agriculture Organization Global Agro-ecological Zones (FAO and IIASA 2021) | 9 km | Potential yield data and land resources (soil, climatology). High-yield, rainfed conditions. Units: 10 kg dry weight per hectare. |
| Regrowth rates Cook-Patton et al. (2020) | 1 km | Linear biomass regrowth rates for secondary forest over first 30 years |
| Travel time to cities (Nelson et al. 2019) | 1 km | Global travel-time accessibility indicators to cities of various populations for the year 2015. Use the most expansive city definition. |
| World Port Index (National Geospatial-Intelligence Agency 2019) | (NA) | Shapefile of the location of world ports |
| City locations data (Akbar et al. 2023) | (NA) | Shapefile of major cities |
| Global Administrative Areas (GADM) (GADM 2022) | (NA) | Country and sub-national boundary polygons; every grid cell is assigned to a country boundary. |
| World Wildlife Fund (Olson et al. 2001) | 1 km | Describes bio- and eco-regions of the world (used to determine "tropical" forest regions) |
| Distributed Active Archive Center for Biogeochemical Dynamics, NASA (Spawn and Gibbs 2020) | 300 m | Measures aboveground biomass in Megagrams Carbon per hectare in 2010. |
| International Union for Conservation of Nature and Natural Resources: Red List of Threatened Species (IUCN 2017) | 5 km | Measures the ranged species rarity of all red list species in each cell. |
| Global Human Settlement Layer (Corbane et al. 2018) | 30m | Maps for 1975 and 1990 on presence of built structures. Derived from Landsat image collections. |

# D Descriptive appendix

Table A.8: Top 50 countries by deforestation emissions, 1982-2016.

| Country | Emissions ($tCO_2$/ha) | Area (mill. ha) | Share (%) | Country | Emissions ($tCO_2$/ha) | Area (mill. ha) | Share (%) |
|---|---|---|---|---|---|---|---|
| Brazil | 61 | 39.54 | 34.4 | Peru | 118 | 0.24 | 0.4 |
| Canada | 44 | 12.05 | 7.6 | Nicaragua | 78 | 0.35 | 0.4 |
| Congo - Kinshasa | 84 | 5.07 | 6.0 | Mexico | 32 | 0.84 | 0.4 |
| Russia | 41 | 8.76 | 5.1 | Papua New Guinea | 165 | 0.16 | 0.4 |
| Paraguay | 42 | 7.38 | 4.4 | Chile | 41 | 0.51 | 0.3 |
| United States | 50 | 5.96 | 4.2 | Malaysia | 112 | 0.18 | 0.3 |
| Bolivia | 67 | 4.22 | 4.0 | South Africa | 27 | 0.71 | 0.3 |
| Indonesia | 108 | 2.37 | 3.6 | Mongolia | 26 | 0.75 | 0.3 |
| Argentina | 29 | 8.40 | 3.4 | Uganda | 42 | 0.42 | 0.3 |
| Zambia | 57 | 3.69 | 3.0 | Malawi | 40 | 0.39 | 0.2 |
| Angola | 60 | 3.38 | 2.9 | India | 52 | 0.29 | 0.2 |
| Mozambique | 42 | 4.69 | 2.8 | France | 46 | 0.31 | 0.2 |
| Australia | 48 | 3.73 | 2.5 | Laos | 71 | 0.20 | 0.2 |
| Tanzania | 40 | 3.27 | 1.9 | Honduras | 82 | 0.17 | 0.2 |
| Madagascar | 58 | 2.00 | 1.7 | Cameroon | 157 | 0.08 | 0.2 |
| Myanmar (Burma) | 76 | 0.94 | 1.0 | Kenya | 44 | 0.23 | 0.1 |
| Congo - Brazzaville | 166 | 0.37 | 0.9 | Ghana | 43 | 0.22 | 0.1 |
| Cambodia | 57 | 0.83 | 0.7 | Belize | 76 | 0.10 | 0.1 |
| Gabon | 211 | 0.22 | 0.7 | Thailand | 48 | 0.16 | 0.1 |
| Colombia | 95 | 0.44 | 0.6 | Côte d'Ivoire | 45 | 0.17 | 0.1 |
| Venezuela | 69 | 0.51 | 0.5 | Nigeria | 43 | 0.15 | 0.1 |
| Vietnam | 57 | 0.58 | 0.5 | Zimbabwe | 31 | 0.19 | 0.1 |
| Ethiopia | 53 | 0.60 | 0.4 | North Korea | 35 | 0.15 | 0.1 |
| Guatemala | 50 | 0.62 | 0.4 | Solomon Islands | 130 | 0.04 | 0.1 |
| China | 20 | 1.53 | 0.4 | Guyana | 114 | 0.04 | 0.1 |

NOTES: Top 50 countries ranked by share of global deforestation emissions, 1982-2016. Emissions calculated as carbon converted to $CO_2$ as in Equation (1). Area shows total deforested area in millions of hectares. Share represents percentage of global deforestation emissions. Only includes areas with forest loss.

Table A.9: Profitability and carbon on deforested land, actual yields.

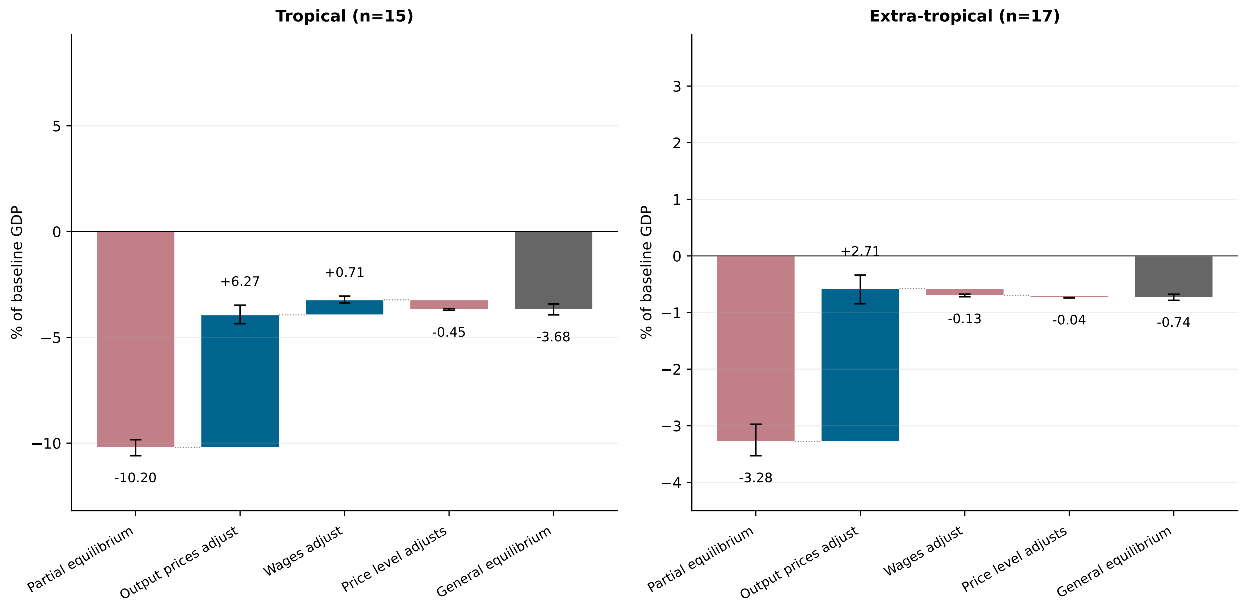
| | (1) Global | (2) Deforested | (3) Focal |
|---|---|---|---|
| Potential $CO_2$, t/ha | 39.19 (0.051) | 55.04 (0.183) | 64.43 (0.353) |
| Travel time to city, hours | 19.49 (0.026) | 7.63 (0.072) | 5.31 (0.111) |
| Profitability, USD/ha/yr | 397.53 (0.924) | 252.17 (1.439) | 243.19 (1.981) |
| Profitability per $tCO_2$, USD/$tCO_2$/yr | 44.05 (0.164) | 8.26 (0.093) | 5.97 (0.062) |
| Forest share, % | 30.29 (0.031) | 48.40 (0.159) | 44.76 (0.263) |
| Cropland share, % | 11.50 (0.018) | 6.36 (0.058) | 5.24 (0.077) |
| Pasture share, % | 20.62 (0.025) | 25.75 (0.144) | 26.88 (0.224) |
| Parcels | 1,588,481 | 41,077 | 13,734 |
| Share of global land, % | 100.00 | 3.10 | 1.20 |
| Share of global emissions, % | 100.00 | 62.15 | 31.72 |
| Crop profitability per $tCO_2$ | 51.12 (84.45) | 16.59 (33.18) | 9.39 (10.34) |
| Pastoral profitability per $tCO_2$ | 10.05 (26.27) | 1.32 (2.30) | 1.80 (1.50) |
| Share below \$3.80/$tCO_2$/yr, % | 41.92 | 80.53 | 83.50 |

NOTES: Each cell reports the area-weighted mean and, in parentheses, its standard error. Columns (2)–(3) limit attention to land whose vegetation cover falls by more than 10% between 1982 and 2016 in Song et al. (2018); column (3) further limits to the focal regions of Brazil, Indonesia, and the Congo River Basin (Cameroon, the Central African Republic, the Republic of the Congo, the Democratic Republic of the Congo, Equatorial Guinea, and Gabon). Profitability is price times yield less freight cost times yield, area-weighted across crops, in USD per hectare per year. Yields are observed harvested yields (Monfreda, Ramankutty, and Foley 2008) averaged across crops using parcel-level planted-area shares. Profitability is therefore per planted hectare and is defined only where some crop is grown. Freight cost per ton is defined in Appendix B. Profitability per $tCO_2$ is annual profitability per ton of potential emissions, computed only on parcels emitting at least one ton of $CO_2$ per hectare so the ratio does not diverge on near-zero-carbon land. Land-use shares are country-biome shares of parcel area and need not sum to one. The final panel reports crop profitability weighted by cropland area and pastoral profitability weighted by pasture area, with standard deviations rather than standard errors in parentheses; the share below \$3.80/$tCO_2$/yr pools the two margins in proportion to their land-use weight, and so is a share of cropland and pasture rather than of all land.

# E Results appendix

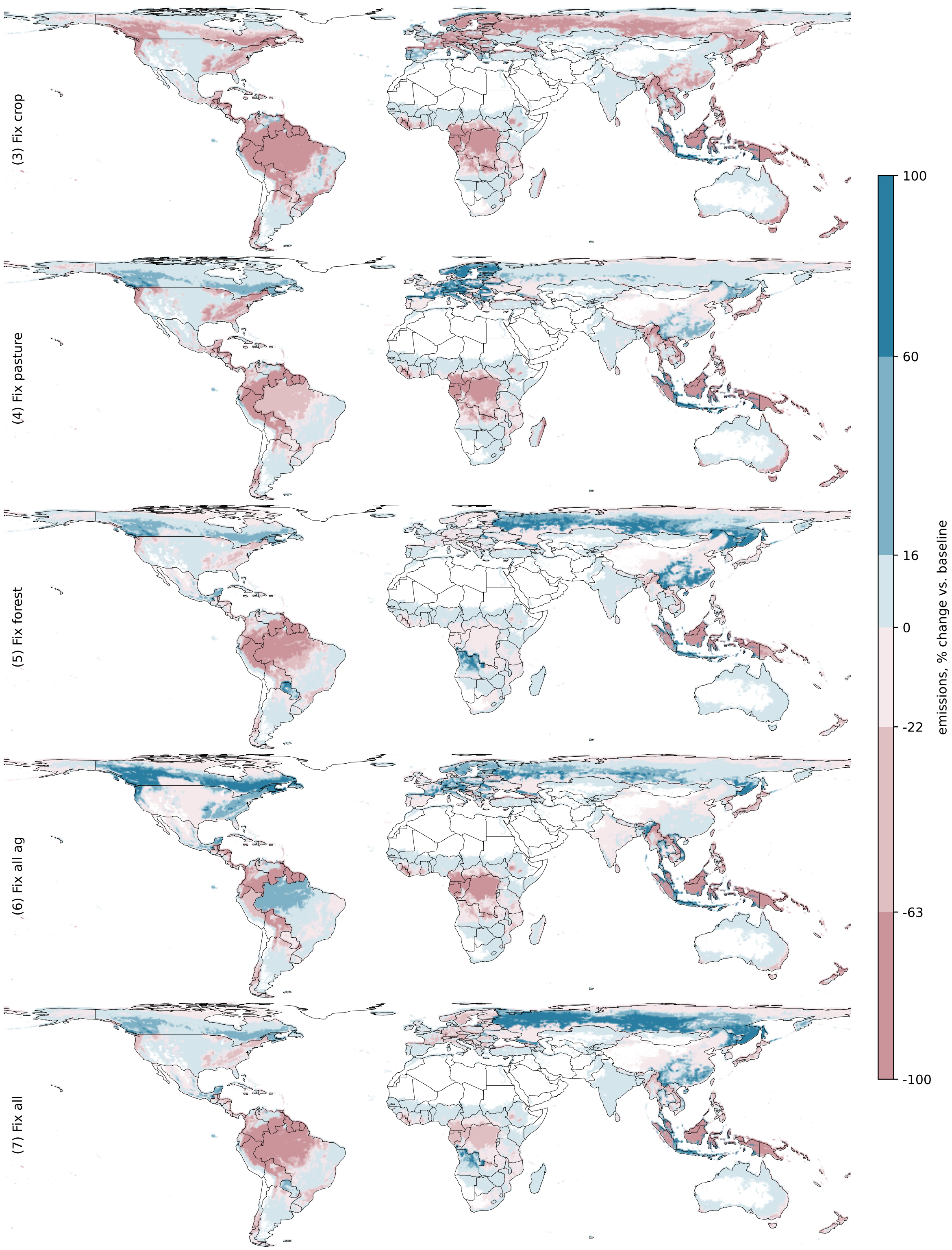
## E.1 Detailed counterfactual results

Figure A.5: Equilibrium price contributions to the change in landowners' surplus, planners' $CO_2$ tax with no transfers.




Notes: Plots components of landowner surplus change as a percent of baseline regional GDP: e.g., tropical, partial equilibrium bar indicates partial equilibrium landowners surplus change as a percent of tropical baseline GDP. Error bars indicate bootstrapped 95% confidence intervals. Tropics include all countries whose area-weighted plot centroid latitude is $\leq 23.5$ in absolute value. Rest-of-World and India (a marginal case) are moved to extra-tropical. Bars do not recompute equilibria with each price fixed; they merely recalculate landowners' surplus with each price held fixed.

Figure A.6: Emissions under constrained planners' problems.




Notes: Blue regions indicate regions which deforest more, increasing emissions relative to status quo. Breaks use a Jenks format with a topcode at 100% increase in emissions. Each subgraph represents a planner setting country-level carbon taxes or subsidies to maximize welfare with a particular constraint: (3) fixes global crop production to status quo; (4) fixes global cattle production to status quo; (5) fixes total forest area to status quo; (6) implements constraints (3) and (4); (7) implements (3), (4), and (5).

Table A.10: Planners' optimal choice of carbon tax, \$/ton across constrained scenarios from Section 6.3.

| | (1) Country quota | (2) Country×biome | (3) Fix crop | (4) Fix pasture | (5) Fix forest area | (6) Fix all ag | (7) Fix all |
|---|---|---|---|---|---|---|---|
| Angola | 8.34 | 7.29 | 3.25 | 2.47 | -3.06 | 1.38 | -1.67 |
| | [6.68, 11.00] | [5.99, 9.11] | [3.07, 3.94] | [0.89, 2.49] | [-3.36, -1.56] | [0.72, 2.28] | [-2.68, -1.21] |
| Australia | 4.21 | 687.04 | 2.22 | 1.74 | -0.07 | 0.28 | 0.19 |
| | [3.40, 5.20] | [515.69, 902.12] | [1.70, 3.09] | [0.58, 1.77] | [-0.16, 0.42] | [0.06, 0.53] | [-0.28, 0.40] |
| Brazil | 2.27 | 2.72 | 3.90 | 0.24 | 0.59 | -0.10 | 0.79 |
| | [1.91, 2.73] | [2.42, 3.11] | [2.80, 4.27] | [0.05, 0.43] | [0.50, 1.41] | [-0.31, 0.17] | [0.44, 1.15] |
| Canada | 3.07 | 7.23 | 2.52 | -0.53 | -0.52 | -2.14 | -0.46 |
| | [2.65, 3.49] | [6.14, 8.61] | [1.69, 3.40] | [-0.81, -0.27] | [-0.71, 0.15] | [-2.47, -1.49] | [-1.22, -0.03] |
| Chile | 2.90 | 5.77 | 3.16 | 1.18 | 0.39 | 0.76 | 0.77 |
| | [2.30, 3.61] | [5.10, 6.38] | [2.99, 3.93] | [0.30, 1.24] | [0.22, 0.81] | [0.38, 1.17] | [0.38, 1.04] |
| China | 10.17 | 43.87 | 4.49 | -1.23 | -4.08 | -0.27 | -2.14 |
| | [8.64, 11.83] | [34.02, 56.33] | [3.95, 5.84] | [-1.83, -0.71] | [-4.47, -2.10] | [-0.90, 0.20] | [-3.61, -1.33] |
| Colombia | 2.55 | 4.05 | 5.21 | 1.42 | 0.85 | 2.65 | 1.69 |
| | [1.95, 3.11] | [3.37, 4.74] | [4.23, 6.34] | [0.99, 1.62] | [0.80, 1.53] | [1.97, 3.20] | [1.27, 1.98] |
| Congo River Basin | 3.48 | 2.95 | 3.49 | 2.21 | 0.03 | 1.57 | 0.50 |
| | [2.84, 4.19] | [2.33, 3.61] | [3.29, 3.86] | [1.34, 2.27] | [-0.09, 0.58] | [1.12, 1.93] | [0.05, 0.75] |
| Algeria | 39.05 | 8.71 | 0.01 | -0.01 | -0.07 | -0.01 | -0.04 |
| | [32.15, 45.19] | [7.82, 9.44] | [0.00, 0.03] | [-0.01, -0.00] | [-0.07, -0.02] | [-0.01, -0.00] | [-0.08, -0.02] |
| Europe | 6.97 | 5.93 | 5.99 | -3.19 | 0.35 | -1.15 | 1.14 |
| | [5.85, 8.45] | [5.19, 6.93] | [4.49, 6.87] | [-3.88, -2.70] | [0.19, 1.06] | [-1.77, -0.69] | [0.65, 1.47] |
| Indonesia | 1.18 | 1.05 | 3.69 | 3.61 | 1.66 | 2.05 | 1.67 |
| | [1.01, 1.37] | [0.90, 1.22] | [2.50, 4.00] | [3.26, 4.15] | [1.64, 2.30] | [1.81, 2.53] | [1.40, 2.44] |

*(continued on next page)*

Table A.10 *(continued from previous page)*

| | (1) Country quota | (2) Country×biome | (3) Fix crop | (4) Fix pasture | (5) Fix forest area | (6) Fix all ag | (7) Fix all |
|---|---|---|---|---|---|---|---|
| India | 3.84 | 136.20 | 0.74 | 1.60 | 0.78 | -1.29 | -0.03 |
| | [2.94, 4.78] | [103.04, 174.36] | [-0.86, 2.35] | [0.49, 1.59] | [0.38, 0.91] | [-1.87, -0.76] | [-0.60, 0.42] |
| Iran | 6.23 | 45638.93 | 0.21 | 0.08 | -0.23 | 0.04 | -0.09 |
| | [4.58, 8.41] | [35267.40, 61684.63] | [0.17, 0.52] | [0.01, 0.09] | [-0.25, -0.05] | [0.02, 0.09] | [-0.20, -0.04] |
| Kenya | 7.11 | 44.02 | 0.20 | 0.06 | -0.07 | -0.00 | -0.03 |
| | [5.77, 8.81] | [32.70, 58.48] | [0.15, 0.58] | [0.00, 0.07] | [-0.07, -0.01] | [-0.01, 0.01] | [-0.08, -0.01] |
| Morocco | 469.07 | 330.60 | 0.00 | -0.00 | -0.01 | 0.00 | -0.00 |
| | [382.23, 574.44] | [246.69, 455.01] | [0.00, 0.00] | [-0.00, -0.00] | [-0.01, -0.00] | [-0.00, 0.00] | [-0.01, -0.00] |
| Madagascar | 2.87 | 4.71 | 1.38 | 1.78 | -0.04 | 0.20 | -0.01 |
| | [2.25, 3.69] | [3.90, 5.47] | [0.89, 2.46] | [0.52, 1.83] | [-0.16, 0.23] | [0.04, 0.52] | [-0.38, 0.16] |
| Mexico | 8.19 | 18.72 | 1.58 | 2.03 | -0.80 | -0.93 | -0.83 |
| | [6.82, 9.44] | [16.05, 21.11] | [0.34, 3.02] | [0.56, 2.15] | [-0.99, -0.15] | [-1.29, -0.59] | [-1.78, -0.40] |
| Mongolia | 10.96 | 7.74 | 0.35 | 0.27 | -0.56 | 0.08 | -0.24 |
| | [8.48, 14.64] | [5.99, 10.46] | [0.29, 0.95] | [0.04, 0.30] | [-0.58, -0.13] | [0.04, 0.21] | [-0.56, -0.11] |
| Malaysia | 1.59 | 1.21 | 4.38 | 3.24 | 1.95 | 3.41 | 2.56 |
| | [1.27, 1.89] | [0.97, 1.44] | [3.93, 5.92] | [2.09, 3.31] | [1.88, 2.27] | [2.52, 4.34] | [2.16, 2.78] |
| Niger | 8028.36 | 3088.21 | -0.00 | -0.00 | -0.00 | -0.00 | -0.00 |
| | [6566.54, 9859.94] | [2453.24, 3884.00] | [-0.00, -0.00] | [-0.00, -0.00] | [-0.00, -0.00] | [-0.00, -0.00] | [-0.00, -0.00] |
| Nigeria | 25.37 | 176.17 | 0.06 | 0.26 | -0.18 | -0.14 | -0.17 |
| | [20.50, 30.97] | [148.03, 204.57] | [-0.30, 0.55] | [0.04, 0.29] | [-0.21, -0.04] | [-0.26, -0.06] | [-0.36, -0.08] |
| Pakistan | 86.02 | 20269.57 | -0.01 | -0.04 | -0.03 | -0.02 | -0.02 |
| | [72.21, 99.73] | [14621.63, 26030.38] | [-0.03, -0.00] | [-0.04, -0.01] | [-0.03, -0.01] | [-0.04, -0.01] | [-0.04, -0.01] |
| Peru | 1.26 | 142.10 | 2.37 | 2.89 | 1.51 | 0.60 | 1.27 |
| | [1.02, 1.59] | [112.06, 180.93] | [1.61, 3.30] | [1.50, 2.93] | [1.36, 1.80] | [0.28, 0.92] | [0.81, 1.53] |

*(continued on next page)*

Table A.10 *(continued from previous page)*

| | (1)<br>Country quota | (2)<br>Country×biome | (3)<br>Fix crop | (4)<br>Fix pasture | (5)<br>Fix forest area | (6)<br>Fix all ag | (7)<br>Fix all |
|---|---|---|---|---|---|---|---|
| Paraguay | 9.41 | 6.78 | 3.44 | 0.52 | -2.21 | 0.60 | -0.82 |
| | [7.95, 11.45] | [5.68, 8.36] | [3.23, 4.63] | [-0.03, 0.63] | [-2.36, -0.75] | [0.26, 1.12] | [-1.81, -0.41] |
| Rest of Amazon | 1.99 | 2.15 | 3.99 | 2.39 | 0.96 | 2.33 | 1.54 |
| | [1.66, 2.34] | [1.92, 2.57] | [3.72, 4.09] | [2.13, 2.54] | [0.94, 1.59] | [2.15, 2.47] | [1.11, 1.80] |
| Russia | 2.64 | 4.23 | 2.76 | -0.09 | -1.04 | -0.39 | -2.01 |
| | [2.21, 3.18] | [3.63, 5.04] | [1.84, 3.53] | [-0.60, 0.16] | [-2.78, -0.91] | [-0.70, -0.32] | [-2.81, -0.95] |
| Rest-of-World | 5.85 | 11.20 | 4.02 | 2.19 | 0.47 | 2.03 | 0.88 |
| | [5.01, 6.49] | [10.13, 12.34] | [3.66, 4.27] | [1.88, 2.34] | [0.21, 0.96] | [1.82, 2.27] | [0.32, 1.05] |
| Thailand | 3.99 | 4.17 | -0.52 | 1.34 | -0.05 | -1.77 | -0.84 |
| | [2.96, 5.17] | [3.34, 4.87] | [-3.73, 1.48] | [0.30, 1.45] | [-0.16, 0.13] | [-3.24, -0.81] | [-1.73, -0.32] |
| Ukraine | 8.33 | 165.85 | 1.23 | -3.06 | -0.30 | -1.03 | -0.17 |
| | [6.60, 11.20] | [122.01, 224.57] | [1.10, 2.01] | [-3.30, -0.69] | [-0.38, -0.07] | [-2.37, -0.49] | [-0.45, -0.02] |
| United States | 3.18 | 17.15 | 1.77 | 1.80 | 0.41 | -0.49 | 0.47 |
| | [2.83, 3.50] | [13.95, 19.85] | [0.88, 3.18] | [1.31, 2.15] | [0.37, 1.25] | [-0.68, -0.18] | [-0.34, 0.75] |
| Vietnam | 4.06 | 6.38 | 0.28 | 1.05 | 0.37 | -1.10 | -0.27 |
| | [3.35, 4.76] | [5.48, 7.64] | [-1.24, 1.85] | [0.22, 1.11] | [0.12, 0.46] | [-1.91, -0.56] | [-0.79, 0.06] |
| South Africa | 28.75 | 535.16 | 0.23 | -0.03 | -0.71 | -0.03 | -0.33 |
| | [23.16, 33.96] | [408.38, 655.32] | [0.19, 0.59] | [-0.04, -0.00] | [-0.73, -0.16] | [-0.09, 0.00] | [-0.72, -0.15] |

Notes: Listed values are point estimates and bracketed values are bootstrap 95% confidence intervals. Values indicate a $CO_2$ tax if $> 0$ and subsidy if $< 0$.Each scenario (1)-(7) runs a constrained planner optimization problem where the planner maximizes welfare with country-specific carbon prices. (1) requires all countries to maintain status quo shares of global emissions; (2) all country-biomes to maintain their share of global emissions; (3) fixes global crop production to status quo; (4) fixes global cattle production to status quo; (5) fixes total forest area to status quo; (6) implements constraints (3) and (4); (7) implements (3), (4), and (5). For (2), taxes are at country-biome resolution and reported values are area-weighted averages.

## E.2 How much does geography drive the price impact of a deforestation carbon tax?

In this section, I vary the data-implied dependence (correlation) between emissions and yields in the data while holding fixed their marginal distributions. Holding fixed marginals ensures that the aggregate potential for production and emissions stays identical across scenarios. Higher correlation between emissions and yields imply that yield-productive land is also high cost, leading to higher prices under the carbon tax.

Define a grid of target within-country correlation between yields and emissions. In each country, yield is re-drawn within country via a Gaussian copula. The *marginal* distribution of productivity is fixed, but the *copula* — the rank-rank relationship between yield and carbon — is not. There are two polar cases: perfect negative dependence — the highest emissions land has the lowest yields — and positive dependence — the highest emissions land has the highest yields. The data has median within-country Spearman correlation of 0.43 — yields and emissions are positively correlated. Figure A.7 shows results for prices and quantities both under the raw copula, with no carbon tax, and with the optimal carbon tax vector in place.

Figure A.7a shows price effects under the two polar cases. As expected, perfect positive dependence (hatched blue) results in larger price effects, while negative dependence (blue) results in lower prices than status quo, bounding the actual estimate (rose). Because all countries redraw simultaneously, in equilibrium these polar cases may not strictly bound the actual price effect — for example, in Europe — because the spillover channel from other countries may dominate.

Globally, 73% of the tax's price effect can be attributed to within-country correlation between yields and emissions, relative to perfect negative dependence. In Indonesia, the joint endowment of yields and emissions explains almost all of the change in equilibrium prices relative to demand-side factors like trade costs and consumer substitution. This "fundamental geographic" component of welfare is consistently large among all three major tropical basins.

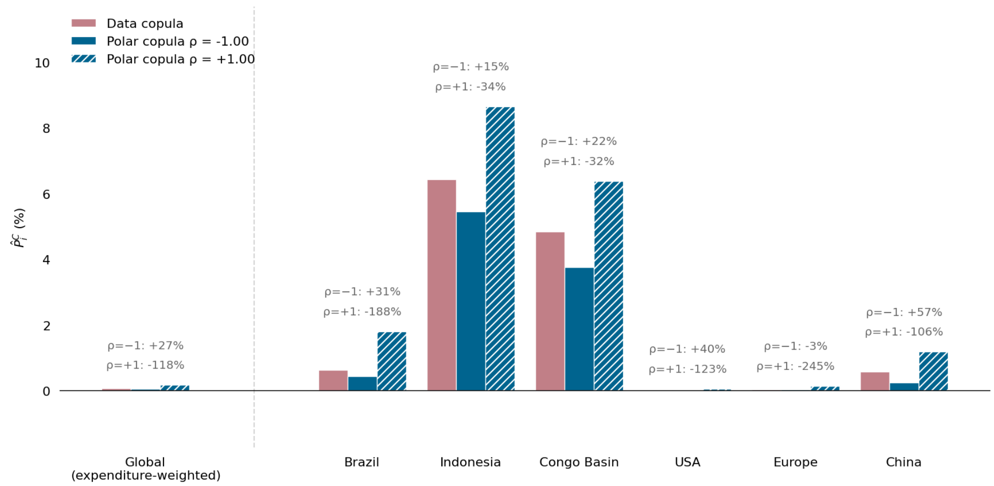
Figure A.7b shows effects on output quantities. Increasing dependence between emissions and yields decreases agricultural quantities relative to the measured first-best, while pastoral production increases to compensate. At perfect positive dependence, food production falls by a further ∼5% and pastoral production rises by 10%.

## E.3 How do common degrees of freedom in $CO_2$ accounting affect results?

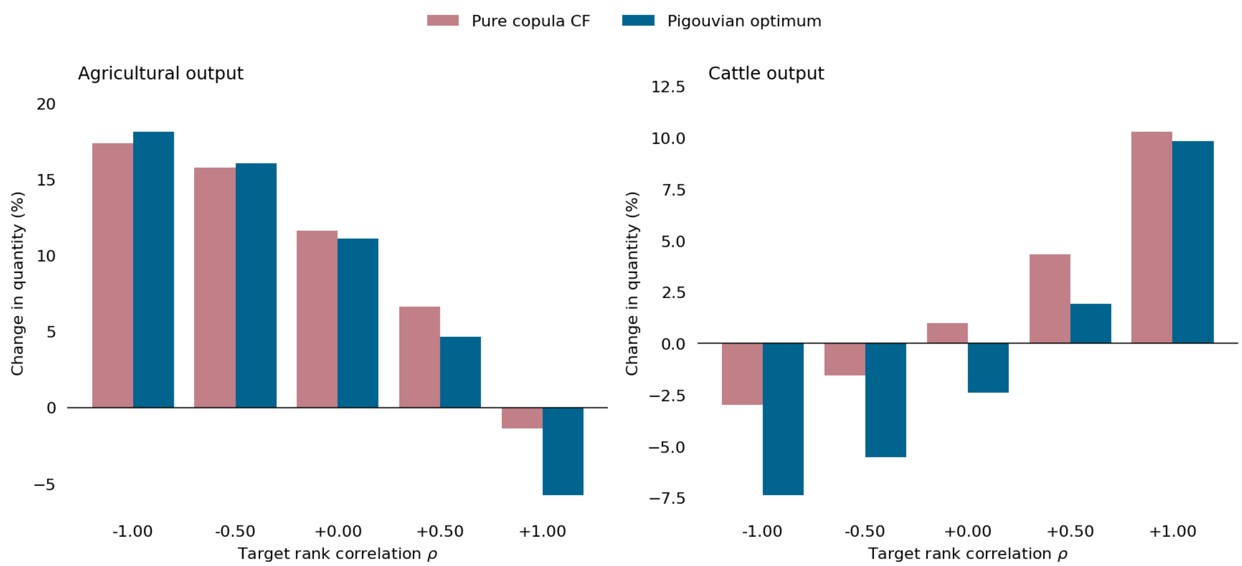
This robustness check varies carbon accounting assumptions. I consider two degrees of freedom: lowering carbon tax burdens to account for the fastest reasonable regrowth rate $r(\omega)$ for a plot of land and raising carbon tax burdens to account for belowground carbon $b(\omega)$ — soil organic carbon, roots and shoots. For regrowth $r(\omega)$ measured in tC/ha/year, I make the bounding assumption

Figure A.7: Sensitivity analysis on the emissions-yield copula.

(a) Price effects under the two "polar" copula cases for selected countries.




(b) Global quantity effects




NOTES: Graph reflects simulations where, for every country, I reorder within-country yields across plots. $x$-axis represents the target correlation $\rho$. Redraws follow an emissions-yield Gaussian copula. In A.7a, Data copula shows the baseline specification while polar copula entries show prices at perfect positive and negative dependence. In A.7b, copula CF compares the no-tax reordered equilibrium to the in-the-data first-best, and Pigouvian optimum compares the country-level optimal tax to the first-best.

that every deforested plot of land is used for one year and then left to regrow at the rate $r(\omega)$. I use data from Cook-Patton et al. (2020) to obtain plot-level regrowth rate estimates based on in situ surveys, and fill missing values with country-biome averages. This data treats regrowth as linear for the first 30 years (other data uses a Chapman-Richards logistic functional form): I track plot-level regrowth until the full $d(\omega)$ regrows, at which point I assume no additional carbon is sequestered. Net sequestration is:

$$d^{net}(\omega) = d(\omega) - \sum_{t=1}^{T} (1.02)^t \frac{44}{12} r(\omega) \text{ where } T : Tr(\omega) = d(\omega)$$

Next, I assume belowground carbon is released when aboveground carbon is burned. This is a bound: IPCC accounting assumes a fraction (often ∼25%) of aboveground carbon is emitted from belowground stocks at the deforestation event (IPCC 2006).

Together, these exercises provide a pessimistic upper bound on deforestation costs (complete belowground loss with no regrowth) and an optimistic lower bound on deforestation costs (complete regrowth with no loss of soil carbon). Each exercise has two effects: it changes the *level* of the welfare cost and the *allocation* of the welfare cost. Belowground carbon tends to be larger in regions with peat deposits (e.g., Indonesia per Hsiao 2026) and permafrost like the far North. Regrowth rates correlate strongly with aboveground carbon but also are high when regions plant high-regrowth carbon sequestering secondary species like eucalyptus or pine in the US. I do not model the choice of regrowing secondary vegetation, taking as given the data which measures the predominant observed revegetation in a region.

Table A.11 shows results. I replicate the first bootstrap draw at the baseline carbon accounting framework in the left column. Adding belowground carbon slightly increases the level of emissions reductions undertaken at additional cost to cattle and crop production. Belowground carbon results in very little change to the spatial allocation of emissions reductions: the Pearson correlation with aboveground stocks is 0.97.

In the right column, I report results with net-of-regrowth accounting. After this adjustment, only 22% of business-as-usual emissions are inefficient relative to the planner's allocation. Remaining reductions are more expensive both in abatement costs (∼\$0.42 more per ton, or 32% more expensive) and in supporting transfers (nearly a 5-fold increase).

Regrowth accounting reallocates emissions reductions northwards. Tropical plots have regrowth medians of 4 or more tC/ha/year, meaning that 86% of tropical aboveground carbon loss is discounted away. Temperate/boreal plots have regrowth medians of 1–2 tC/ha/yr, so only ∼65% is discounted. Relative damages fall more sharply in the fast-regrowing tropics, so the planner rationally reallocates whatever abatement remains toward the temperate/boreal margin where the discount bites less. Gross transfers increase because the trade-exposed (see Figure 2) global northern countries require larger levels of transfers to rectify price changes.

Table A.11: First-best allocation under different carbon accounting strategies.

| Metric | Baseline | With belowground | Net of regrowth |
|---|---|---|---|
| Welfare gain (%) | 0.04 | 0.05 | 0.00 |
| Welfare gain (\$/ton $CO_2$) | 2.5 | 2.553 | 2.082 |
| Crop production change (%) | -1.75 | -2.09 | -0.29 |
| Cattle production change (%) | -4.05 | -5.11 | -0.68 |
| Emissions change (%) | -56.44 | -59.92 | -21.56 |
| Gross transfers per $tCO_2$ abated (\$/$tCO_2$) | 6.61 | 5.57 | 30.63 |

NOTES: Each column reports the utilitarian planner's first-best allocation, implemented as a global optimal carbon tax with access to lump-sum transfers $\mathcal{F}_i$. BASELINE is the first bootstrap draw of Table 4, which uses Equation (1) to calculate emissions. WITH BELOWGROUND adds soil organic carbon plus roots and shoots and assumes complete release at the clearing event. This is deliberately a bound rather than a physical model, since IPCC accounting attributes root-shoot emissions of roughly 47% of aboveground carbon to the event itself (IPCC 2006). NET OF REGROWTH credits the plot with carbon resequestered after a single year of use, using plot-level regrowth rates $r(\omega)$ (tC/ha/yr) from Cook-Patton et al. (2020), with missing plots filled by country-biome means; regrowth is linear until cumulative regrowth reaches $d(\omega)$ and zero thereafter, discounted at 2% annually.

## E.4 What are the co-benefits between biodiversity protection and deforestation $CO_2$ reductions?

In this section, I perturb the first-best allocation to marginally tax losses of threatened species habitat. This uncovers the local complementarity (co-benefits) between the optimal $CO_2$-driven policy and a policy which additionally values threatened species loss. This exercise is illustrative and makes no efficiency claim: threatened species habitat is only one dimension of biodiversity loss (Baylis, Garcia, and Heilmayr 2026).

To quantify this local complementarity of optimal carbon policy for biodiversity, I introduce a biodiversity price $t^s$. The tax is denominated in dollars per hectare of threatened animal habitat. For example, a potential hectare of agricultural land with three threatened species faces a biodiversity-related tax burden of $3t^s$. Starting at the planner's optimal solution with transfers, I calculate equilibrium outcomes for a grid of values of $t^s$:

$$\pi_\epsilon^h(\omega; t_i, t^s) = \pi_\epsilon^h(\omega; t_i, 0) - \frac{t^s}{P_i^C}\text{species}(\omega) \quad h \in \{A, P\}$$

In this framework, co-benefits of carbon for biodiversity protection arise from the spatial endowments of carbon and threatened species. If highly valuable threatened species are largely found in carbon dense regions, then marginal pricing of biodiversity loss will have limited effects on land use after carbon is already optimally priced. I again emphasize the illustrative nature of this exercise: the social value of "biodiversity" varies across species and geography. While biodiversity may offer certain global public goods (e.g., option value for future medical innovation), it is also certainly local. However, biodiversity objectives have been coordinated at global scale since the Ramsar Convention through land protection efforts (Grupp et al. 2026).

Figure A.8 illustrates results. In Figure A.8a, the welfare cost of marginal biodiversity prices rises sublinearly. If threatened species have a positive value, the gains from pricing are positive and, up to a value of $5/ha, growing in the value of species. This is a global projection: locally varying biodiversity values may induce a different cost structure depending on whether more socially valuable species are located in high-abatement cost regions with valuable agriculture. In Figure A.8b, I present a measure of the co-benefits of emissions reductions for biodiversity pricing. The average elasticity of emissions reductions to threatened species habitat loss reductions is between 0.55-0.85 and increasing in the biodiversity price. This suggests that the two objectives are locally complementary at the social optimum I outlined in the main text.

### E.5 Varying the social cost of carbon.

I recover an average abatement cost curve under various social costs of carbon using an unrestricted planner with access to a carbon tax instrument and cross-country transfers. In Figure A.9, I recover a convex abatement cost curve for deforestation prevention at global scale. The vertical dashed line indicates the focal social cost of carbon in this paper.

Per Lemma 1, the low average abatement cost at every tax rate implies that nearly all abatement by tonnage occurs at relatively low marginal abatement cost. This result is entirely consistent with the tail of low-cost abatement in summary statistics from Table 1. There are large gains from very small carbon taxes.

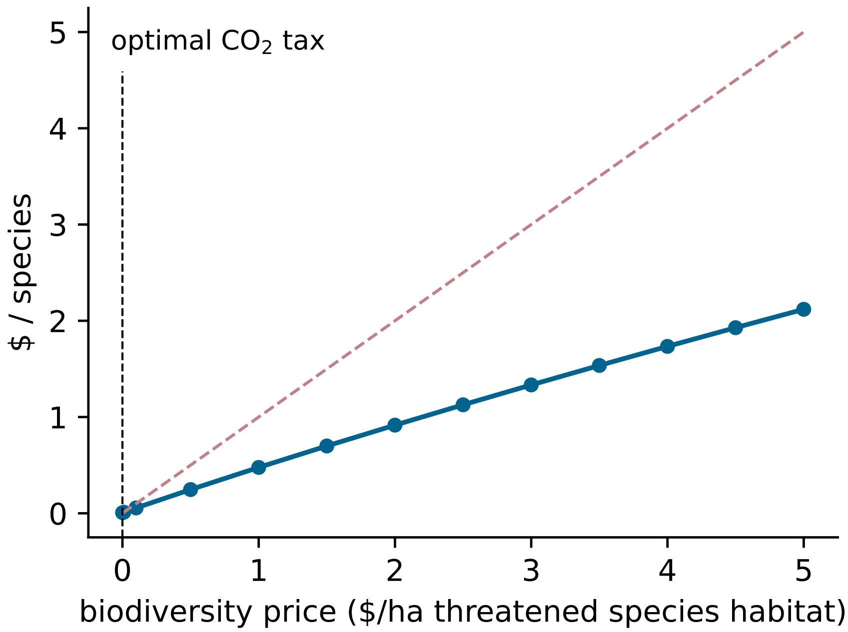
### E.6 Non-utilitarian social planners.

The main text focuses on a utilitarian planner who uses a carbon tax and transfer instruments to obtain global first-best. This presentation choice does not affect core quantitative findings. To demonstrate, I present results for the Negishi solution, which decentralizes the planner allocation by solving for planner weights $\psi_i$ from the planner's objective (A.1) such that transfers are unnecessary. I also present results for a planner who uses transfers to instead equalize the welfare gains from the tax across countries, so every country experiences the same proportional increase from the policy.

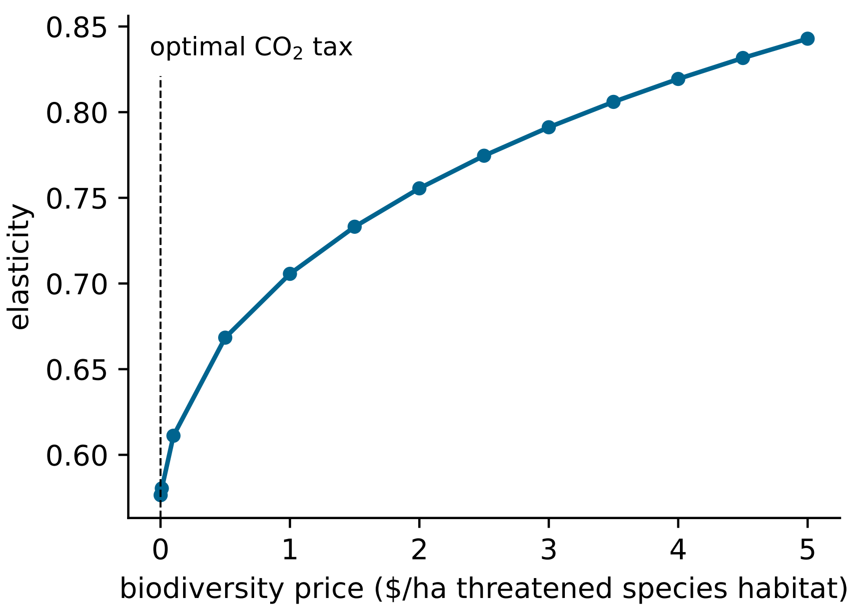
In Table A.12, all three planner's regimes are presented. Welfare gains are nearly identical across mechanisms (though, to be precise: total welfare is the utilitarian metric, so total utilitarian welfare is highest under the utilitarian solution). The utilitarian planner uses transfers to sustain a slightly higher emissions reduction at great cost to global crop supply. The cost of equity in welfare terms is on average 0.0022% of global GDP per year (6% of utilitarian gains), and transfers are modest

Figure A.8: Results of marginal biodiversity pricing

(a) Welfare

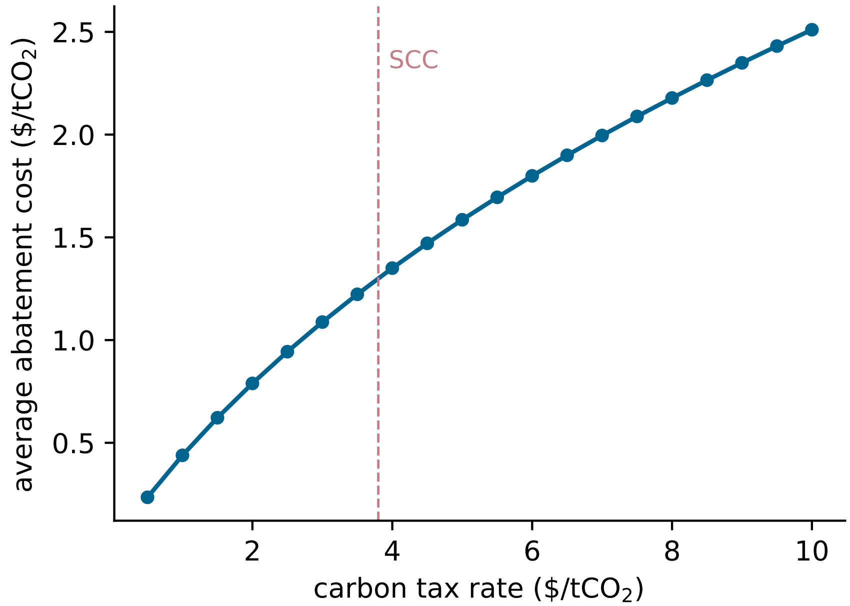



(b) Emissions-species tradeoff




NOTES: Origin is the planner's optimal tax with access to transfers. $x$-axis reflects $t^s$, the price per hectare of threatened species habitat loss imposed by the planner. Top shows welfare costs per species without including any benefits from biodiversity in welfare, with a dashed 45 degree line. Bottom shows cross-elasticity of emissions percent change in the percent loss in threatened species habitat.

Figure A.9: Average abatement costs when varying the planner's $CO_2$ valuation




NOTES: $x$ axis plots the optimal carbon tax rate under successive marginal damages values $\kappa$ in (13) when a planner has access to carbon taxes and transfers. Under Proposition 1, the $x$-axis is just $\kappa$. Dashed line at $3.80/t$CO_2$/year is the value used in the main text. $y$-axis plots the average abatement cost defined in Lemma 1.

relative to the utilitarian solution. All three allocations are feasible with positive weights on all countries $\psi_i > 0$.

### E.7 Imperfect information and the value of granularity.

Suppose the planner does not observe plot-level carbon measurements, instead relying on country-level average carbon. The FAO Forest Resources Assessment and the IPCC both report regional woody biomass averages: both are used in offset accounting documents. Coarse measurement can prevent gains from carbon trade as policy no longer discriminates between high- and low-emissions land within a country.

I evaluate policy where a planner chooses a tax rate subject to the constraint that the tax is applied at country-level average $CO_2$ intensities,

$$t \frac{1}{N_i} \sum_{\omega \in \Omega_i} d(\omega)$$

The naive policy sets the Pigouvian tax $t = \kappa$ from Proposition 1 as usual. The ex post optimal

Table A.12: Optimal policy under different social planner weights

| | Negishi | Utilitarian | Equal-gain |
|---|---|---|---|
| Welfare gain (%) | 0.0347 | 0.0364 | 0.0342 |
| | [0.0334, 0.0357] | [0.0351, 0.0374] | [0.0330, 0.0353] |
| Welfare gain (\$/ton $CO_2$) | 2.477 | 2.500 | 2.457 |
| | [2.448, 2.514] | [2.470, 2.536] | [2.427, 2.495] |
| Emissions reduction (%) | 54.27 | 56.44 | 53.98 |
| | [52.97, 55.20] | [55.14, 57.36] | [52.67, 54.93] |
| Crop production change (%) | -0.267 | -1.753 | -0.079 |
| | [-0.468, -0.105] | [-1.996, -1.569] | [-0.291, 0.095] |
| Cattle production change (%) | -3.781 | -4.053 | -3.722 |
| | [-3.958, -3.474] | [-4.255, -3.740] | [-3.897, -3.411] |
| Gross transfers (\$B) | 0.0 | 74.3 | 12.0 |
| | [0.0, 0.0] | [72.2, 76.0] | [11.6, 12.3] |

NOTES: First column decentralizes planner allocation with 0 transfers $\mathcal{F}_i$. Second column is the main text specification: a utilitarian planner with access to both transfers and a carbon tax. Third column is a planner who prefers all countries experience an equal proportional change to welfare with access to both transfers and a carbon tax.

policy allows the planner to set an optimal tax rate, which will generically deviate from the Pigouvian rule because the targeting of the Pigouvian instrument is blunted. Per (15), the gains from taxation are *quadratic* in the emissions of land — capturing the upper tail is essential to gains from carbon trade.

Before presenting results, it is worth commenting that this exercise is stylized: it imagines a planner who can in real time shade their tax rate to account for the blunt targeting of the instrument.

Results are in Table A.13. Column (1) replicates the socially optimal full information-environment outcome from Table 4 for the central bootstrap draw. Column (3) shows the naive policy, just applying the same tax without adjusting targeting, creates a welfare *loss* — \$4.78 of welfare loss per ton abated. Strikingly, the level of emissions reductions is similar, but these reductions are completely spatially mistargeted.

The ex post optimal tax in column (2) is \$0.75/t$CO_2$, much lower than the social value of $CO_2$. Without, granular measurement, 68% of efficient $CO_2$ emissions reductions cannot occur. Remaining emissions reductions are more costly (\$0.46 more per ton) because of poor targeting, and overall the carbon tax achieves only 26% of first-best welfare. Put differently, the planner would pay \$20.8 billion for a technology which provides accurate $CO_2$ measurement.

## E.8 Free riding under alternative willingness to pay.

The importance of free riding in Figure 6a is driven in part by the "polluter pays," GDP-based approach to allocating the global social cost of carbon $\kappa$ to individual country-level willingness to pay $\chi_i\kappa$. If forest basins have a higher willingness to pay for carbon, the role of free riding will fall: they internalize more of the *benefit* of the carbon tax.

This section explores the local sensitivity of the free riding share of planner welfare gains which is driven by the specific choice of willingness to pay in each of the three large forest basins: Brazil, the Congo River Basin, and Indonesia. To test sensitivity, I reassign willingness to pay for $CO_2$ in these regions to total $\alpha\kappa$, allocating by GDP shares within $\alpha\kappa$. Other countries' willingness to pay for $CO_2$ comprises the residual $(1-\alpha)\kappa$, also allocated on a GDP-share basis. As $\alpha$ grows, the major tropical forest basins capture a larger share of the benefits of carbon taxation, while the remaining set of countries experiences a small incentive to tax $CO_2$. At each value of $\alpha$, I recompute the Nash and Nash-Internalized equilibrium decomposition from 7.1.

Figure A.10 shows that as the tropical band internalizes more of the social cost of $CO_2$, the overall gap between the planner and the non-cooperative solution shrinks. However, even when the forest basins internalize the full social cost of carbon collectively, the free riding gap remains large. Welfare costs from cross-country spillovers are a growing share of the total gap in $\alpha$ only because the free riding gap falls: they remain constant in levels.

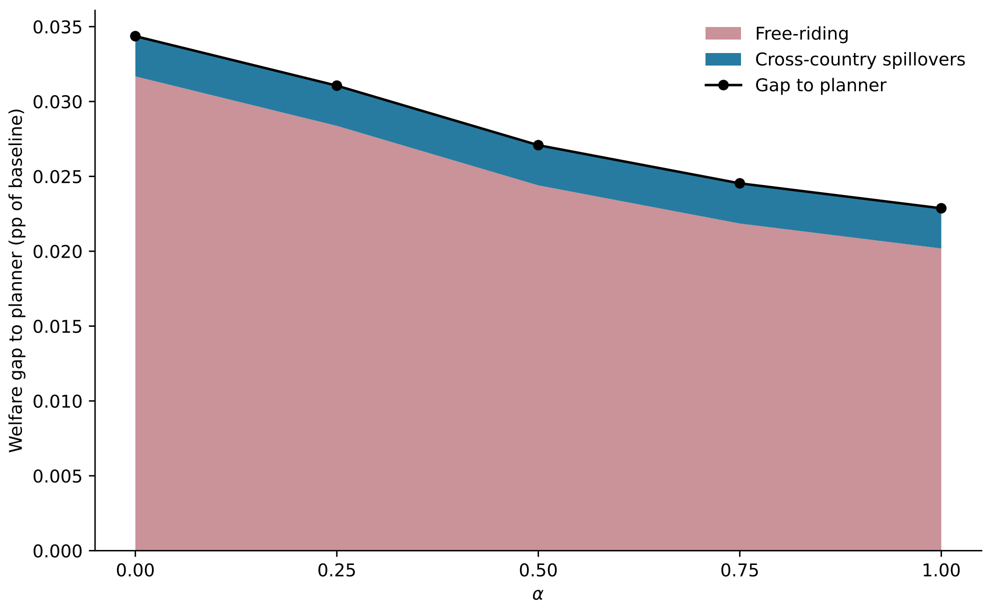
## E.9 Country-level tax rates under alternative demand elasticities.

I present two alternative demand-side parameterizations in Table A.14. All three columns recompute the planner's solution without access to transfers, as by construction the utilitarian planner can use transfers to smooth any income implications of alternative demand parameters.

In Column (2), I maintain $\sigma^P$ and $\sigma^K$ as in Table 3, but set $\sigma^A = 8$, a much more elastic value used in Caliendo and Parro (2015). Welfare, production, and emissions are all fairly unaffected by the alternative parameterization.

In Column (3), I maintain all the values of $\sigma^A$ and $\sigma^P$ as in Table 3 but allow for gross complementarity between meat and crop production. With gross complementarities, the planner still abates a similar amount of emissions. However, the planner reduces crop production more and sustains more meat production. Intuitively, when consumers experience higher relative cattle prices to crop prices, gross complementarity suggests they demand less food production in equilibrium. This force puts downward pressure on the incentive to reallocate land towards crop production. Welfare gains are higher because prices do not rise as much.

Figure A.10: Welfare gap between Nash and planner under varying country-level social cost allocation rules.




NOTES: $x$ axis plots the share $\alpha$ of the global social cost of carbon $\kappa$ which I allocate to the three major forest basins: Indonesia, Brazil, and the Congo River Basin. Within these basins, $\alpha\kappa$ is divided up on a GDP basis; among other countries, $(1-\alpha)\kappa$ is divided the same way. I resolve the Nash and Nash-internalized equilibrium at 5 grid points using the definition described in 7.1. The planner's solution is always the same.

### E.10 Nonhomothetic preferences.

I explore how results change when, in the outer nest of my demand system, consumers trade off food and other goods according to nonhomothetic preferences. I adopt a simplified preference based on Comin, Lashkari, and Mestieri (2021) in the outer nest in place of the homothetic preference reported in the main text. The representative consumer trades off the food composite $K$ and the non-food composite $M$ according to a nonhomothetic constant elasticity of substitution aggregator, defined implicitly by:

$$\sum_{h\in\{K,M\}} \left(\theta_j^h\, X_j^h\, U_j^{-\varepsilon^h}\right)^{\frac{\sigma-1}{\sigma}} \;=\; 1, \tag{A.10}$$

where the new parameters $\varepsilon^h > 0$ are the income-elasticity parameters for each sector $h$.

Income-elasticity parameters are identified only up to a common location, so I set $\varepsilon^M \equiv 1$. In the three-sector system of Comin, Lashkari, and Mestieri (2021), $\varepsilon^A = 0.11$. This parameter does not map directly to my setting: I do not model a services sector.

The solution to the consumer cost-minimization problem $\min_{\{X_j^h\}} \sum_h P_j^h X_j^h$ subject to (A.10) satisfies

$$X_j^h = (\theta_j^h)^{\sigma-1}\,(P_j^h)^{-\sigma}\,Y_j^\sigma\,U_j^{\varepsilon^h(1-\sigma)},$$
$$\lambda_j^h \;\equiv\; \frac{E_j^h}{Y_j} = (\theta_j^h)^{\sigma-1}\left(\frac{P_j^h}{Y_j}\right)^{1-\sigma} U_j^{\varepsilon^h(1-\sigma)}.$$

Define the unit expenditure $P_j^C \equiv \bar{Y}_j/U_j$, where $\bar{Y}_j$ is the amenity- and transfer-adjusted expenditure of the welfare appendix. Unlike the homothetic case, $P_j^C$ depends on the reference utility level, so welfare comparisons across counterfactuals must be stated as equivalent variation at a common reference $U_j$. The marginal utility of income is $\frac{\partial U_j}{\partial \bar{Y}_j} = \frac{1}{\bar{\varepsilon}_j\, P_j^C}$, where $\bar{\varepsilon}_j$ is the expenditure-weighted average income elasticity. In Appendix A.1, every occurrence of $\psi_j/P_j^C$ in the planner's first-order conditions is therefore replaced by $\psi_j/(\bar{\varepsilon}_j P_j^C)$. The Pigouvian benchmark is unchanged, but transfers must accommodate nonhomotheticities.

Table A.15 asks whether the income-inelasticity of food overturns any of the results presented under homothetic preferences. On efficiency it does not. Aggregate abatement is unchanged. There is slight substitution away from cattle into more food consumption. Income effects thus have little effect on aggregates.

Two objects differ. Gross transfers fall from \$6.62 to \$5.80 per $CO_2$ abated. And the incidence of the policy is less regressive: the rank correlation of transfer burden $\mathcal{F}_i/Y_i^0$ with baseline income

Table A.13: Taxing $CO_2$ under imperfect measurement.

| Scenario<br>Information Environment | Baseline optimal<br>Full<br>(1) | Ex post optimal<br>Partial<br>(2) | Naive<br>Partial<br>(3) |
|---|---|---|---|
| Optimal tax ($/t$CO_2$) | 3.80 | 0.75 | 3.80 |
| Welfare gain ($/t$CO_2$) | 2.500 | 2.034 | -4.782 |
| $\Delta$ Emissions (%) | -56.44 | -17.94 | -55.75 |
| $\Delta$ $Q_{ag}$ (%) | -1.75 | -1.17 | -6.98 |
| $\Delta$ $Q_{cattle}$ (%) | -4.05 | -6.48 | -24.06 |
| Welfare gain ($ billion) | +28.06 | +7.26 | -53.01 |

NOTES: Full information environment replicates the first bootstrap draw of Table 4, where planner taxes plots on $d(\omega)$, the welfare-relevant emissions quantity. Partial information environment — planner taxes plots on country-level average of plot-level emissions $\frac{1}{N_i}\sum_{\omega\in\Omega_i} d(\omega)$. Baseline optimal replicates utilitarian optimal Pigouvian tax with access to transfers. Ex post optimal shows welfare-maximizing tax when planner constrained to tax under partial info environment. Naive shows what happens when planner continues to tax at baseline optimal tax but on imperfectly measured tax base.

Table A.14: Sensitivity analysis on demand parameters

| | Baseline<br>(1) | $\sigma^A = 8$<br>(2) | $\sigma^K = 0.5$<br>(3) |
|---|---|---|---|
| Welfare gain (%) | 0.03 | 0.04 | 0.03 |
| Welfare gain ($/ton $CO_2$) | 2.476 | 2.490 | 2.474 |
| Crop production change (%) | -0.27 | -0.25 | -0.79 |
| Cattle production change (%) | -3.78 | -3.84 | -3.01 |
| Emissions change (%) | -54.29 | -55.99 | -54.00 |

NOTES: (1) presents results of a random bootstrap draw from baseline specification reported in Table 4. (2) shows results for the same bootstrap draw when $\sigma^A = 8$, the value which is reported in Caliendo and Parro (2015) Table 1. (3) shows results for the same bootstrap draw when $\sigma^K = 0.5$.

falls from +0.48 to +0.28, and the rank correlation of crop losses with baseline income falls from +0.25 to $\sim 0$.

Table A.15: Comparing utilitarian planner's solution under homothetic (main paper) vs. non-homothetic (robustness).

| Metric | CES | NHCES | Ratio (NH/CES) |
|---|---|---|---|
| Global abatement (%) | -56.44 | -56.45 | 1.000× |
| Global $\hat{Q}^A$ (%) | -1.75 | -1.45 | 0.826× |
| Global $\hat{Q}^P$ (%) | -4.05 | -4.27 | 1.054× |
| Gross transfers $\lvert\mathcal{F}\rvert/2$ (\$B) | 74.25 | 65.22 | 0.878× |
| Transfers per $tCO_2$ abated (\$/t) | +6.615 | +5.809 | 0.878× |
| Welfare per $tCO_2$ (\$/t) | +2.500 | +3.118 | 1.247× |
| *Incidence: Spearman rank correlation with baseline income* $Y_i^0$ | | | |
| Transfer share: $\rho(\mathcal{F}_i/Y_i^0,\, Y_i^0)$ | +0.481 | +0.275 | -0.205 |
| Emissions: $\rho(\hat{D}_i,\, Y_i^0)$ | -0.251 | -0.282 | -0.031 |
| Crop loss: $\rho(\hat{Q}_i^A,\, Y_i^0)$ | +0.253 | +0.021 | -0.233 |
| Cattle loss: $\rho(\hat{Q}_i^P,\, Y_i^0)$ | -0.043 | -0.006 | +0.037 |

NOTES: Utilitarian planner implements a global optimal carbon tax with access to lump-sum transfers $\mathcal{F}_i$. CES column replicates the first bootstrap draw of Table 4. NHCES refers to the calibration of Comin, Lashkari, and Mestieri (2021) with two sectors at their estimated parameter values, $\varepsilon^A = 0.11$ (agriculture, a necessity) and $\varepsilon^M = 1$ (manufacturing, homothetic). Incidence block: positive $\rho$ means rich countries do better than low-income (regressive); negative $\rho$ means low-income countries do better (progressive). The NH−CES column reports the *difference* (not a ratio) since correlations bracket to $[-1, 1]$.

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